

Field Deployment Handbook

Getting Started

To get started, you must at least go through NavWiz and DFleet training.

Source of Patch Files

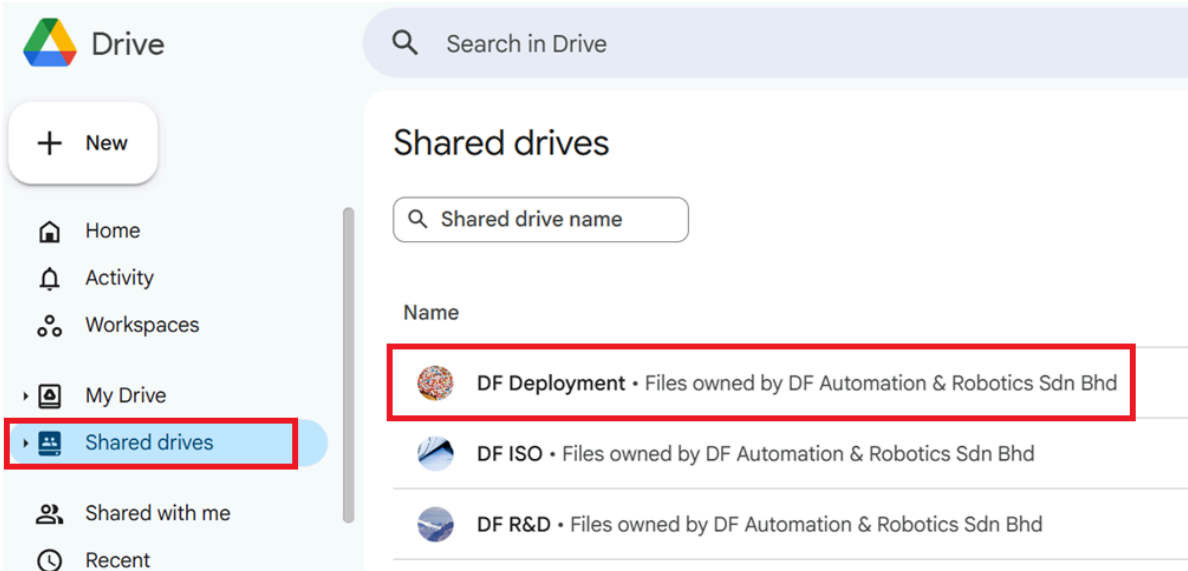
- Always get the official ISO and patch files from <https://releases.dfautomation.com/>

Source of Product Specs

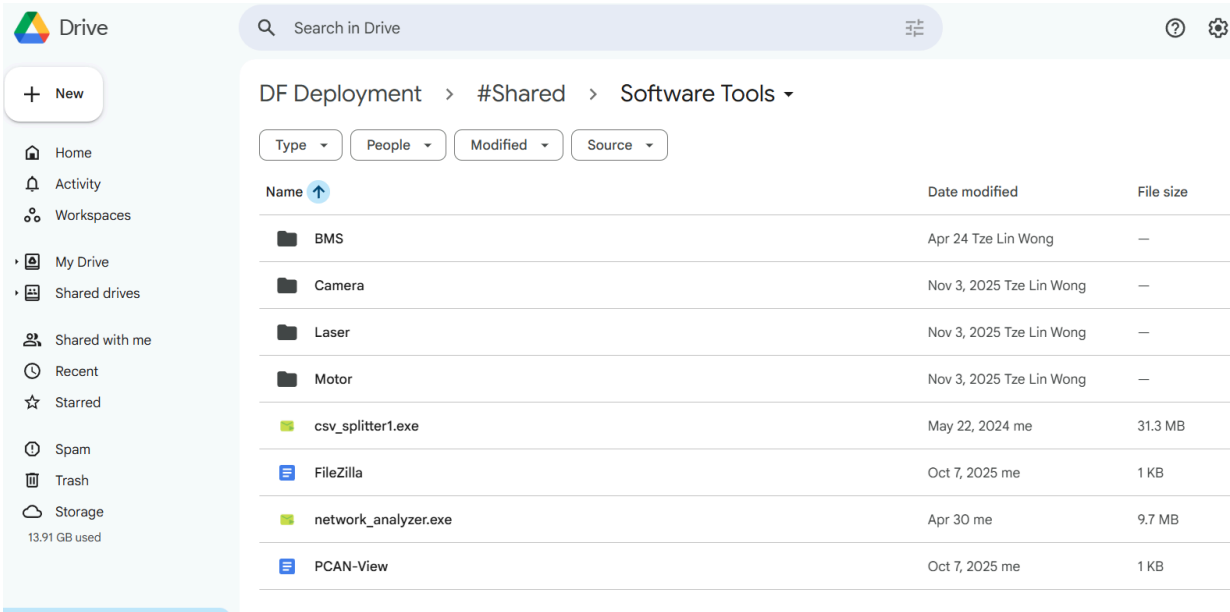
- ?

Shared Drive File Structure

- You should have access to this Shared Drive:



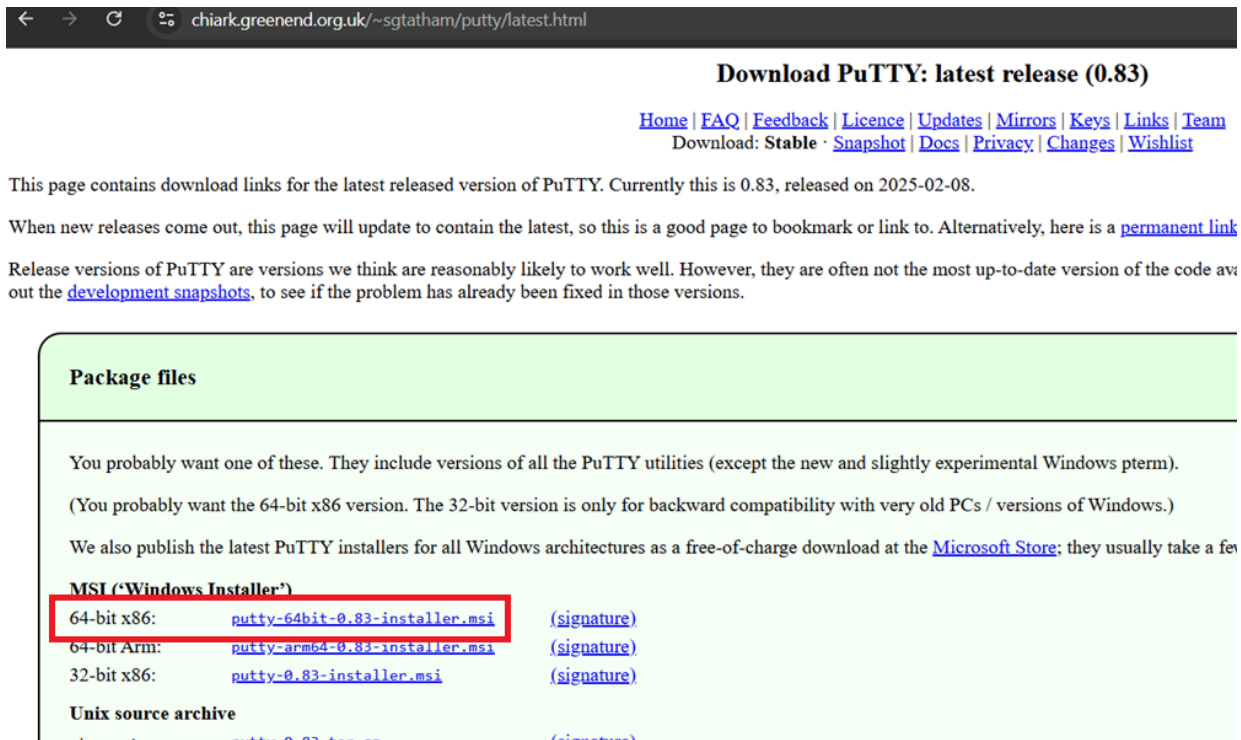
- Software Tools



Essential Linux and CLI Fundamentals

Remote Access via SSH

- To access the terminal of a robot's IPC or a Virtual Machine (VM) remotely, engineers use the SSH protocol via PuTTY.
- Download PuTTY from here: <https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>



The screenshot shows the PuTTY download page. At the top, it says "Download PuTTY: latest release (0.83)". Below this are links for Home, FAQ, Feedback, Licence, Updates, Mirrors, Keys, Links, and Team. There are also links for Download: Stable, Snapshot, Docs, Privacy, Changes, and Wishlist. The page states that the current version is 0.83, released on 2025-02-08. It mentions that when new releases come out, the page will update. It also notes that release versions are chosen for reliability, but development snapshots are available for the most up-to-date code. A section titled "Package files" lists various download options. A red box highlights the "MSI ('Windows Installer')" section, which includes links for 64-bit x86, 64-bit Arm, and 32-bit x86 installers, each with a signature link. Below this is a section for "Unix source archive".

Download PuTTY: latest release (0.83)

[Home](#) | [FAQ](#) | [Feedback](#) | [Licence](#) | [Updates](#) | [Mirrors](#) | [Keys](#) | [Links](#) | [Team](#)
Download: [Stable](#) · [Snapshot](#) | [Docs](#) | [Privacy](#) | [Changes](#) | [Wishlist](#)

This page contains download links for the latest released version of PuTTY. Currently this is 0.83, released on 2025-02-08.

When new releases come out, this page will update to contain the latest, so this is a good page to bookmark or link to. Alternatively, here is a [permanent link](#)

Release versions of PuTTY are versions we think are reasonably likely to work well. However, they are often not the most up-to-date version of the code available out the [development snapshots](#), to see if the problem has already been fixed in those versions.

Package files

You probably want one of these. They include versions of all the PuTTY utilities (except the new and slightly experimental Windows pterm).
(You probably want the 64-bit x86 version. The 32-bit version is only for backward compatibility with very old PCs / versions of Windows.)

We also publish the latest PuTTY installers for all Windows architectures as a free-of-charge download at the [Microsoft Store](#); they usually take a few minutes to download.

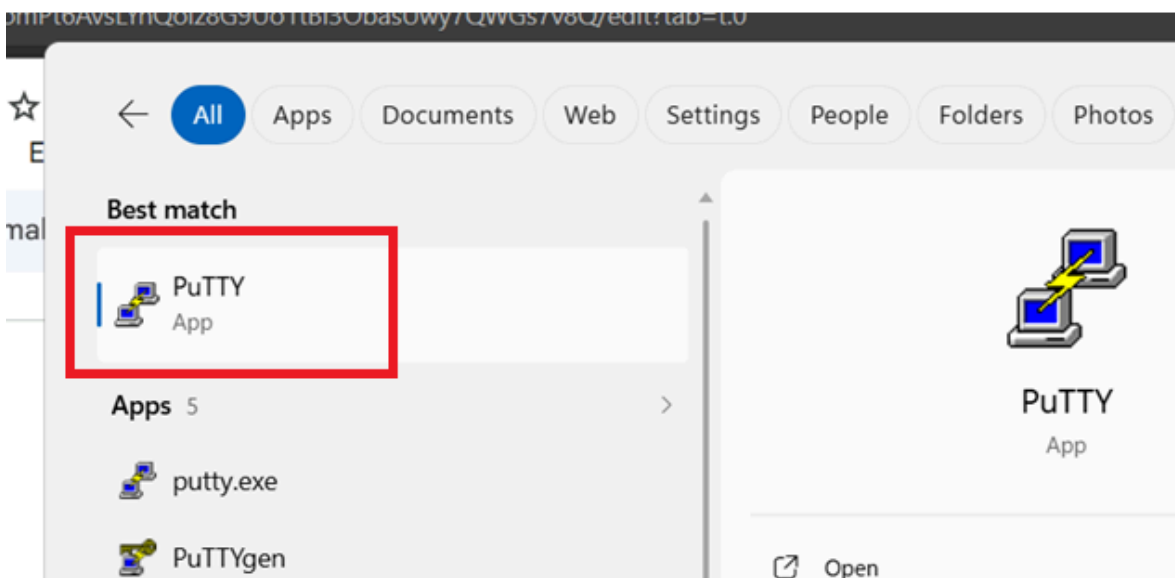
MSI ('Windows Installer')

64-bit x86:	putty-64bit-0.83-installer.msi	(signature)
64-bit Arm:	putty-arm64-0.83-installer.msi	(signature)
32-bit x86:	putty-0.83-installer.msi	(signature)

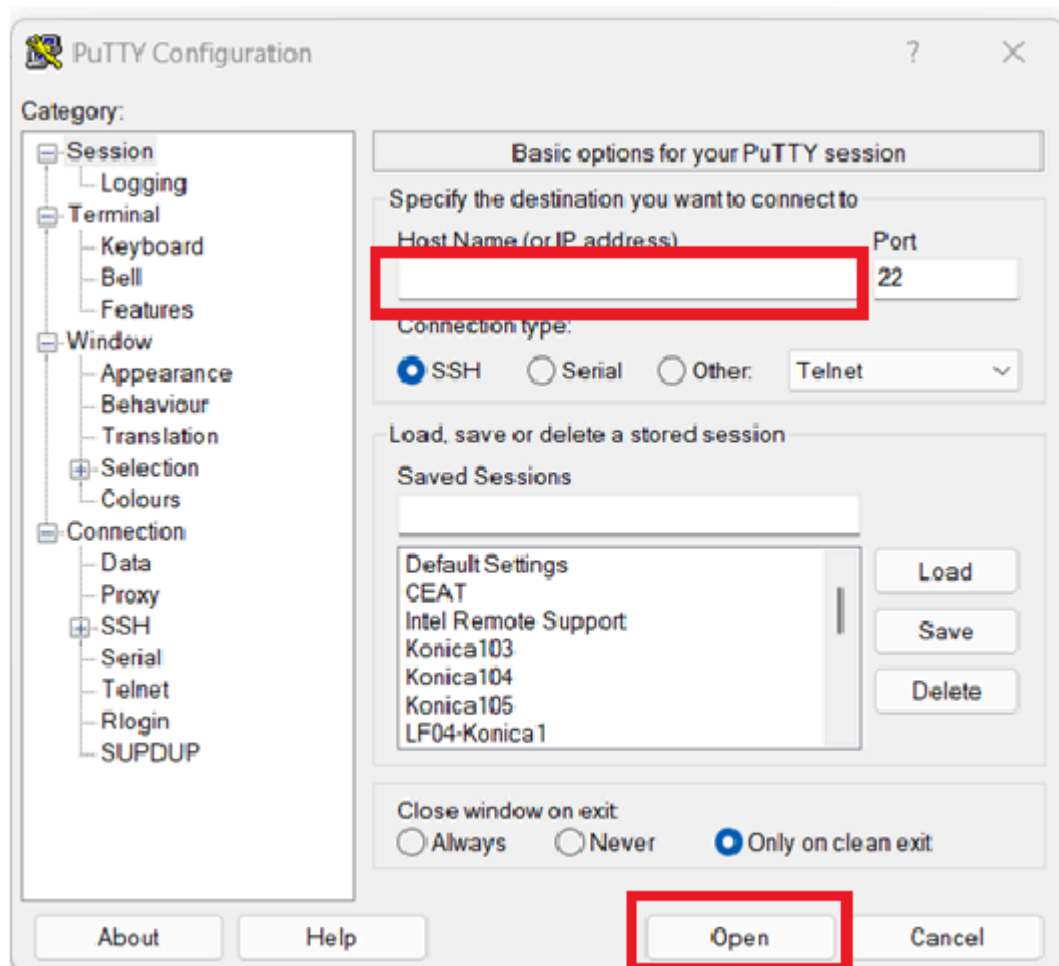
Unix source archive

64-bit x86:	putty-0.83.tar.gz	(signature)
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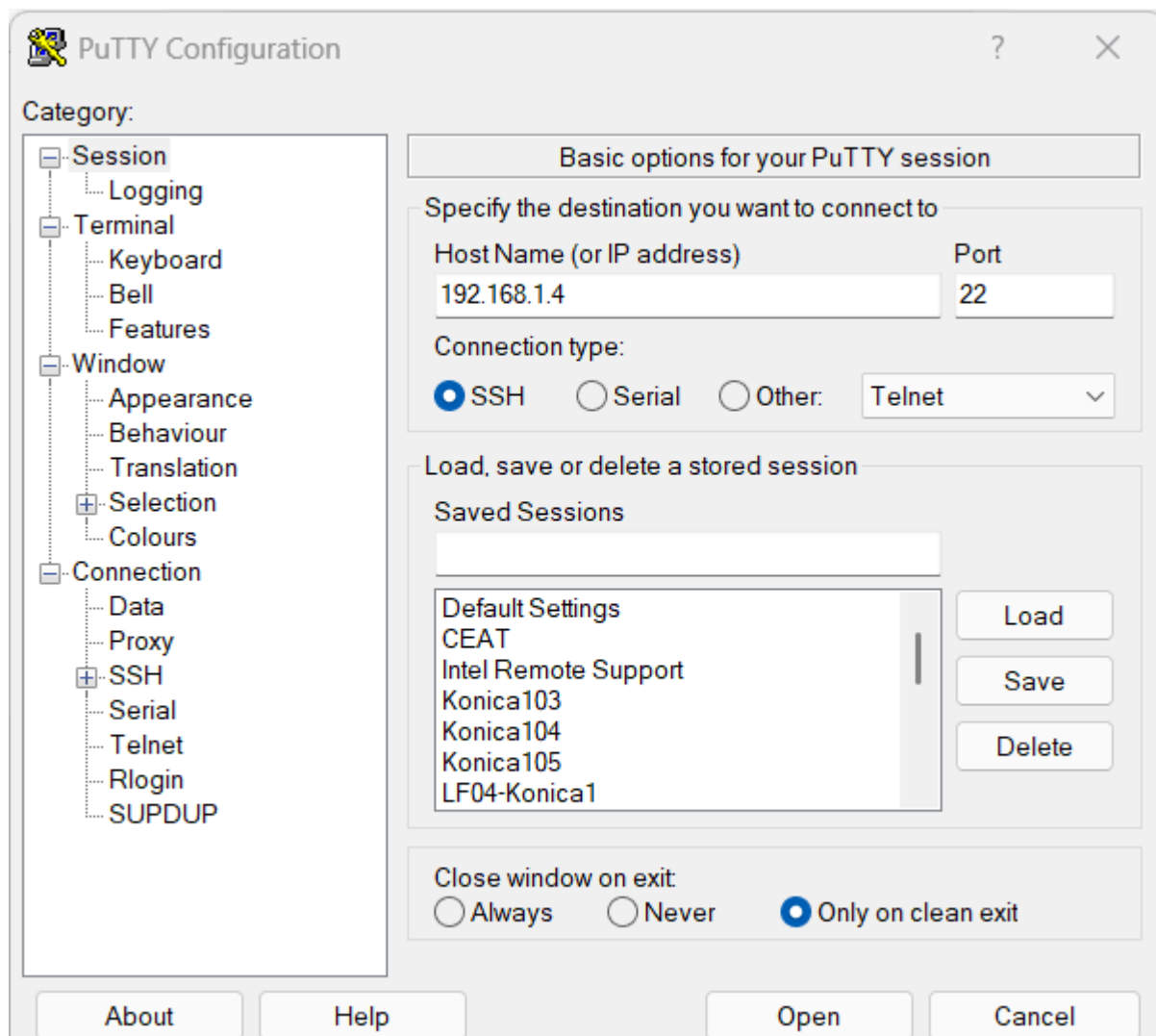
- Before using PuTTY, **kindly ensure** your laptop is connected to the same network as the robot's IPC.
- To use PuTTY, run the app.



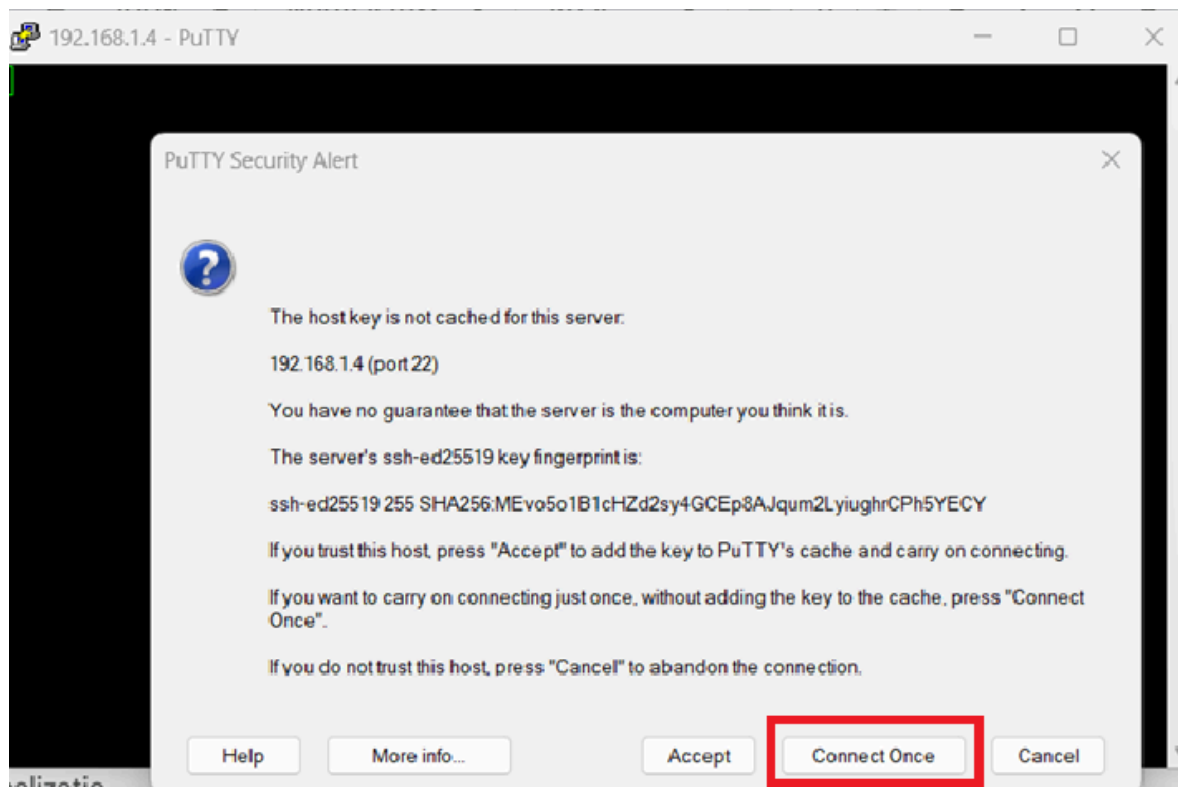
- Key in the ip address of the device that you want to SSH, and then click Open.



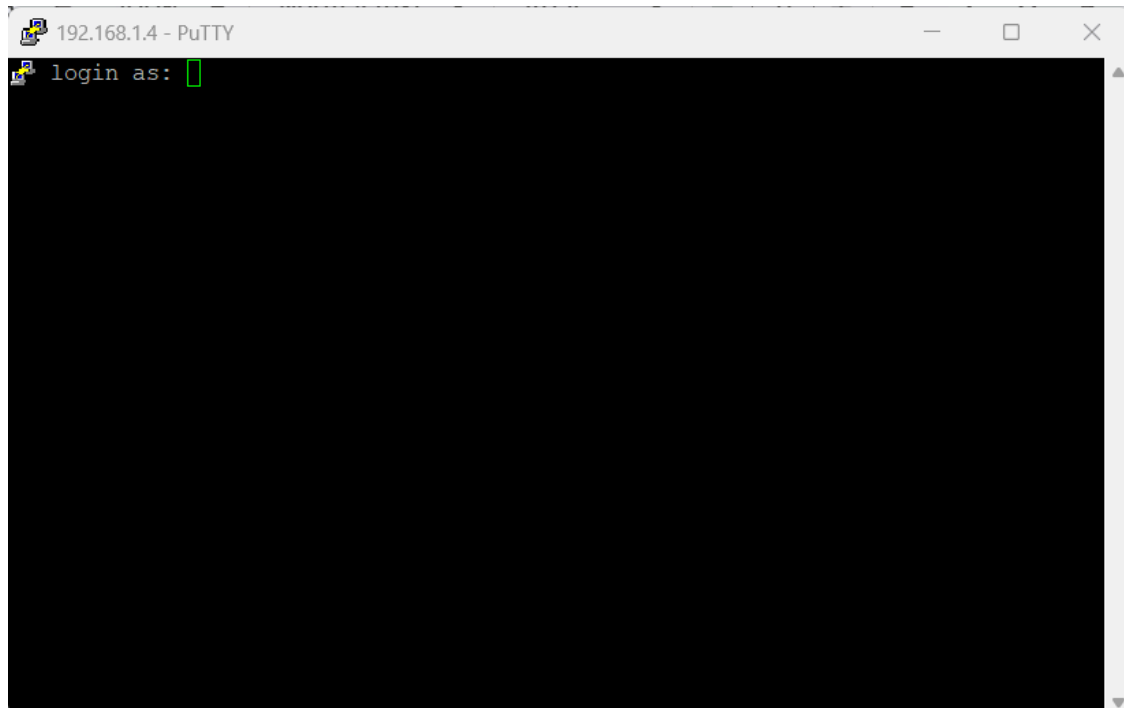
- For example, I want to SSH to a device with IP address 192.168.1.4



- Then this window will pop up, and click “Connect Once”:



- Then you will need to provide the username and password of that device:



- Standard Credentials: Use Username: df and Password: df

Opening web browser when DFleet/NavWiz is in Kiosk mode

- To open a web browser, simply press Ctrl + N

Switching between Kiosk display and Terminal Display

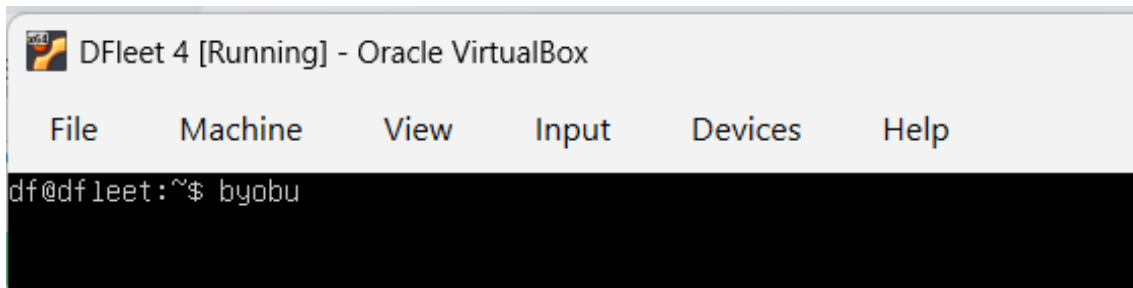
- By default the DFleet server and NavWiz IPC will display the kiosk (the UI interface you see). However, we can actually switch between kiosk mode and terminal mode by using the keyboard.
- To enter terminal mode, press Ctrl + Alt + F3 , then it will prompt for username and password:

```
Ubuntu 20.04.6 LTS dfleet tty3
dfleet login:
```

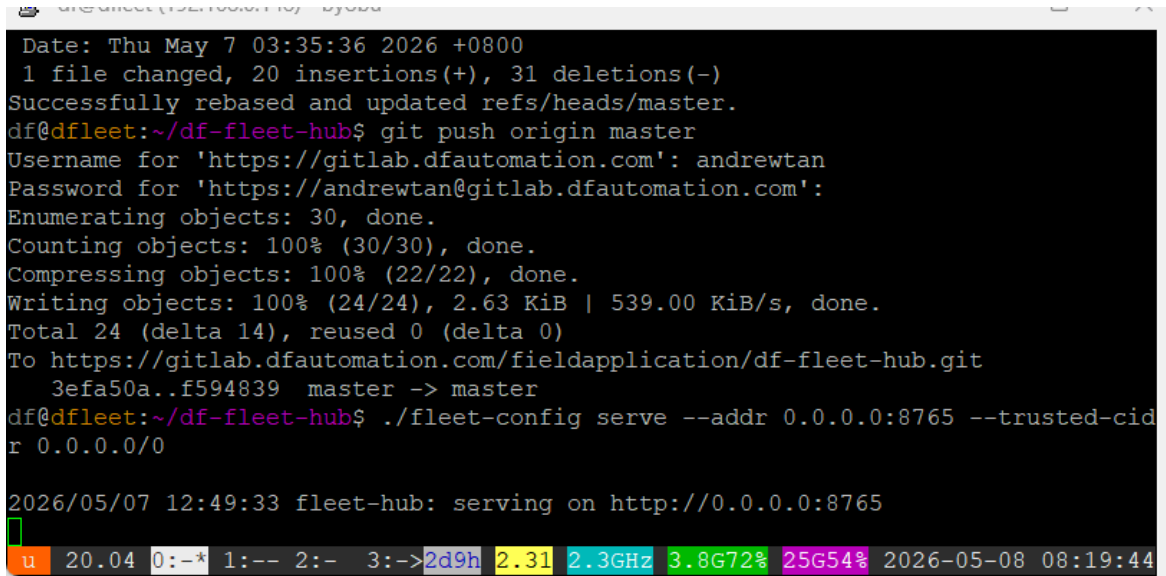
- To switch back to kiosk display:
 - For NavWiz, press Ctrl + Alt + F7
 - For DFleet, press Ctrl + Alt + F2

Using Byobu

- Byobu is a text-based window manager and terminal multiplexer that is by default installed on many Unix distributions, so our DFleet server NavWiz IPC will have it usually.
- This is really needed to know because sometimes you may need to run multiple terminals.
- To run byobu, basically just type byobu and press enter.



- In byobu mode, you can open multiple tabs.
 - To open a new tab, press F2
 - To switch to previous tab, press F3
 - To switch to next tab, press F4



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UNIX Basic Commands

- pwd command
 - Syntax
`pwd`
 - To check current directory
 - Example:

```
df@dfleet:~$ pwd
/home/df
```

- ls command
 - Syntax
`ls [options]`
 - To list files and directories present in the current working directory
 - Example:

```
df@dfleet:~$ ls
agvccs_4.0.0~rc1+nightly.20260129.193310~de7b48e-0focal_amd64.deb
agvccs-rs_4.0.0~rc1+nightly.20260129.194315~1416614_amd64.deb
agvccs-rs_4.0.0~rc1+nightly.20260203.042522-b9568a2_amd64.deb
dfleet
'dfleet-file '
navwiz-fms_4.0.0~rc1+nightly.20260111.172744~bb0ce0e-0focal_amd64.deb
```

- Example to list everything including hidden files:

```
df@dfleet:~$ ls -a
.
..
agvccs_4.0.0~rc1+nightly.20260129.193310~de7b48e-0focal_amd64.deb
agvccs-rs_4.0.0~rc1+nightly.20260129.194315~1416614_amd64.deb
agvccs-rs_4.0.0~rc1+nightly.20260203.042522-b9568a2_amd64.deb
.bash_history
.bash_logout
.bashrc
.byobu
.cache
.config
dfleet
'dfleet-file '
navwiz-fms_4.0.0~rc1+nightly.20260111.172744~bb0ce0e-0focal_amd64.deb
.profile
.ssh
.sudo_as_admin_successful
.viminfo
.Xauthority
```

- mkdir command

- Syntax

```
mkdir [directory_name]
```

- To make directory

- Example:

```
df@dfleet:~$ mkdir example_folder
df@dfleet:~$ ls
agvccs_4.0.0~rc1+nightly.20260129.193310~de7b48e-0focal_amd64.deb
agvccs-rs_4.0.0~rc1+nightly.20260129.194315~1416614_amd64.deb
agvccs-rs_4.0.0~rc1+nightly.20260203.042522-b9568a2_amd64.deb
dfleet
'dfleet-file '
example_folder
navwiz-fms_4.0.0~rc1+nightly.20260111.172744~bb0ce0e-0focal_amd64.deb
```

- cd command

- Syntax

```
cd [directory_name]
```

- To change directory

- Example:

```
df@dfleet:~$ cd example_folder/
df@dfleet:~/example_folder$ pwd
/home/df/example_folder
```

- Example to go to previous directory:

```
df@dfleet:~/example_folder$ pwd
/home/df/example_folder
df@dfleet:~/example_folder$ cd ..
df@dfleet:~$ pwd
/home/df
```

- touch command

- Syntax

```
touch [file_name]
```

- To create an empty file.

- Example:

```
df@dfleet:~/example_folder$ touch example.txt
df@dfleet:~/example_folder$ ls
example.txt
```

- cp command

- Syntax

```
cp [directory/file_name_copied] [directory/file_name_pasted]
```

- To copy a file and paste it.

- Example:

```
df@dfleet:~/example_folder$ cp ./example.txt ./example02.txt
df@dfleet:~/example_folder$ ls
example02.txt  example.txt
```

- Take note, the “.” above means current directory

- rm command

- Syntax

```
rm [directory/file_name]
```

- To delete a file. Use this carefully!

- Example:

```
df@dfleet:~/example_folder$ ls
example02.txt  example.txt
df@dfleet:~/example_folder$ rm example02.txt
df@dfleet:~/example_folder$ ls
example.txt
```

- nano command

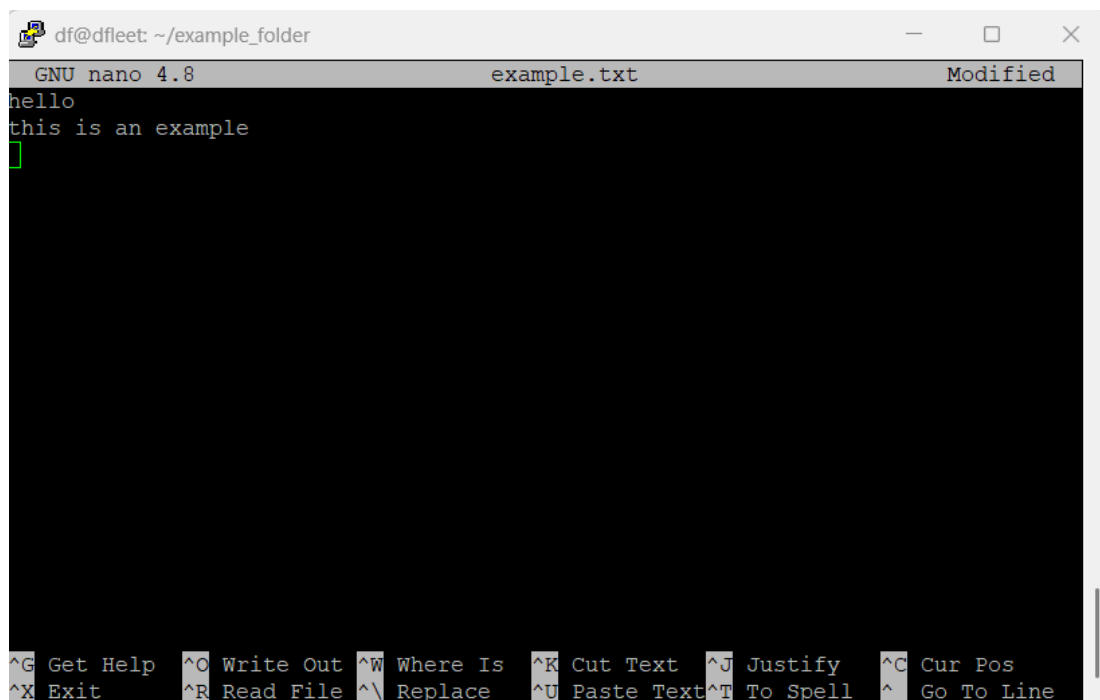
- Syntax

```
nano [directory/file_name]
```

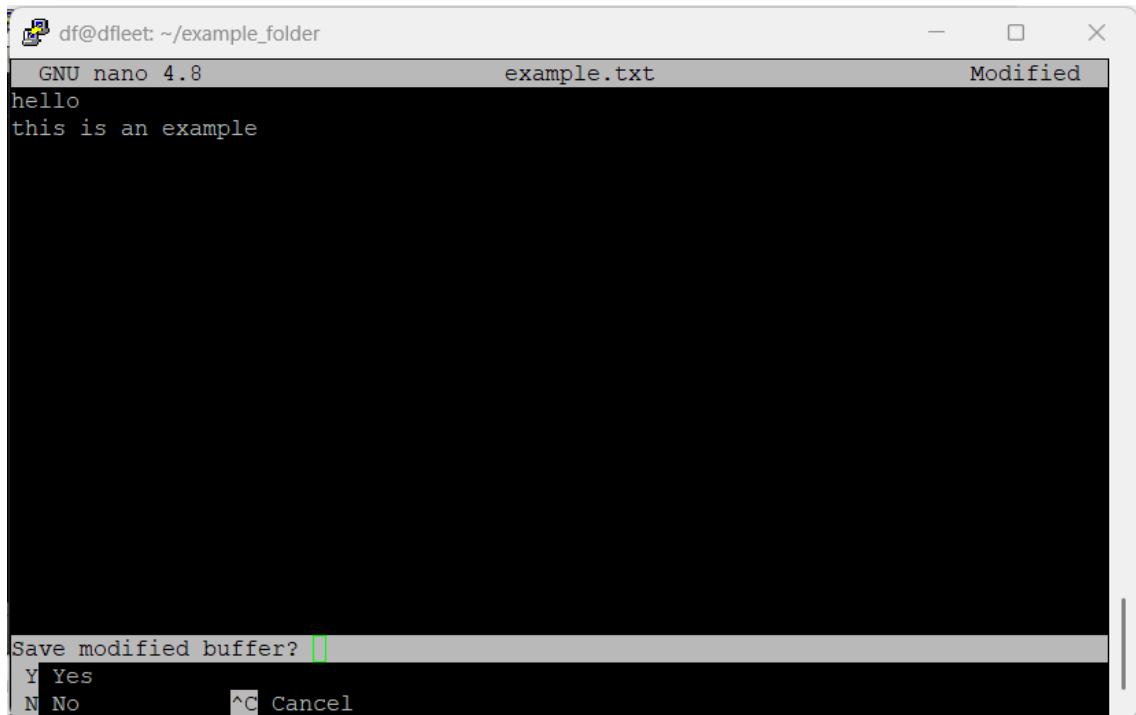
- Nano is actually an editor tool, which is built-in installed in most of the Linux distributions. With this tool, we can edit files.

- Example:

```
df@dfleet:~/example_folder$ nano ./example.txt
df@dfleet:~/example_folder$
```

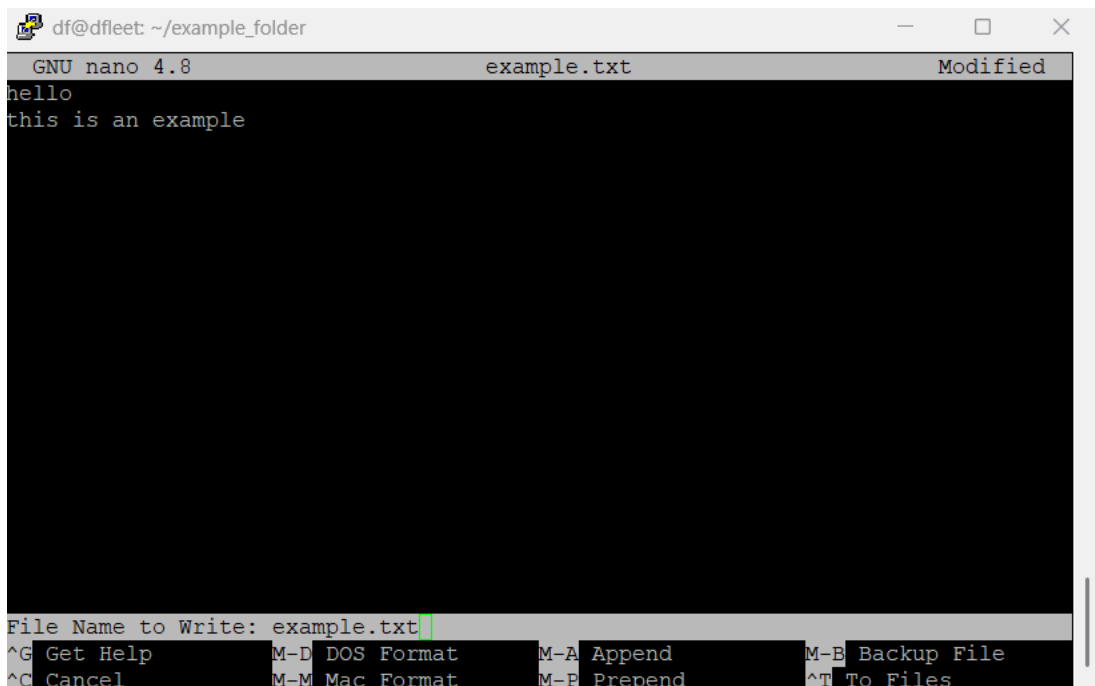


- After editing, we need to “Ctrl + X” to exit the editor. Then it will prompt us to ask whether we want to save the modified buffer. Press “y” to yes.



The screenshot shows a terminal window with the title bar 'df@dfleet: ~/example_folder'. The nano editor is open, editing 'example.txt'. The file content is 'hello' and 'this is an example'. The status bar at the top indicates 'GNU nano 4.8', 'example.txt', and 'Modified'. At the bottom, a prompt 'Save modified buffer?' is displayed with a green cursor. Below the prompt are three options: 'Y Yes', 'N No', and '^C Cancel'.

- After pressing yes, it will ask for the new file name. Press enter without editing to reuse the file name.



The screenshot shows the same terminal window as before. The nano editor is still open. The status bar at the bottom now displays 'File Name to Write: example.txt' with a green cursor. Below this, a list of nano editor shortcuts is shown: '^G Get Help', '^C Cancel', 'M-D DOS Format', 'M-M Mac Format', 'M-A Append', 'M-E Prepend', 'M-B Backup File', and '^T To Files'.

- less command

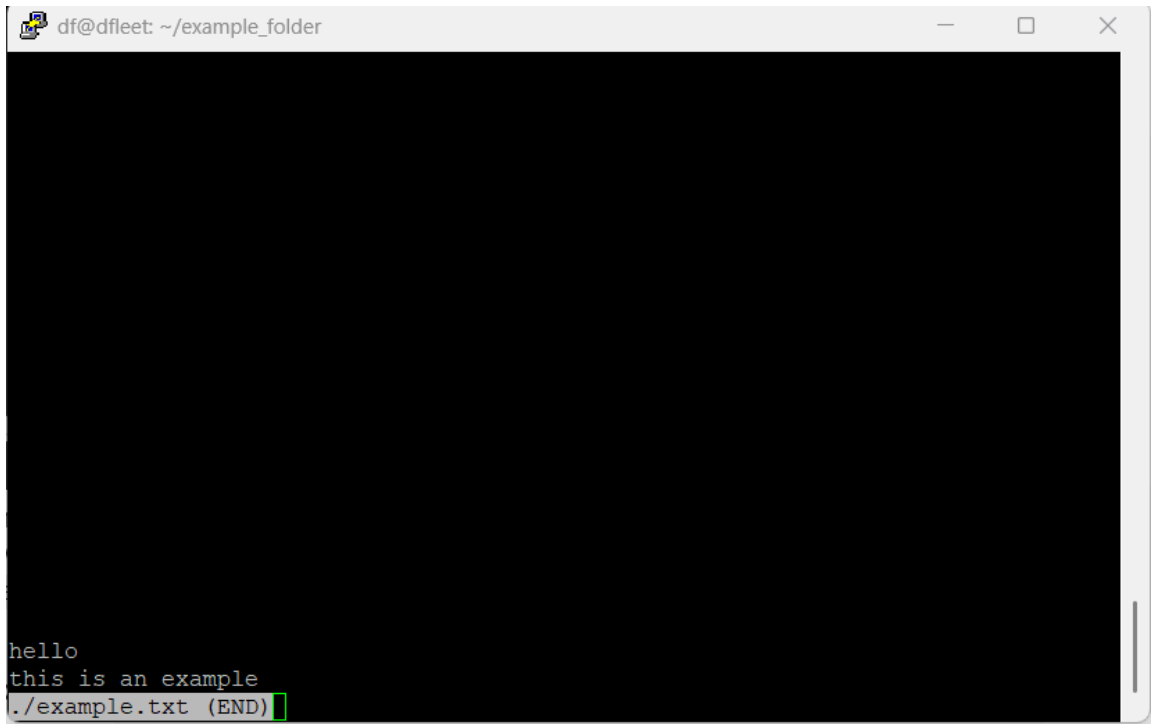
- Syntax

```
less [directory/file_name]
```

- To show the content of a file.

- Example:

```
df@dfleet:~/example_folder$ less ./example.txt
df@dfleet:~/example_folder$
```

A terminal window titled 'df@dfleet: ~/example_folder' showing the output of the 'less' command. The terminal displays the text 'hello' and 'this is an example' on separate lines. The cursor is on the third line, which is './example.txt (END)'.

```
df@dfleet: ~/example_folder
hello
this is an example
./example.txt (END)
```

- Press Q to exit less command.
- Tips:
 - If the content is very long, and you want to directly go to the last page, use Shift + G
 - If the content is keep updating, and you want to follow the latest page, use Shift + F
- sudo command
 - Syntax

```
sudo [UNIX command]
```
 - To give sudo permission to perform actions.
 - Example:

```
df@dfleet:~/example_folder$ sudo nano /etc/network/interfaces
```
- scp command
 - Syntax

```
scp [source_file] [username]@[ip_address]:[destination_path]
```
 - scp stands for Secure Copy Protocol. It allows you to transfer files between your laptop and a remote device (such as an AMR's IPC or a DFleet VM) directly through the terminal, without needing FileZilla.
 - It uses the same SSH connection as PuTTY, so the same credentials and port apply.
 - When to use scp vs FileZilla:
 - Use FileZilla when you need to browse folders visually or transfer multiple files at once.
 - Use scp when you are already in a terminal and need to transfer a single file quickly, or when FileZilla is unavailable.
 - Copying a file FROM your current device TO a remote device
 - Syntax:

```
scp [source_file] [username]@[ip_address]:[destination_path]
```
 - Copying a file FROM your current device TO a remote device
 - Syntax:

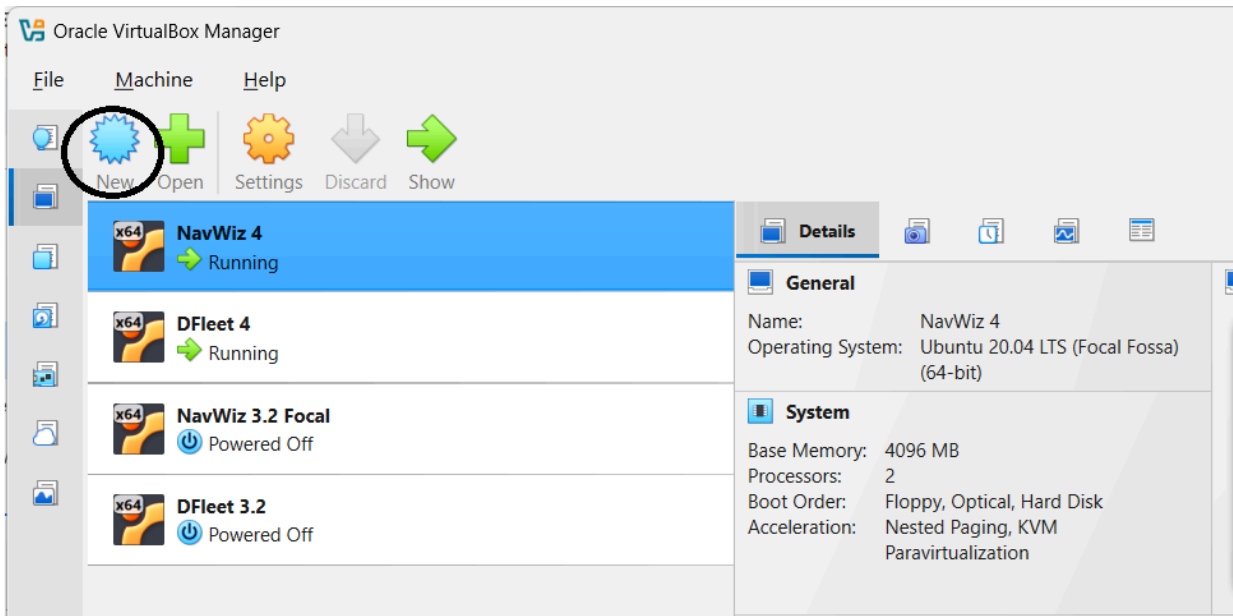
```
scp [username]@[ip_address]:[source_file] [destination_path]
```

- Example — downloading a log file from the IPC to your current local folder:
 - `scp df@192.168.1.50:/var/log/syslog ./`

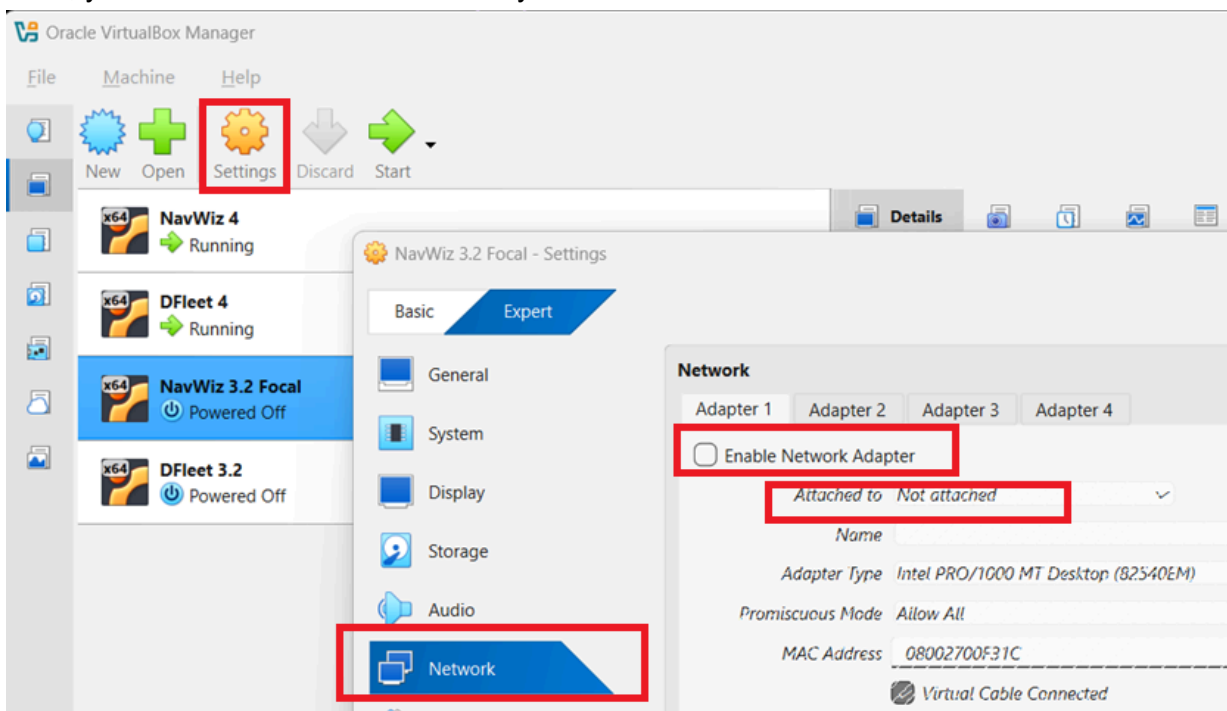
Setting Up VM

Setting Up NavWiz 4.0 VM

- Download and install Oracle Virtualbox from <https://www.virtualbox.org/wiki/Downloads>
- Download NavWiz ISO from: <https://releases.dfautomation.com/#navwiz-4.0.0>
- Add new VM on Oracle:

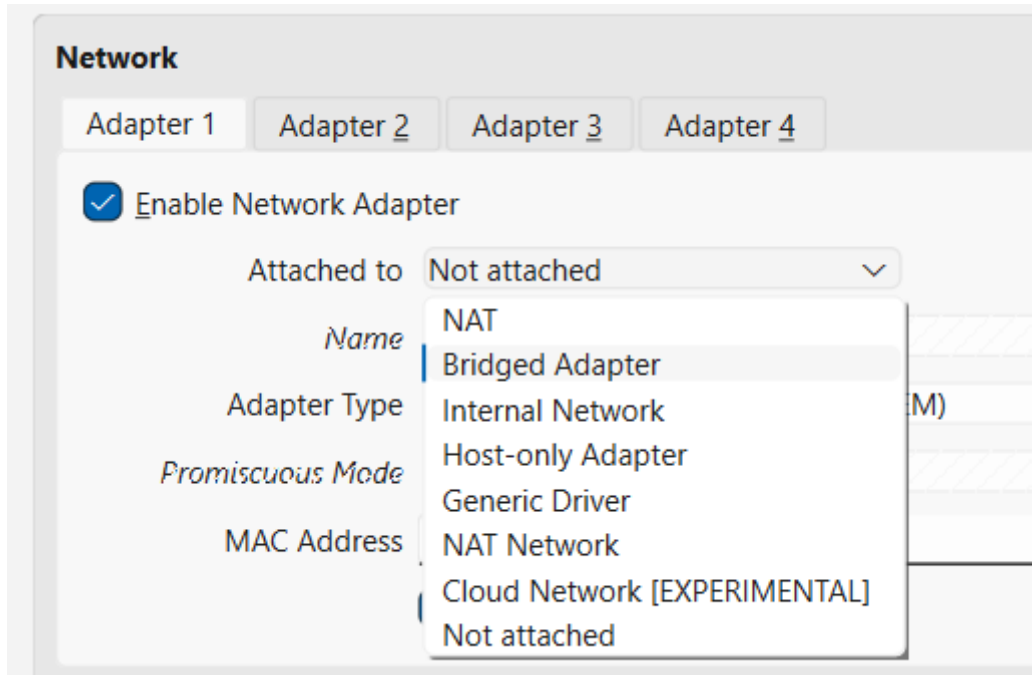


- Select the ISO for NavWiz 4.0
- Then, you need to turn off network ability for that VM



- Start the VM and install.
- After installation, turn off the VM and enable back the network.
 - Use bridge adapter to let the VM has access to internet

- Use host-only Adapter if you do not need the VM to have access to internet



- After that, run the VM, and then Ctrl + Alt + F3 to enter terminal mode.
- Then, edit the network interface file:

```
sudo nano /etc/network/interfaces
```

- Then add these lines:
 auto enp0s3
 iface enp0s3 inet dhcp
 post-down ip addr flush dev \$IFACE

```
GNU nano 4.8 /etc/network/interfaces Modified
# interfaces(5) file used by ifup(8) and ifdown(8)
# Include files from /etc/network/interfaces.d:
source-directory /etc/network/interfaces.d

auto enp0s3
iface enp0s3 inet dhcp
    post-down ip addr flush dev $IFACE
```

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo
 ^X Exit ^R Read File ^_ Replace ^U Paste Text ^T To Spell ^_ Go To Line M-E Redo

- Ctrl + X and y and Enter to save the changes.
- Then, run this to restart :

```
sudo nohup sh -c "ifdown enp0s3 && ifup enp0s3"
```

- Then run ifconfig and you should see the ip address.

```
df@navwiz:~$ sudo nohup sh -c "ifdown enp0s3 && ifup enp0s3"
nohup: ignoring input and appending output to 'nohup.out'
df@navwiz:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.1.164  netmask 255.255.255.0  broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe69:9ea2  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:69:9e:a2  txqueuelen 1000  (Ethernet)
    RX packets 2375  bytes 158054 (158.0 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 45  bytes 5944 (5.9 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

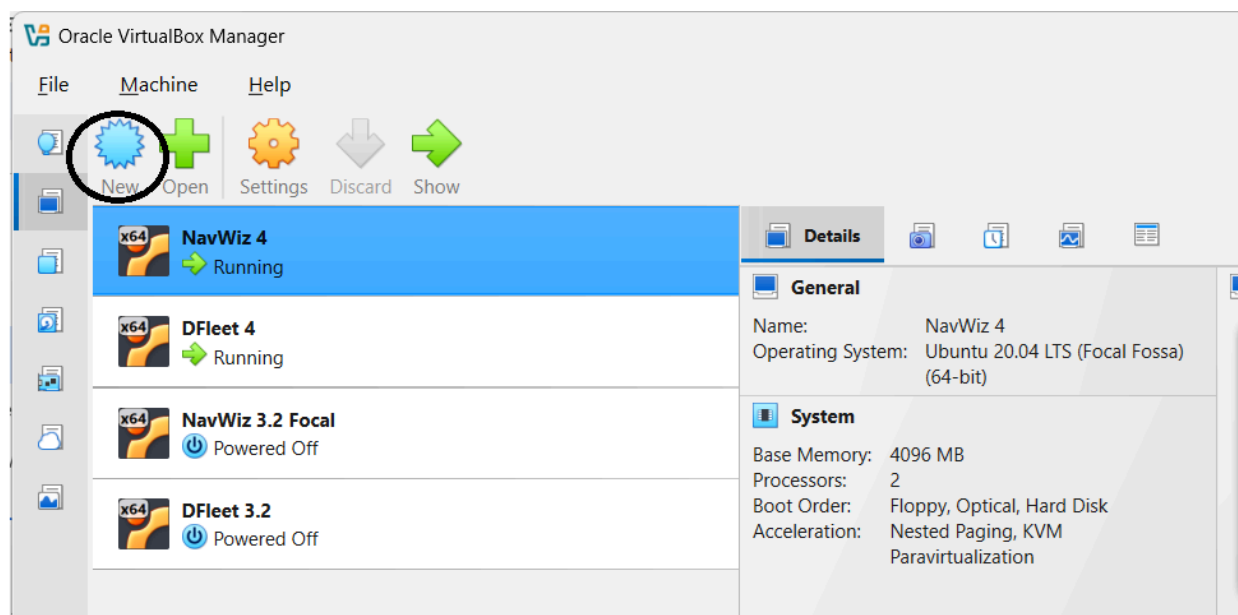
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 18709  bytes 4216570 (4.2 MB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 18709  bytes 4216570 (4.2 MB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

df@navwiz:~$ _
```

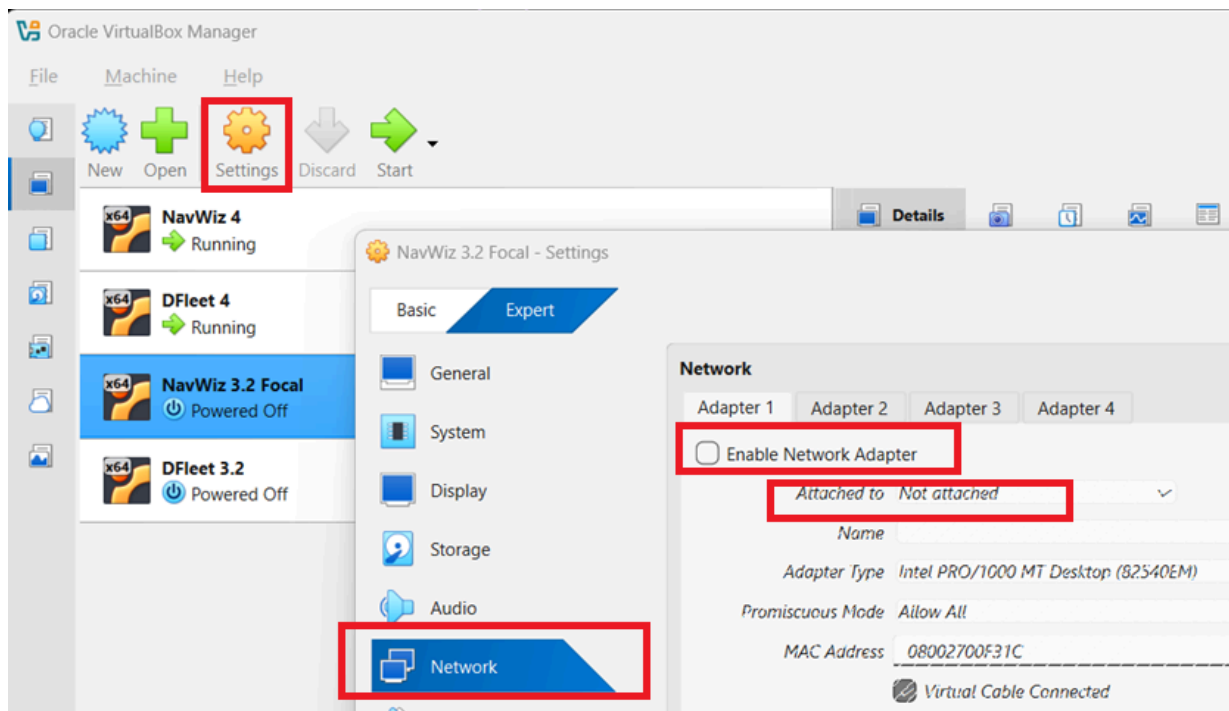
- Accounts:
 - Username: supervisor , password: dfautomation
-

Setting Up DFleet 4.0 VM

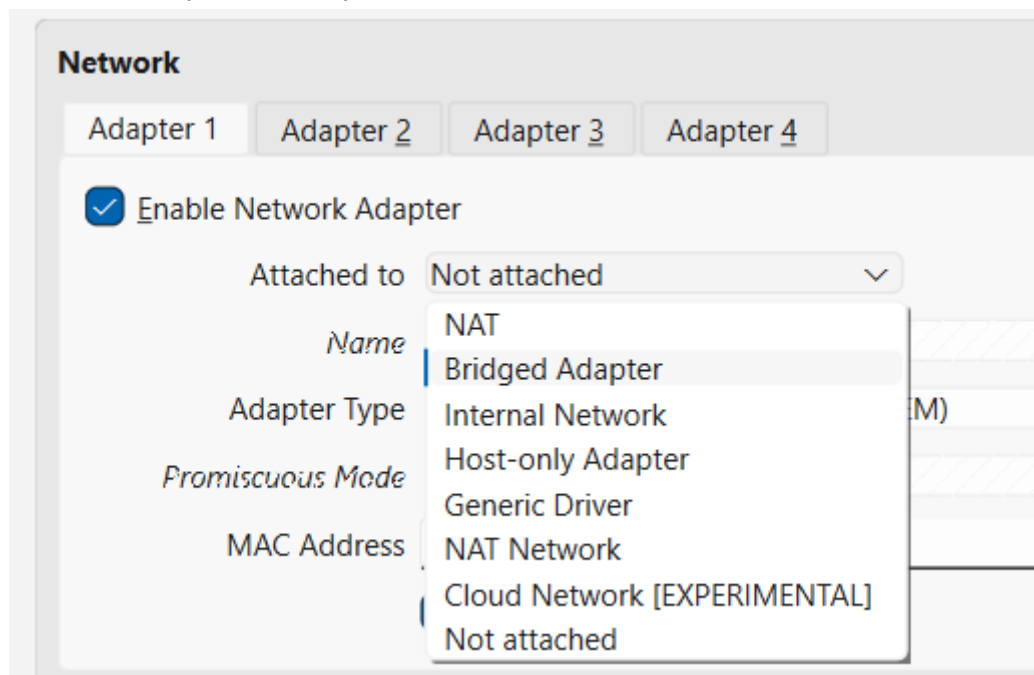
- Download and install Oracle Virtualbox from <https://www.virtualbox.org/wiki/Downloads>
- Download latest nightly DFleet ISO from: <https://releases.dfautomation.com>
- Download latest patch, the two files here:
 - <https://releases.dfautomation.com/dfleet/nightly/4.0.0~rc1+nightly.20260521/debs/>
- Add new VM on Oracle:



- Then, you need to turn off network ability for that VM



- Start the VM and install.
- After installation, turn off the VM and enable back the network.
 - Use bridge adapter to let the VM has access to internet
 - Use host-only Adapter if you do not need the VM to have access to internet



Network Fundamentals

What Is a Network, and Why Does It Matter for AMR Deployment?

Think of a network like a road system. Every device (AMR, laptop, DFleet server, Wi-Fi access point) is like a building with an address. For two buildings to send packages to each other, they need to be on the same road system, and each must have a unique address — otherwise the package gets lost or delivered to the wrong place.

In AMR deployments, the AMR, the DFleet server, and your laptop must all be on the same network and have unique addresses. If any one of them has the wrong address, or is on a different "road", they cannot communicate — and the robot will disconnect or fail to receive commands.

Understanding IP Addressing

- Every device on a network is assigned an IP address — a unique numerical label that identifies it, similar to a house address.
 - IPv4 addresses follow the format XXX.XXX.XXX.XXX, for example: 192.168.1.4
 - Each of the four numbers (called octets) ranges from 0 to 255.
 - The first three octets usually identify the network (e.g. 192.168.1), and the last octet identifies the specific device (e.g. .4).
 - Two devices can only talk to each other directly if they are on the same network (same first three octets in a typical /24 subnet).
 -  192.168.1.4 and 192.168.1.50 — can communicate directly
 -  192.168.1.4 and 192.168.2.50 — different network, cannot communicate directly without a router
- An IP address is a unique numerical label assigned to each device on a network.

Subnet Mask, Gateway, and DNS — What Do These Mean?

These three settings appear whenever you configure a static IP. Here is what each one does, in plain language:

Subnet Mask

- The subnet mask tells a device which part of the IP address is the "network" and which part is the "device".
- The most common subnet mask in deployments is 255.255.255.0, also written as /24.
- In plain language: all devices sharing the same first three octets (e.g. 192.168.1.x) are on the same network and can reach each other directly.
- Why it matters: If the subnet mask is wrong on an AMR, the AMR may think other devices are on a different network and refuse to communicate — even if the IP addresses look correct.

Gateway

- The gateway (also called the default gateway) is the address of the router — the device that connects your local network to other networks or the internet.
- Example: If your network is 192.168.1.x, the gateway is typically 192.168.1.1 (the router).
- Why it matters for AMR deployment:
 - For AMR-to-DFleet communication within the same site network, the gateway is usually not involved — both devices talk directly.
 - However, if the DFleet server or AMR needs to reach the internet (e.g. for updates, remote diagnostics), the gateway must be correctly set.
 - If the gateway is misconfigured, the AMR can still talk to devices on the same subnet, but cannot reach anything outside — including cloud services or remote support tools.

DNS (Domain Name System)

- DNS is like a phone book for the internet. It converts human-readable names (e.g. diagnostic.dfautomation.com) into IP addresses that computers understand.
- A common DNS server address is 8.8.8.8 (Google's public DNS).
- Why it matters for AMR deployment:

- If DNS is not set or is wrong, the AMR or DFleet server may fail to resolve domain names — for example, it may be unable to connect to the diagnostics portal or download updates, even if the internet connection itself is working.
- DNS is usually only relevant if the device needs internet access. In a fully isolated site network, DNS may not matter

Example Static IP Configuration.

Setting	Example
IP Address	192.168.1.50
Subnet Mask	255.255.255.0
Gateway	192.168.1.1
DNS	8.8.8.8

Static IP vs DHCP

- **DHCP (Dynamic Host Configuration Protocol):** The router automatically assigns an IP address to any device that connects. It is convenient, but the IP address can change every time the device reconnects.
- **Static IP:** A fixed IP address manually set on the device. It never changes.
- **For AMR deployments, always use Static IP for AMRs and the DFleet server.**
 - If an AMR's IP keeps changing, DFleet will lose track of it and show it as disconnected.
 - If the DFleet server's IP changes, none of the AMRs will know where to find it.
- To configure static IP on NavWiz, go to Config Panel → Network Settings and enter the IP address, subnet mask, gateway, and DNS

Finding the Network Interface Name

- A network interface is the hardware port or Wi-Fi chip on a device that connects it to the network. Think of it as the "socket" your network cable or Wi-Fi signal plugs into.
- Each interface has a name assigned by the operating system. This name is not always the same across different AMRs or IPCs. Common names include:
 - wlp3s0, wlan0 — Wi-Fi interfaces
 - eth0, enp2s0, eno1 — Wired (Ethernet) interfaces
- Before running any network command, always check which interface name applies to your device:
 - ip addr

or

 - ifconfig
- Look for the interface that has an IP address assigned. Example output:

```
2: wlp3s0: <BROADCAST,MULTICAST,UP,LOWER_UP> inet 192.168.1.50/24 brd 192.168.1.255
scope global wlp3s0
```

- In this example, the Wi-Fi interface name is wlp3s0 and the assigned IP is 192.168.1.50.

- Whenever a guide or command references wlp3s0, replace it with your actual interface name found using ip addr.

Common Ports Used in Deployment

A port is like a specific door on a building. Even if two devices can reach each other by IP address, a service will only respond if you knock on the correct port number.

Service	Port	Protocol
SSH (PuTTY)	22	TCP
FileZilla (SFTP)	22	TCP
DFleet Web UI	80 / 8080	TCP
NavWiz Web UI	80	TCP

- Ensure these ports are **not blocked by firewalls** on either the laptop or the robot's network.

Port Forwarding — Concept and Use Case

What is Port Forwarding?

- Normally, devices inside a private network (e.g. 192.168.1.x) are not reachable from outside that network — the router blocks incoming connections by default for security.
- Port forwarding is a setting that says: "If a connection arrives at port X on this device, forward that connection to port Y on another device."
- Think of it like a hotel receptionist. The hotel has one main entrance (one IP address). When a guest arrives asking for Room 302 (port 80), the receptionist directs them to that specific room. Without the receptionist, the guest cannot get in.
- In AMR deployments, port forwarding is most commonly used in two practical situations:

Use Case 1 — Remote SSH Access Into the Site Network

- When the DF support team needs to SSH into an AMR from outside the customer's site (e.g. from the office), they cannot reach the AMR directly because the AMR has a private IP address.
- The solution is to configure port forwarding on the customer's router:
 - The router has a public IP address (reachable from anywhere over the internet), e.g. 203.0.113.10
 - The AMR has a private IP inside the customer network, e.g. 192.168.1.50
 - The router is configured to forward: any connection arriving on port 2222 → send it to 192.168.1.50 on port 22
 - The remote engineer then connects using:
 - `ssh df@203.0.113.10 -p 2222`
- This reaches the AMR's IPC as if they were physically on-site.
- Key notes:
 - Always use a non-standard external port (e.g. 2222, 2250) instead of the default port 22. This reduces exposure to automated internet scanners.
 - For multiple AMRs, assign a different external port to each:
 - AMR 1: port 2221 → 192.168.1.50:22
 - AMR 2: port 2222 → 192.168.1.51:22

- DFleet: port 2223 → 192.168.1.100:22
- This must be arranged with the customer's IT team — they control the router and must configure the forwarding rules.
- Port forwarding for remote support should be documented, and disabled after the support session where possible.

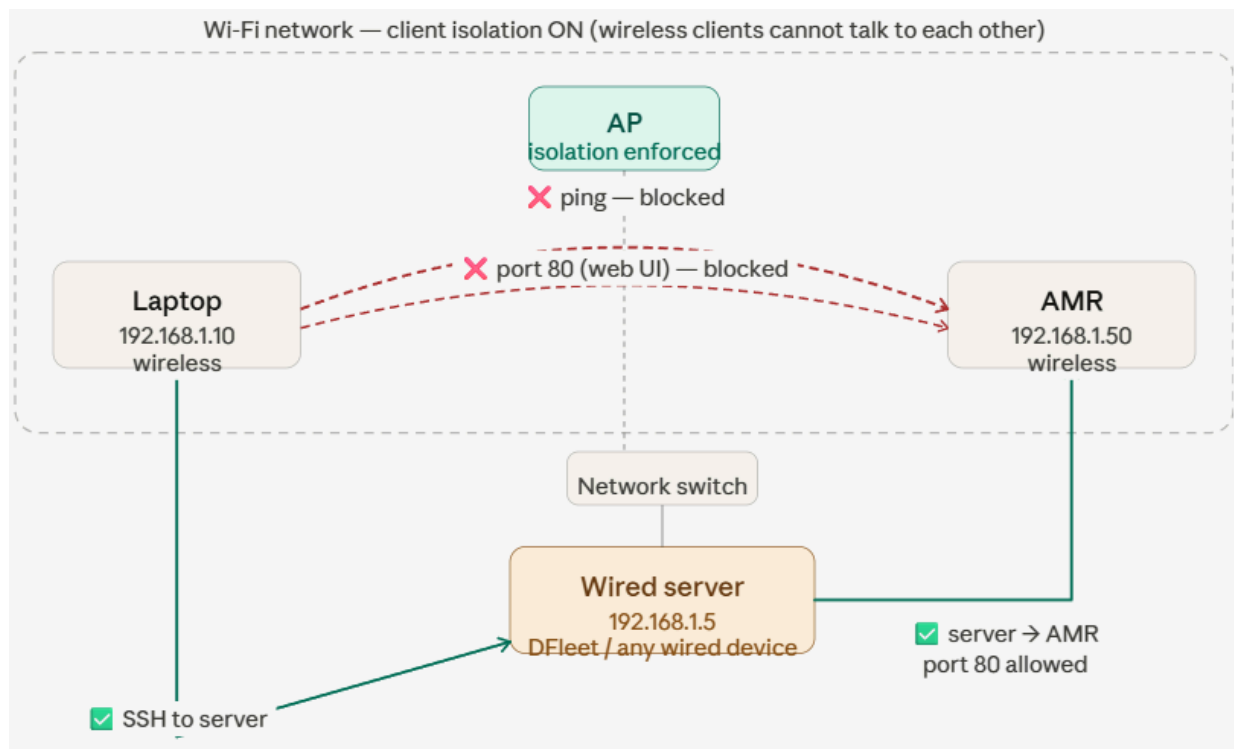
Use Case 2 — Local Port Forwarding via PuTTY (On-Site Access)

- This is a different but equally important technique that FAEs use directly on-site.
- What it does: Instead of forwarding through a router, PuTTY creates a tunnel through an existing SSH connection. Traffic you send to a port on your own laptop (e.g. localhost:8080) gets carried inside the SSH session and delivered to a target device as if the request came from the AMR itself.
- Why this matters in practice: You open <http://localhost:8080> in your browser on your laptop, and it actually shows the web UI of the AMR.
- How to set it up in PuTTY:
 - Open PuTTY and enter the IP of the device you are SSH-ing into
 - Before clicking Open, go to Connection → SSH → Tunnels
 - Fill in:
 - Source port: 8080 (any unused port on your laptop)
 - Destination: 192.168.1.50:80 (the AMR IP and the port of its web UI)
 - Click Add
 - Click Open to start the SSH session
 - Now open your browser and go to <http://localhost:8080> — this will show the AMR's NavWiz web UI

Scenario A — Same SSID But Cannot Reach AMR Directly (Client Isolation)

- This is a situation you may encounter on-site. Your laptop and the AMR are both connected to the same Wi-Fi network (same SSID, same subnet) — but you cannot reach the AMR at all. Depending on how strictly client isolation is enforced, you may experience different symptoms:

Symptom	What it means
Browser cannot load AMR web UI, but ping works	Partial isolation — ICMP allowed, TCP port 80 blocked
Ping fails AND browser cannot load web UI, but SSH works	Stronger isolation — most traffic blocked except established sessions
Ping fails, browser fails, AND SSH fails	Full isolation — all wireless-to-wireless traffic blocked

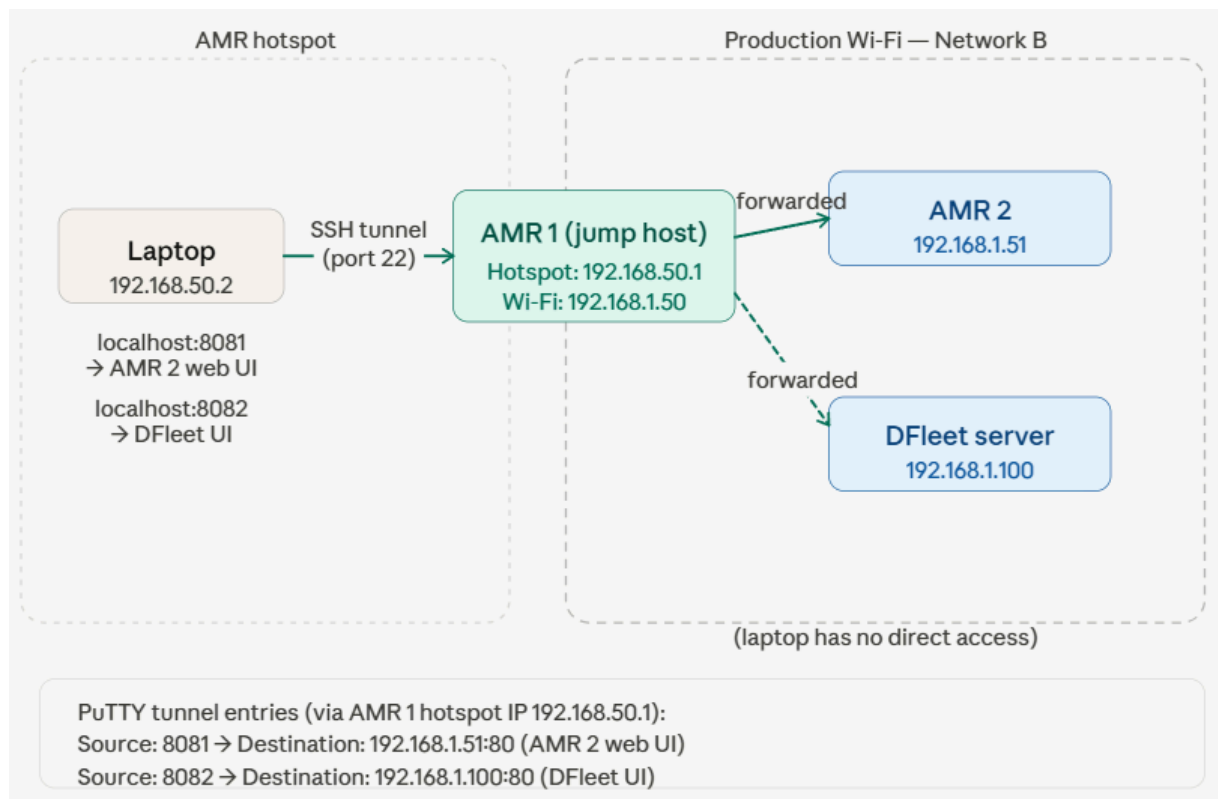


- Why does this happen?
 - Many enterprise and managed Wi-Fi access points have a feature called client isolation (sometimes called AP isolation). When enabled, devices connected to the same Wi-Fi SSID are prevented from communicating directly with each other. This is a deliberate security feature commonly used in offices, hotels, and factories to stop guests from accessing each other's devices.
 - The critical thing to understand is: client isolation only blocks lateral traffic between wireless clients. A device that is connected via a wired network cable — such as the DFleet server, a site PC, or a network switch port — is typically not subject to the same restriction. Wired devices can usually still reach the AMR freely.
- The fix depends on what you can still reach:
 - If SSH to the AMR still works:
 - Set up a local port forward through the SSH session to the AMR itself. In PuTTY:
 - SSH target: AMR IP (192.168.1.50)
 - Tunnel: Source 8080 → Destination 192.168.1.50:80
 - Open `http://localhost:8080` in your browser
 - This works because the web UI traffic rides inside the already-established SSH session, rather than as a new direct connection to port 80.
 - If even ping and SSH to the AMR are blocked:
 - Full client isolation is in effect.
 - Your laptop cannot initiate any connection to the AMR directly. In this case, look for a wired device on the same network that you can reach — for example the DFleet server, a site workstation, or any machine connected via Ethernet to the switch.
 - Use that wired device as your jump host — SSH into it, then set up a port forward from your laptop through it to the AMR:
 - SSH target: wired device IP (e.g. 192.168.1.5 — the DFleet server)
 - Tunnel: Source 8080 → Destination 192.168.1.50:80
 - Open `http://localhost:8080` in your browser
 - Because the wired device is not a wireless client, it is not subject to client isolation. It can reach the AMR freely, and your laptop's tunnel rides through the wired device to get there.
 - Why does this work even when ping fails?

- The wired server sits on the same switch as the access point's uplink. Traffic from your laptop to the wired server travels up through the AP to the switch, then down to the server — this is not a wireless-to-wireless path, so it is not blocked. The server then forwards the traffic down to the AMR through the switch — again, not a wireless-to-wireless path. Client isolation only intercepts traffic that travels laterally between wireless clients.
 - If no wired device is available:
 - You will need to request the customer's IT team to either:
 - Disable client isolation for the deployment VLAN, or
 - Provide a temporary wired connection between your laptop and the switch directly.
- In short: if direct browser access fails but SSH works, set up a local port forward in PuTTY as described above, then open `http://localhost:8080` in your browser instead.
- If even SSH is blocked, then client isolation is fully enforced and you will need to request the customer's IT team to either:
 - Disable client isolation for the deployment VLAN, or
 - Create a wired connection between your laptop and the switch instead

Scenario B — Using an AMR as a Jump Host (Hotspot + Wi-Fi Simultaneously)

- Some AMR models support running a Wi-Fi hotspot and a Wi-Fi client connection at the same time — that is, the AMR is simultaneously:
 - Connected to the site network (Network B) via its Wi-Fi card
 - Broadcasting its own hotspot that your laptop can connect to
- This is useful when you cannot get your laptop onto the site network directly (e.g. no IT approval, VLAN restrictions, or the network is managed separately).
- How it works:
 - Connect your laptop to the AMR's hotspot. The AMR assigns your laptop an IP on its hotspot network, e.g. 192.168.50.2, and the AMR's hotspot interface is 192.168.50.1
 - SSH into the AMR via its hotspot IP: `ssh df@192.168.50.1`
 - Set up local port forwards in PuTTY pointing to other devices on Network B:
 - Source: 8081 → Destination: 192.168.1.51:80 (AMR 2 web UI)
 - Source: 8082 → Destination: 192.168.1.100:80 (DFleet web UI)
 - Open `http://localhost:8081` in your browser to access AMR 2's web UI
 - Open `http://localhost:8082` to access the DFleet web UI

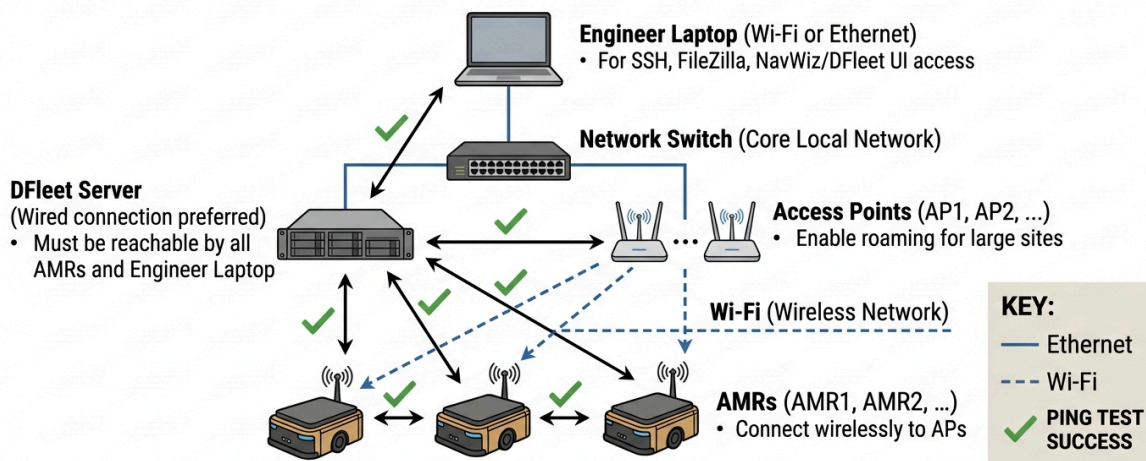


- Why this works:
 - The AMR has a foot in both networks at the same time. When PuTTY sends traffic through the SSH tunnel asking for 192.168.1.51:80, the AMR receives that request from inside the tunnel and forwards it out through its own Wi-Fi connection to Network B. From the perspective of Network B devices, the request comes from the AMR — not from your laptop.
- Think of the AMR as a receptionist at the border between your hotspot network and the site network. Your laptop hands the receptionist a message ("get me the web UI from AMR 2"), and the receptionist walks into the site network, retrieves it, and hands it back to you.
- Limitations to be aware of:
 - This only works if the AMR model supports concurrent hotspot + Wi-Fi client. Verify this before relying on it for deployment.
 - Your laptop's internet access will not work while connected to the AMR hotspot, unless the AMR is also configured to share its connection.
 - This technique is primarily for read access and troubleshooting. Do not use it for large file transfers (e.g. full map uploads) as performance will be limited.
 - If the site network enforces client isolation between AMRs as well, some forwarded connections may still be blocked.

Network Topology in a Typical Deployment

- A typical deployment network consists of:
 - **Access Points (APs):** Wireless infrastructure that AMRs connect to via Wi-Fi. Multiple APs are used for large sites to enable roaming.
 - **DFleet Server:** Connected to the same network (wired preferred) and reachable by all AMRs.
 - **AMRs:** Connect wirelessly to the APs. Each AMR must be able to ping the DFleet server and all other AMRs on the network.
 - **Engineer Laptop:** Connected to the same network for SSH, FileZilla, and browser-based access to NavWiz/DFleet UI.

TYPICAL DEPLOYMENT NETWORK TOPOLOGY



RULE OF THUMB: All devices (Laptop, DFleet Server, AMRs) must successfully ping each other before deployment.

Checking Network Connectivity

- Check your own IP address and interface name:
 - ip addr
- **Ping test:**

```
ping [ip_address]
```
- Example: `ping 192.168.1.4`
- A successful ping confirms basic IP-level reachability between devices.
- **Check IP address of the current device:**
 - **On Windows command prompt:** `ipconfig`
 - **On Linux terminal:** `ifconfig`
- **Check routing table:**
 - `route -n`
- **Check open ports and active connections:**
 - `netstat -tulnp`

Wi-Fi Fundamentals for AMR Deployments

- AMRs typically connect over **2.4 GHz or 5 GHz Wi-Fi**.
 - **2.4 GHz:** Longer range, better wall penetration, but more interference and lower throughput.
 - **5 GHz:** Shorter range, less interference, higher throughput. Preferred for high-bandwidth or latency-sensitive environments.
- During site survey, verify **signal strength (RSSI)** at all planned AMR operating areas. A minimum of **-65 dBm** is recommended for stable DFleet connectivity.
- To check signal level on the AMR terminal:
 - `iwconfig`
 - or
 - `iw dev [network interface] link`
- For sites with multiple access points, ensure:
 - All APs broadcast the **same SSID** (network name)

- APs support **fast roaming / 802.11r** where possible, so the AMR transitions between APs without dropping its DFleet connection
- The AMR is **not locked to a specific AP's MAC address** in its configuration

Checking Wi-Fi Stability During Deployment

- Navigate the AMR across all planned operating areas while monitoring for DFleet disconnections.
- Upload DFleet backup and NavWiz diagnostics to the Diagnostics Visualizer at <https://diagnostic.dfautomation.com/> or <https://hub.dfautomation.com/diagnostic/>
- In Table View, monitor these columns:
 - **DFleet Status**
 - **AP MAC Address**
 - **Signal Level**
- If **AP MAC Address** changes frequently, it indicates the AMR is roaming between APs. Correlate this with any DFleet disconnection events.

Network Troubleshooting Quick Reference

Symptom	First Check	Action
Cannot SSH to AMR	Same subnet?	Check laptop and AMR IP addresses
DFleet shows AMR disconnected	Ping AMR from DFleet server	Check Wi-Fi signal and AP coverage
FileZilla cannot connect	Port 22 open?	Verify SFTP service is running on AMR
AMR cannot reach DFleet	Gateway configured?	Check static IP settings on AMR
AMR roaming causes disconnection	AP signal overlap	Tune AP placement or roaming thresholds
Hidden SSID not connecting	scan_ssid set?	Edit wpa_supplicant.conf, set scan_ssid=1
Cannot reach internet or cloud services	Is DNS set?	Check DNS entry in static IP config
Remote SSH not working from outside site	Is port forwarding configured?	Confirm router forwarding rules with customer IT team
Browser cannot load AMR web UI, ping works	Partial client isolation?	Set up PuTTY local port forward: SSH to AMR, tunnel localhost:8080 → AMR IP:80
Ping fails AND browser fails, but SSH to AMR works	Stronger client isolation?	Set up PuTTY local port forward through the SSH session to AMR — do not attempt direct browser access
Ping fails, browser fails, AND SSH to AMR fails	Full client isolation in effect	Find a wired device on the same network (e.g. DFleet server). SSH into it, then set up port forward:

		localhost:8080 → AMR IP:80 through that wired device
No wired device available, full isolation in effect	No local jump host exists	Request customer IT team to disable client isolation on deployment VLAN, or provide a direct wired connection to the switch
Cannot access Network B devices but AMR hotspot is available	AMR supports concurrent hotspot + Wi-Fi?	Connect laptop to AMR hotspot, SSH to AMR hotspot IP, set up port forwards to Network B devices through the AMR

Reserved IP Ranges — Do Not Conflict

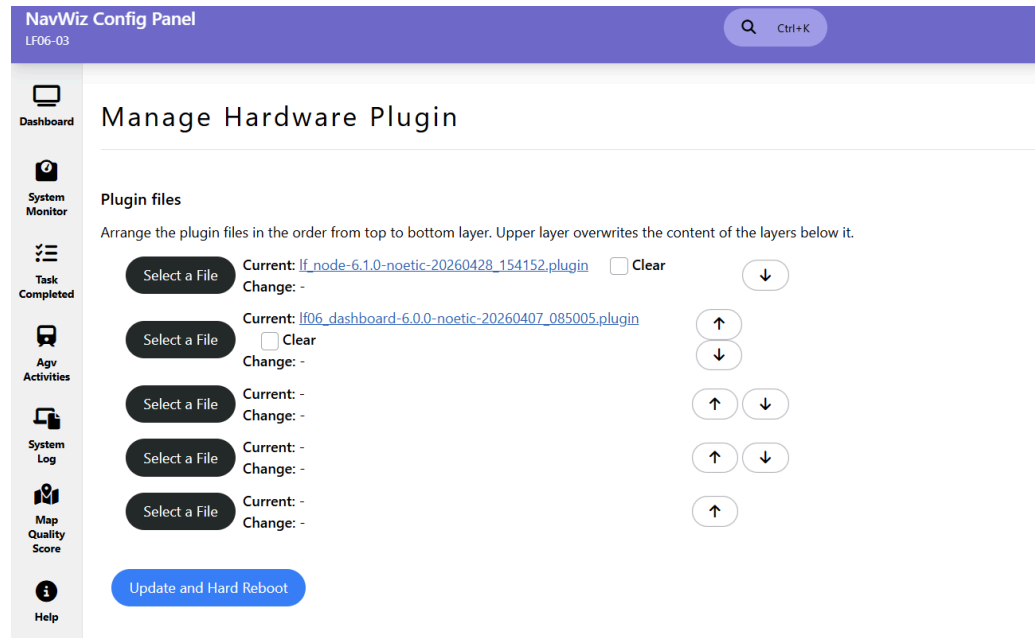
- Avoid assigning AMR or DFleet server IPs that conflict with:
 - **192.168.0.1** — Common router/gateway default
 - **169.254.x.x** — Link-local addresses (indicates DHCP failure)
 - **127.0.0.1** — Loopback address (localhost only)
- Always align the IP plan with the customer's IT team before deployment to avoid IP conflicts with existing infrastructure.

Using nmap for Network Discovery

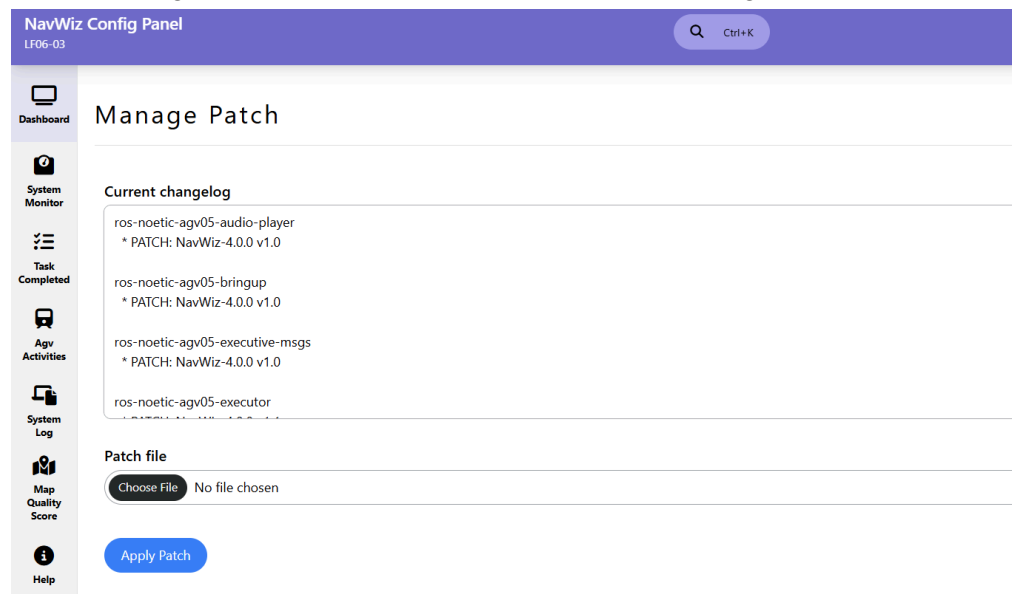
- nmap is useful for discovering devices on a network and identifying hostnames.
- Install if not present:
 - `sudo apt install nmap`
- Scan for all active devices on the subnet:
 - `nmap -sn 192.168.1.0/24`
- Identify the hostname and OS of a specific device:
 - `nmap -O 192.168.1.68`
- This is particularly useful when investigating unexpected shutdowns or identifying unknown devices on the network (refer to AMR Shutdown troubleshooting).

Understanding the Fundamentals of NavWiz DFleet

- Other than NavWiz/DFleet fundamental training, you should also know what role each component plays.
 - NavWiz
 - The high level general software for all DF AMRs
 - Hardware plugin
 - The hardware related software file that differentiates the hardware integration of AMRs (for example, some AMRs having hooking motor, some AMRs having lifter mechanism, led control is also done by hardware plugin)



- Firmware
 - The low-level software of mctrl board, which controls how mctrl board behaves.
- Patch File
 - The .tar zip file consists of many .deb files. When applying patch, these new .deb files (with bug fix or new feature etc) will overwrite the original .deb files of NavWiz



- Robot Config
 - Robot Config is the file for configuring the Robot Type, Network Configuration and Environment Variables.

Preset Name	IP Address	Subnet Mask	Options	Operation
Bottom Lidar	192.168.11.1	255.255.255.0	☐ Autoneg Off (100 Mbit/sec)	↑ ↓ 📄 ✕
Livox	192.168.12.1	255.255.255.0	☐ Autoneg Off (100 Mbit/sec)	↑ ↓ 📄 ✕

Retrieving Logs

How to download NavWiz diagnostics?

The following steps illustrate how to download NavWiz diagnostics:

- Log in to the NavWiz User Panel (AM801) as an administrator.
- Click on the **Config Panel** link in the top right corner.
- In the **NavWiz Config Panel**, click on the **System Log** link in the left sidebar.
- Under the **Download Log** section, select **Diagnostics** from the log type dropdown menu.
- Click the **Download** button to retrieve the logs.

The **NavWiz Config Panel** shows the **System Log** section with the following log types: Main log files, Peripheral log files, Custom log, Audi log, and **Diagnostics**.

The **NavWiz Config Panel** also shows the **Download Log** section with the following options:

- Start Date: 2025-03-17
- End Date: 2025-03-17
- Download button

The **NavWiz Config Panel** displays the **Diagnostics** table with the following data:

#	Name	Modified Time	File Size	Operation
1	20250318.csv	1:47:02 a.m.	15.2 MB	[Icon]
2	20250317.csv.1.gz	11:39:39 p.m. 17/03/2025	9.6 MB	[Icon]
3	20250318.csv.1.gz	11:39:39 p.m. 18/03/2025	3.0 MB	[Icon]
4	20250314.csv.1.gz	2:04:48 a.m. 14/03/2025	0.1 MB	[Icon]

How to download NavWiz logs?

The screenshot shows the NavWiz Config Panel interface. The top header includes the title 'NavWiz Config Panel' and 'RG AMR/0007'. The left sidebar contains navigation icons for Dashboard, System Monitor, Task Completed, Agv Activities, System Log (highlighted with a red box), Map Quality Score, and Help. The main content area is titled 'System Log' and lists four options: Main log files, Peripheral log files, Custom log, and Diagnostics. Below this is the 'Download Log' section, which contains a form with 'Start Date' and 'End Date' fields, both set to '2026-04-09', and a 'Download' button. Red boxes highlight the 'System Log' menu item, the date fields, and the 'Download' button.

How to download DFleet diagnostics?

The first screenshot shows the DFleet interface. The 'Report' menu is highlighted with a red box and a circled '1'. The 'Report' dropdown menu is open, and 'System Log' is highlighted with a red box and a circled '2'. The second screenshot shows two file index pages. The left page is titled 'Index of /logs/' and lists various log files. The 'diagnostics/' directory is highlighted with a red box and a circled '3'. The right page is titled 'Index of /logs/diagnostics/' and shows a file named '20260317.csv' highlighted with a red box and a circled '4'.

How to get the syslog of DFleet?

- Use filezilla to connect and the download /var/log/syslog

Applying Patches

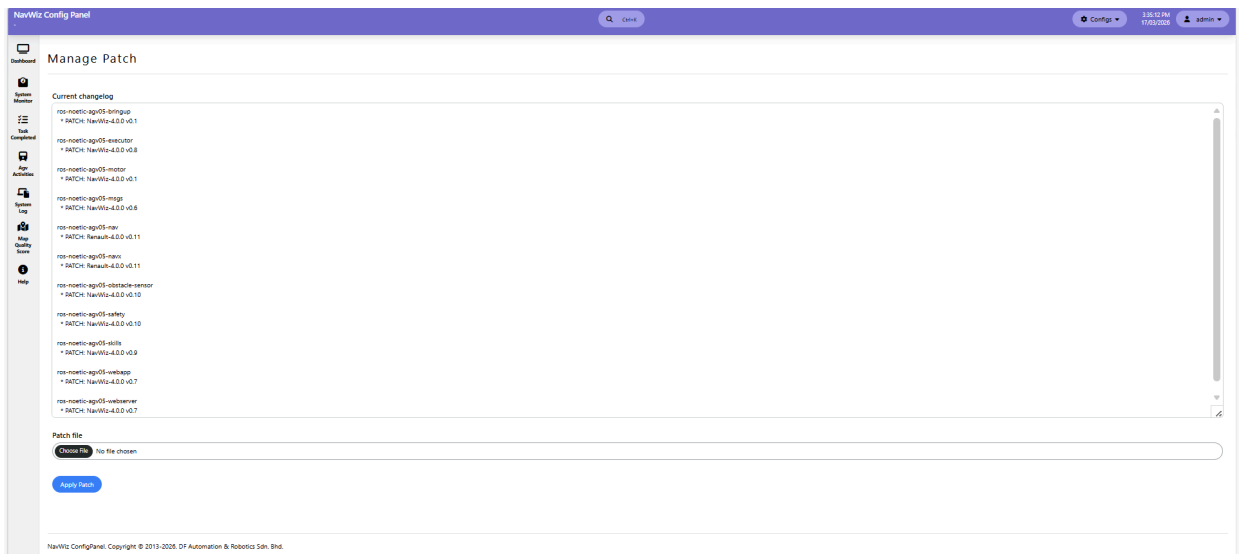
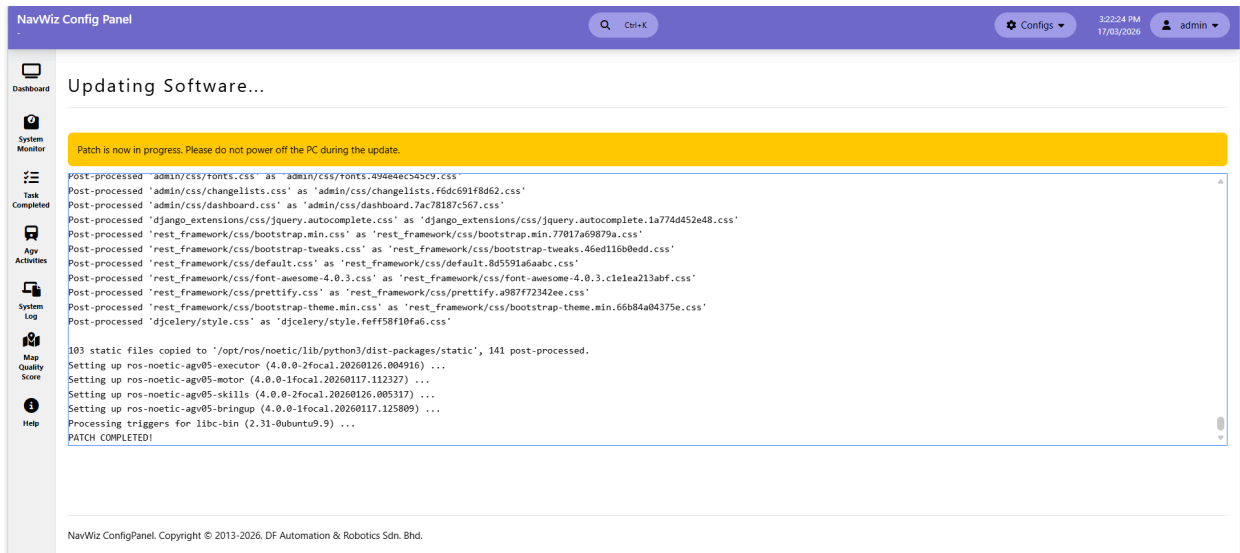
How to apply a NavWiz patch?

The screenshots illustrate the process of applying a patch in the NavWiz system:

- NavWiz User Panel:** The top interface showing the user profile icon (1) and the 'Config Panel' option in the dropdown menu (2).
- NavWiz Config Panel - Help Page:** The 'Help' icon in the sidebar (3) and the 'Change license key' button (4).
- NavWiz Config Panel - Manage Patch:** The 'Choose File' button to select a patch file (5) and the 'Apply Patch' button (6).

NavWiz ConfigPanel. Copyright © 2013-2026, DF Automation & Robotics Sdn. Bhd.

-
- Example after patch installed:



How to apply a DFleet patch?

- There are two types of files: .deb and .bin.
- Transfer the patch file to DFleet at **/home/df** using FileZilla. ([refer to this](#))
- Go to DFleet terminal then
 - .deb (*file name may vary)
 - `sudo dpkg -i {FILE NAME}`
 - Example: `sudo dpkg -i ./agvccs_3.1.0-st-davinci-zoei+248-g6ee5e02c-0bionic_amd64.deb`



- .bin (*file name must be **agvccs**)
 - `sudo rm /usr/bin/agvccs`
 - `sudo cp /home/df/agvccs /usr/bin/`
 - `sudo chmod a+x /usr/bin/agvccs`

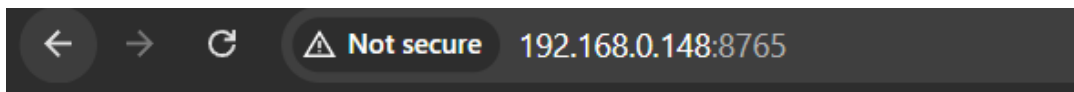
Deployment Tools

Diagnostics Visualizer

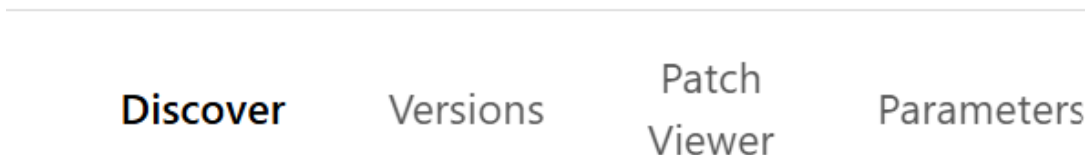
- Should be accessible through <https://diagnostic.dfautomation.com/> or <https://hub.dfautomation.com/diagnostic/>
-

Fleet Hub

- Get the latest version of deb file from <https://wiki.dfautomation.com/deployment-guide/deployment-tools>
- Transfer the file to DFleet using Filezilla.
- To install the fleet hub, simply change directory to that folder and run:
 - `sudo dpkg -i [filename]`
- To run fleet hub, simply run the following command in byobu mode:
 - `fleet-config serve --addr 0.0.0.0:8765 --trusted-cidr 0.0.0.0/0`
- Then you can now access port 8765 of the IP address to access the fleet hub. For example, I installed fleet hub on a DFleet with IP 192.168.0.148, so to access the fleethub, i need to access 192.168.0.148:8765

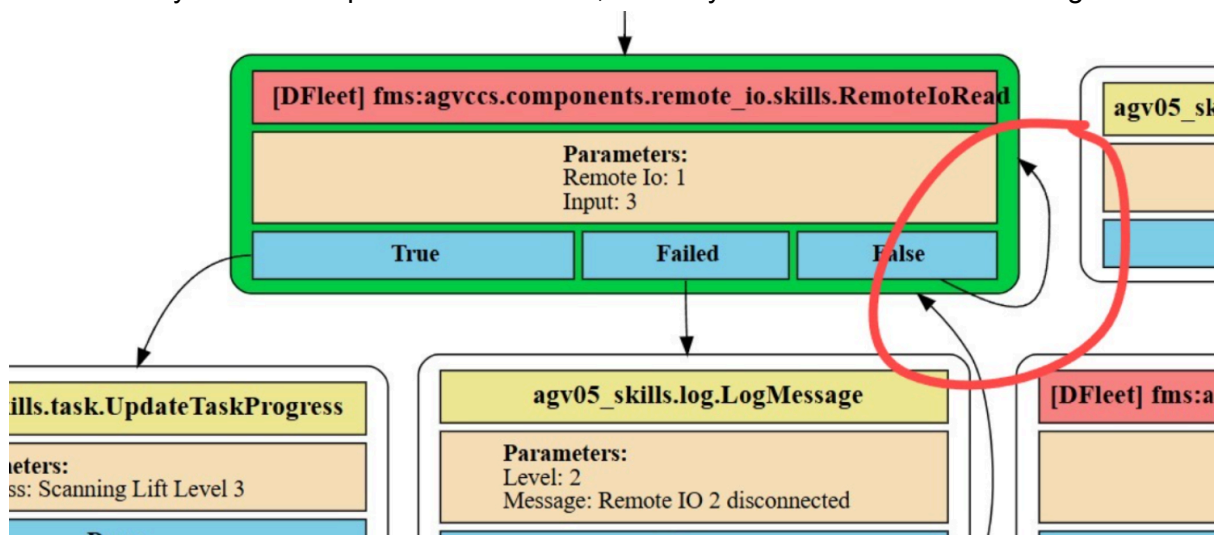


fleet-hub NavWiz AMR configuration



Task Template Guidelines

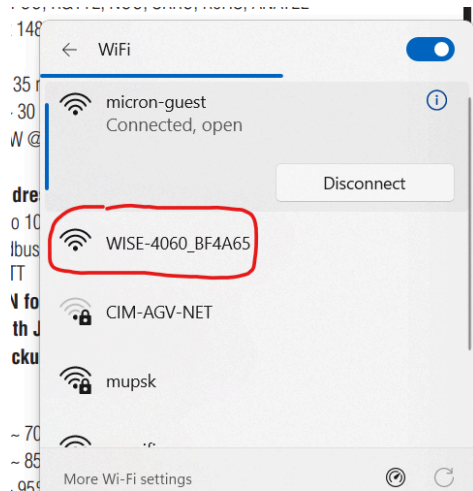
- **Never** directly wire the output back to the skill, this may cause the executor to hang.



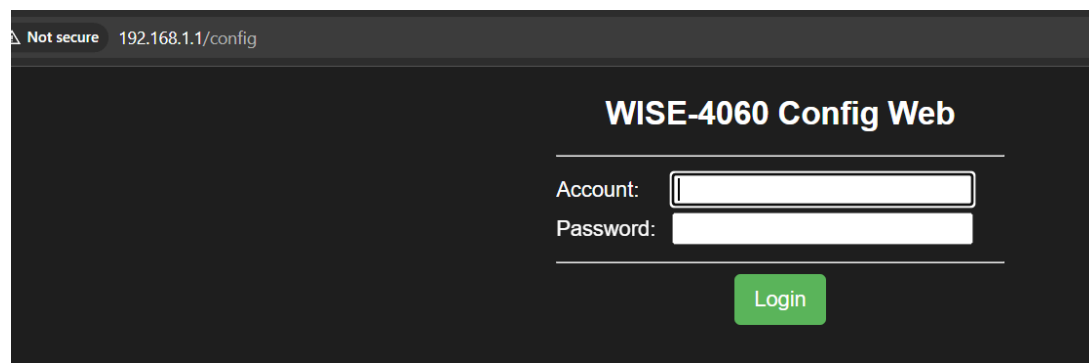
- Always play music before navigate
-

WISE4060 Wireless Remote IO Setup

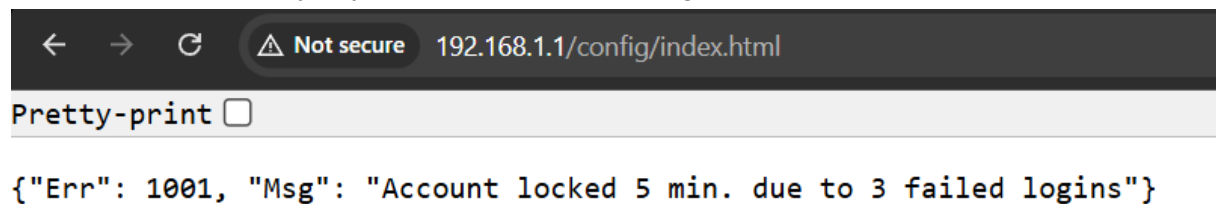
- For WISE-4000 Wireless Series, the default operation mode in normal mode is AP Mode



- The default IP of the WISE 4060 is 192.168.1.1



- Be careful not to simply key in password, else it will get locked



- They changed to 11111111 at inhouse
- Upon successful login:

WISE-4060

Information

Copy Info to Clipboard

Module

Model Name: WISE-4060

Customized Name: WISE-4060

UUID: WISE-4060_74FE48BF4A65

Location:

Description:

Working Mode: Normal Mode

Go to Configuration

- Mark down the MAC Address of the Remote IO:

Network Information

WLAN Mode: AP Mode

Mac: 74-FE-48-BF-4A-65

IP: 192.168.1.1

Subnet: 255.255.255.0

Gateway: 192.168.1.1

DNS: 0.0.0.0

IP Mode: ☒ Static ☐ DHCP

Go to Configuration

-

Miscellaneous How Tos

How to clear preventive maintenance?

- The maintenance popup mainly is because the preventive maintenance date is due or current mileage exceeds the Preventive Maintenance Mileage.
- To reset, go to the Maintenance Info page:

Configs

Maintenance Info

Robot Configuration

White Label

Webhooks

Rest API

Backup Restore

Network Connection

Date Time

User

Group

Permissions

Identity Provider

Agv

Audio

I/O

Obstacle Profiles

Detection Profiles

Maps

Parameters

Task Template

- Click on "Edit"

NavWiz Config Panel

Zoeimini

Q Ctrl+K

Configs

2:15:20 PM
12/06/2026

admin

Maintenance Info

AMR & Parts

Preventive Maintenance

Service Logs

View in DF Hub

Edit

Next P.M. Due Date 2027-03-02

Next P.M. Mileage (m) 1000000

Current Mileage (m) 14777

- Update the next P.M. Due Date and Next P.M. Mileage:

NavWiz Config Panel

Zoeimini

Q Ctrl+K

Configs

2:16:05 PM
12/06/2026

admin

Maintenance Info

AMR & Parts

Preventive Maintenance

Service Logs

View in DF Hub

Submit

Cancel

Next P.M. Due Date 02/03/2027

Next P.M. Mileage (m) 1000000

Current Mileage (m) 14777

Maker Key

- But in order to make the changes, you need the maker key:

NavWiz Config Panel

Zoeimini

Q Ctrl+K

Configs

2:16:05 PM
12/06/2026

admin

Maintenance Info

AMR & Parts

Preventive Maintenance

Service Logs

View in DF Hub

Submit

Cancel

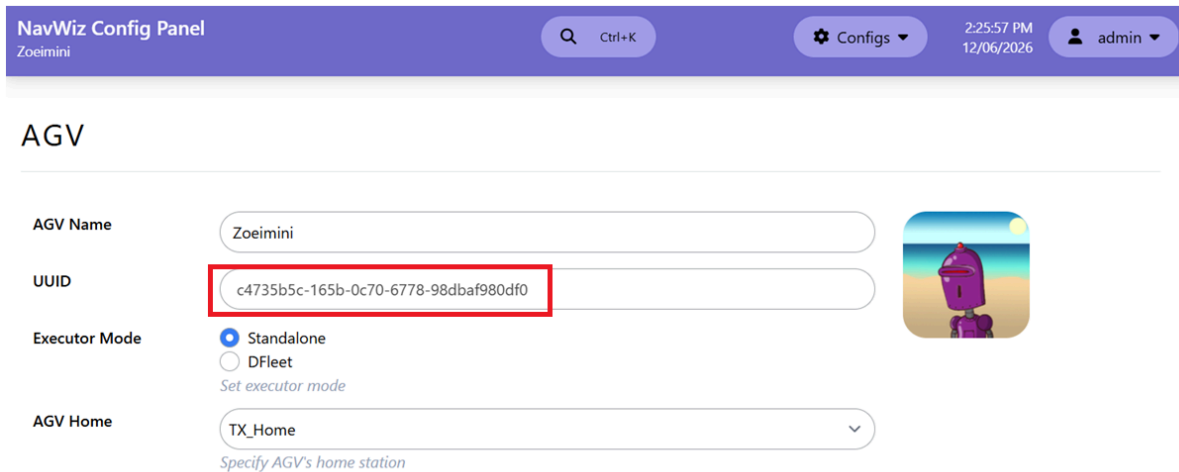
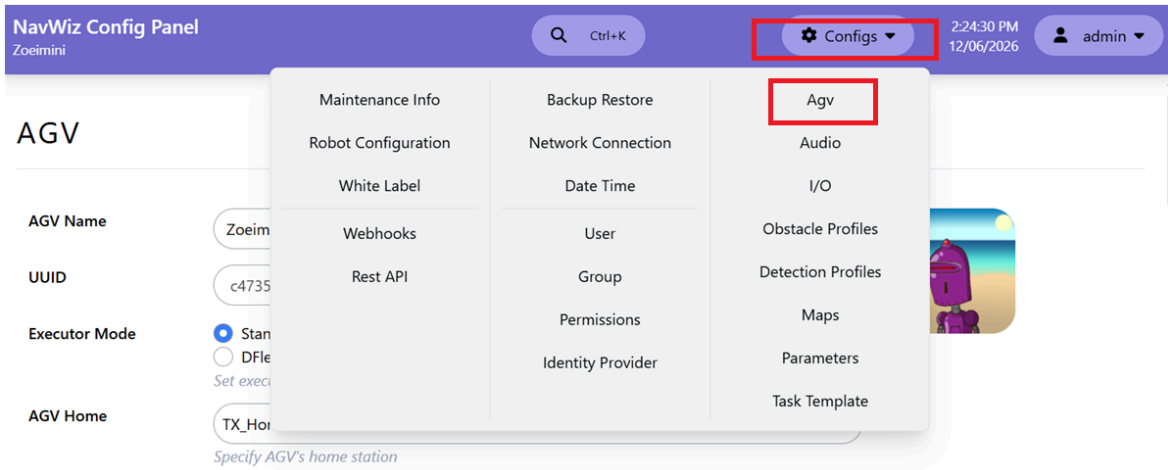
Next P.M. Due Date 02/03/2027

Next P.M. Mileage (m) 1000000

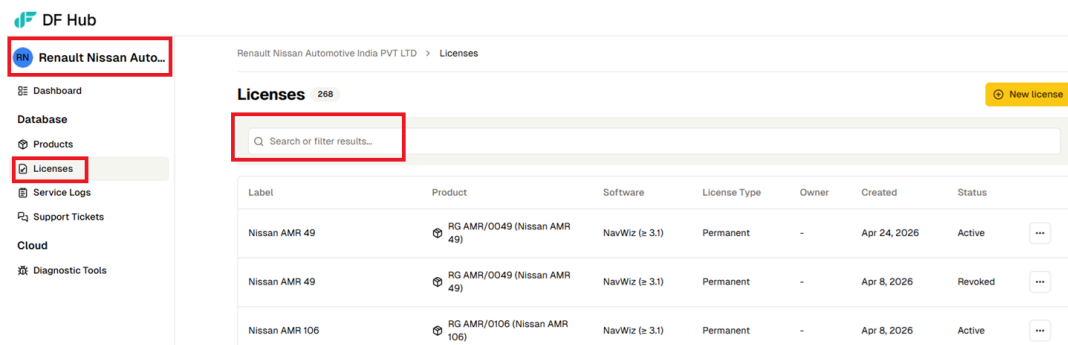
Current Mileage (m) 14777

Maker Key

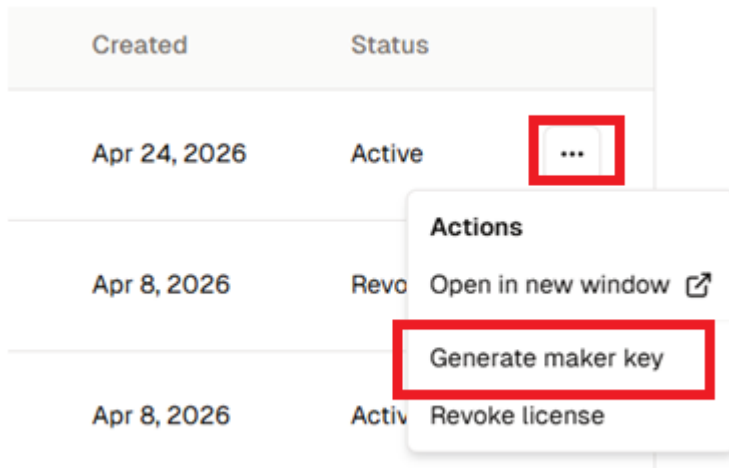
- To get maker key, you need to know the UUID of the AMR:



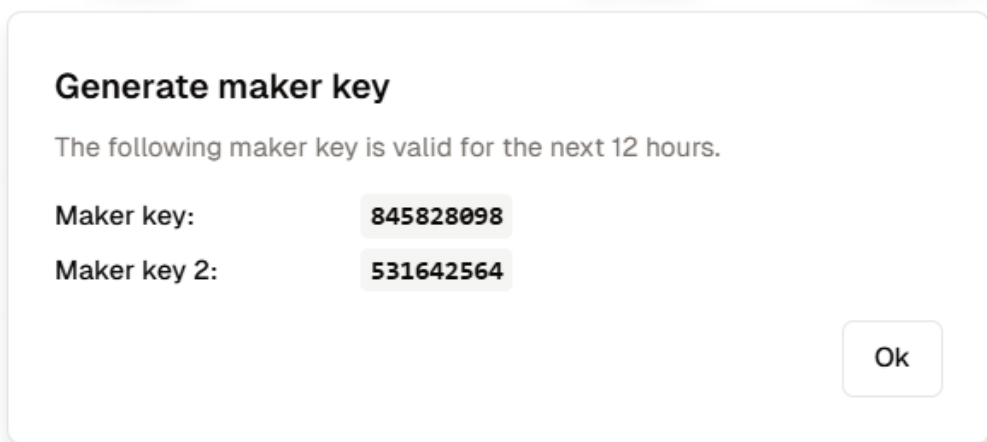
- Access DF Hub, and access the list of licenses in the organization, and search for license for the UUID:



- Then generate maker key:



- Then copy the maker key:



•

How to upgrade NavWiz 3.2 to NavWiz 4.0?

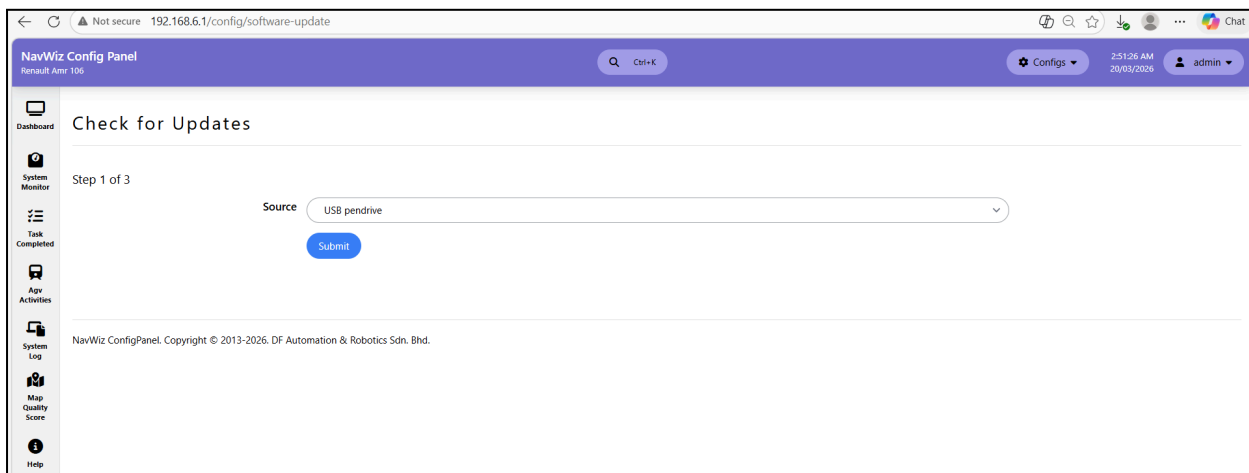
1. Better if you can backup all these:
 - a. Diagnostic
 - b. Log
 - c. Screenshot of assembly info (serial number and manufacturing date), millage and existing license page
 - d. Hardware Plugin
 - e. Screenshot of patch
 - f. Network configuration screenshot (WiFi and LAN)
 - g. Wifi p12 cert
 - h. Backup restore
 - i. Robot configuration file
 - j. Mysql
 - k. /etc/udev/rules.d/*
 - l. Workspace (if the amr is running workspace overlay)
 - i. To check this, check /usr/bin/agv05_env

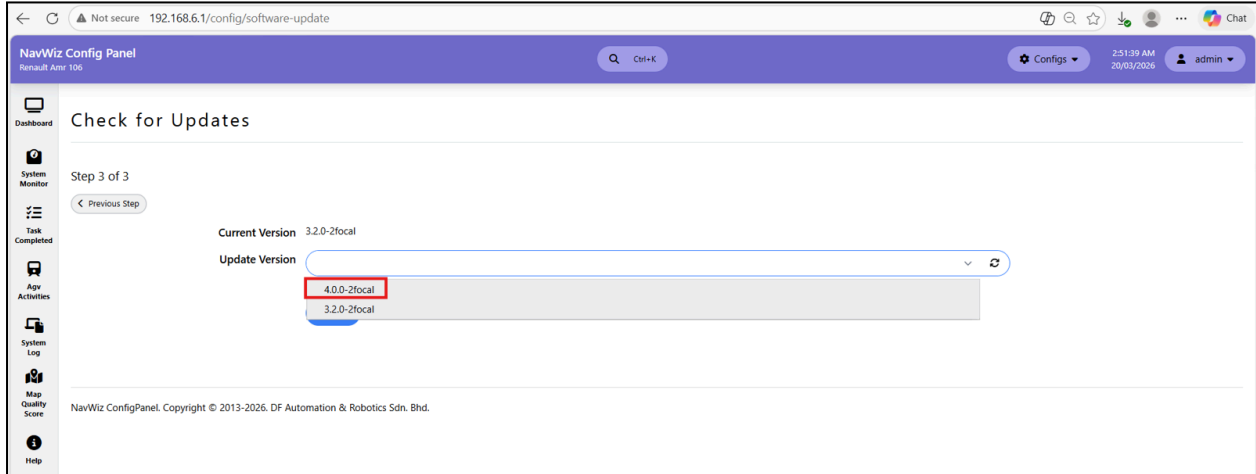
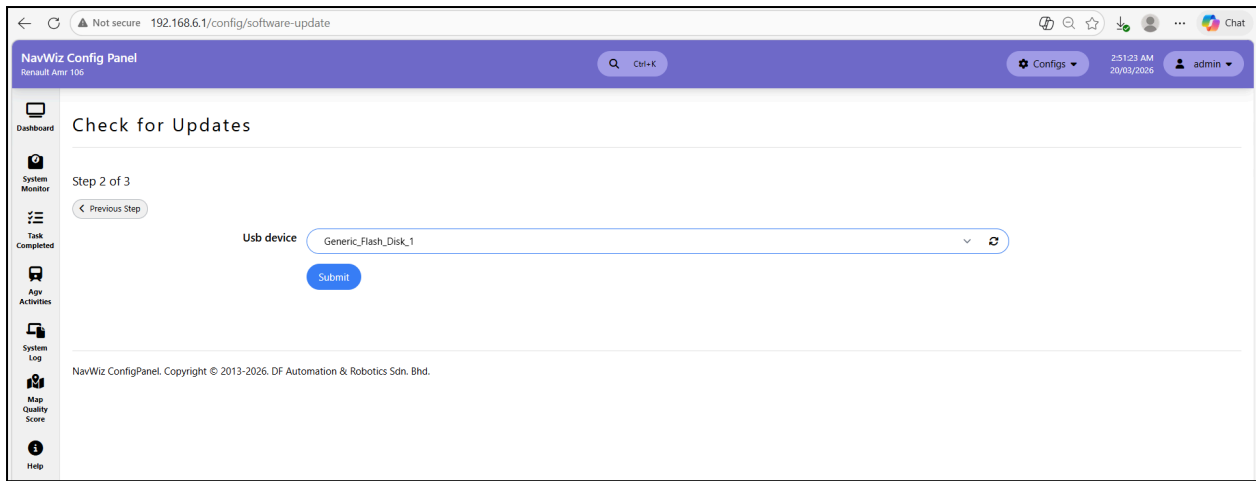

```
df@navwiz:~$ less /usr/bin/agv05_env
```
 - ii. Check whether there is /home/df/ws/devel/[setup.sh](#) at this section:

```
# Activate the ROS workspace.
export HOME
. $VENV/setup.sh
```

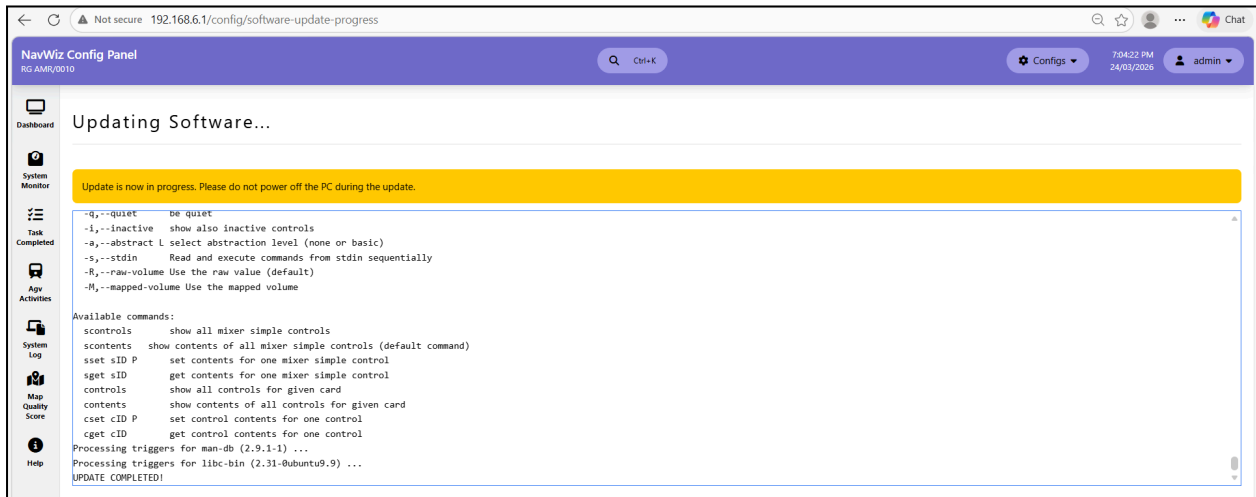
iii.

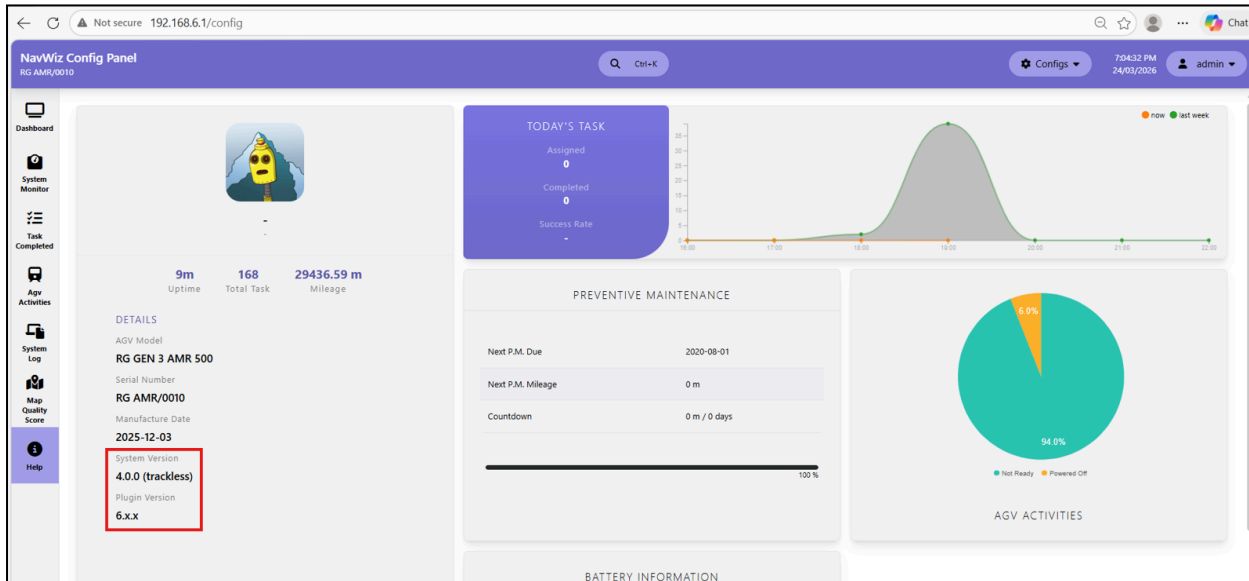
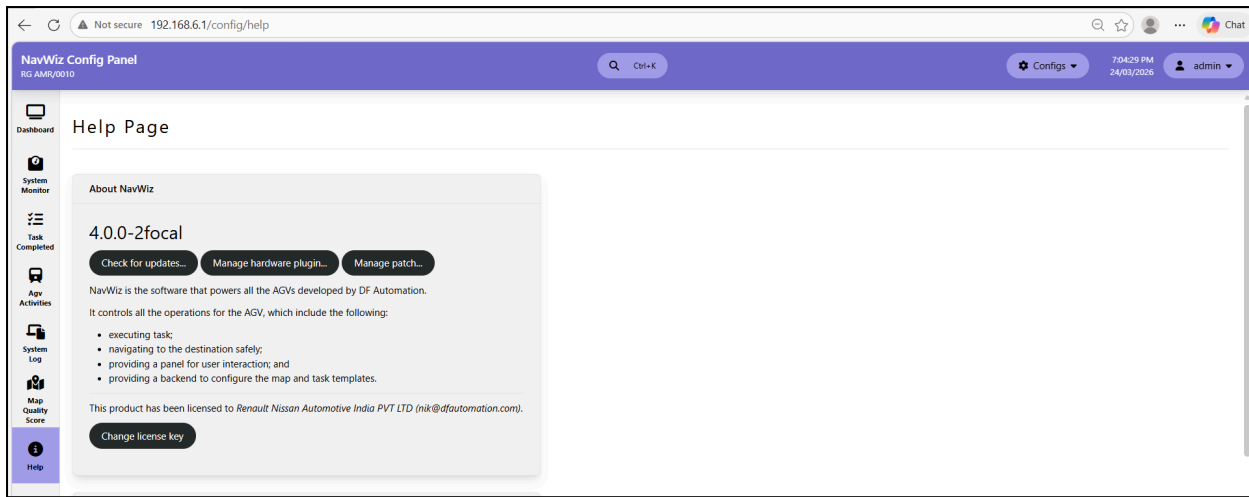
2. For diagnostic(#1), log(#2), mysql(#8) and udev(#9) backup, MAKE SURE THE ROBOT TASK RUNNER IS STOPPED, then ssh into that robot, then
 - a. `sudo supervisorctl stop all`
 - b. `mkdir ~/backup_beforeUpgrade`
 - c. `mkdir ~/archive_260202`
 - d. `find /var/lib/navwiz/.ros/diagnostics -type f ! -newermt "2025-12-01" -exec sudo mv -t ~/archive_260202 {} +`
 - e. `sudo tar --same-owner -czvpf ~/backup_beforeUpgrade/custom.tar.gz /var/lib/navwiz/media/log`
 - f. `sudo tar --same-owner -czvpf ~/backup_beforeUpgrade/roslog.tar.gz /var/lib/navwiz/.ros`
 - g. `sudo tar --same-owner -czvpf ~/backup_beforeUpgrade/varlog.tar.gz /var/log`
 - h. `sudo tar --same-owner -czvpf ~/backup_beforeUpgrade/udev.tar.gz /etc/udev/rules.d`
 - i. `sudo mysqldump -p --all-databases > ~/backup_beforeUpgrade/all_db_backup.sql`
 - j. `cp /etc/wpa_supplicant/wpa_supplicant.conf ~/backup_beforeUpgrade/.`
 - k. `cp /etc/network/interfaces ~/backup_beforeUpgrade/.`
 - l. **<TAKE NOTE:** copy wifi cert p12 file to ~/backup_beforeUpgrade folder>
 - m. `sudo ln -sT /home/df/backup_beforeUpgrade /var/log/navwiz/backup_beforeUpgrade`
 - n. `sudo supervisorctl restart all`
 - o. **<TAKE NOTE:** Filezilla download the files inside folder ~/backup_beforeUpgrade>
3. Plug the NavWiz bootable installer USB pendrive into the AMR USB port.
4. Go to NavWiz Config Panel → Help → Check for Updates.
5. Follow the steps and choose the correct settings.





6. After the update is completed, the help page will show the NavWiz version while the dashboard page will show the NavWiz system version updated to version 4 and the plugin version as 6.x.x.





How to upgrade NavWiz without USB?

-

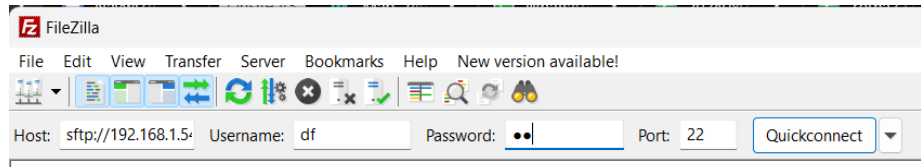
How to upgrade DFleet 3 to DFleet 4?

- First of all, you need to backup all the necessary files:
 - <https://wiki.dfautomation.com/technical-committee/software/how-to-guides/dfleet-data-migration-guide>
- You should backup:
 - License page screenshot
 - Network Configuration screenshot
 - Backup Restore
- If DFleet 3 is bionic and DFleet 4 is focal, then you can only fresh install.
- If DFleet 3 is focal, then you can sudo dpkg -i the deb files
-

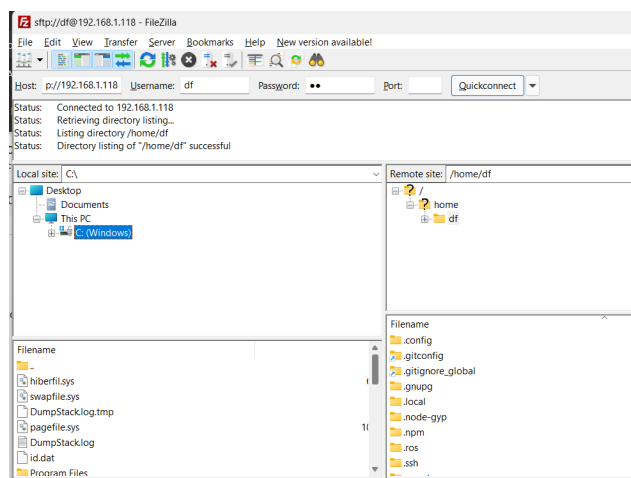
How To Transfer File From Laptop To IPC/VM?

- Requirements
 - Install FileZilla Client from <https://filezilla-project.org/>
 - Both Laptop and the device should be connected under the same network.

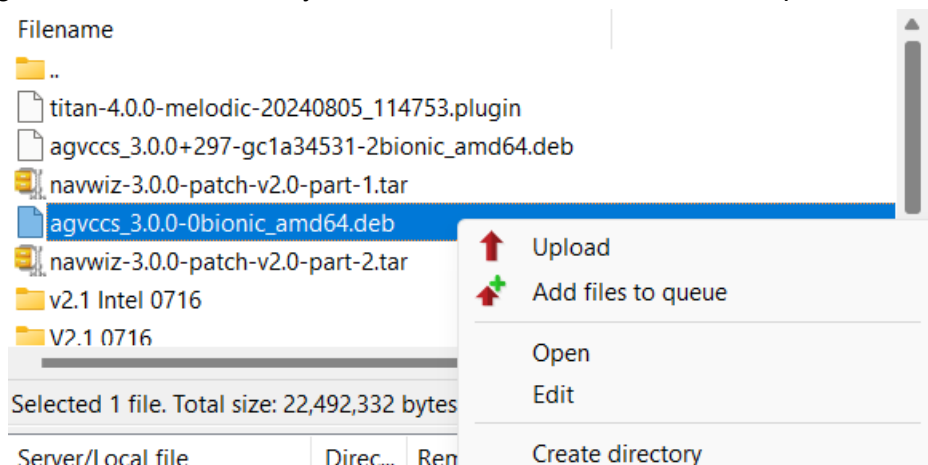
- Connect to the IPC/VM
 - Open FileZilla
 - Fill in the top 4 columns
 - Replace the IP address to IP address of the IPC (remember to add sftp:// in front)
 - Example: sftp://192.168.1.12
 - Username: df
 - Password: df
 - Port:22



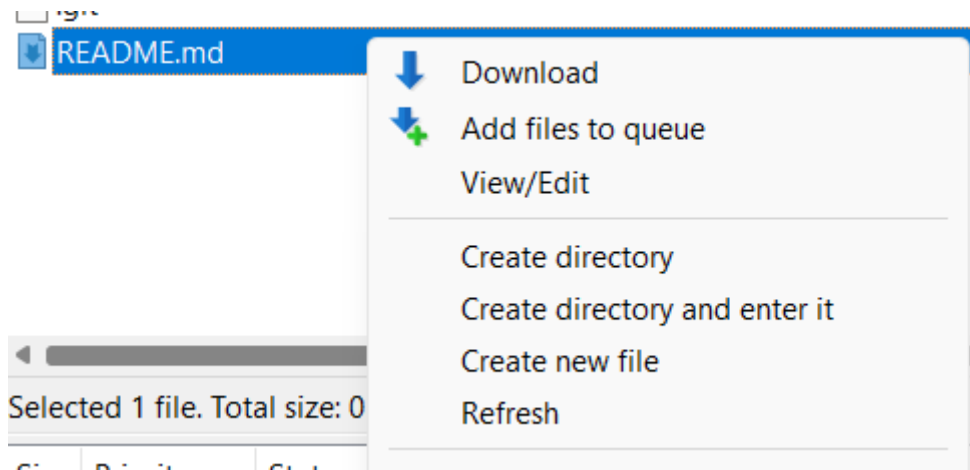
- Click Quickconnect
- Then the left hand side is the directory of your laptop, right hand side is the directory of the IPC.



- Transfer file from Laptop to IPC
 - Right click on the file that you want to transfer, and then click Upload.



- Transfer file from IPC to Laptop
 - Right click on the file that you want to transfer, and then click on Download:



How to check the skillset difference between AMRs?

- Go to /admin page and then got to agvs page

Django administration

Site administration

AGVCCS	
Agv statistics	+ Add Change
Agvs	+ Add Change
Asset 3ds	+ Add Change
Carriers	+ Add Change

Recent
My action
None av

Django administration

Home > Agvccs > Agvs

Select agv to change

Action: Go 0 of 6 selected

☐	AGV	UUID	HOME	ACTIVE
☐	SIM 01	00000000-0000-0000-0001-000000000000	Home1	✓
☐	SIM 02	00000000-0000-0000-0002-000000000000	Home2	✓
☐	SIM 03	00000000-0000-0000-0003-000000000000	Home3	✓
☐	SIM 04	00000000-0000-0000-0004-000000000000		✗
☐	SIM 05	00000000-0000-0000-0005-000000000000		✗
☐	SIM 06	00000000-0000-0000-0006-000000000000		✗

6 agvs

- Then, click on the AMR and you can see its skillset structure

Django administration

Home › Agvccs › Agvs › SIM 01

Change agv

Token: 27046af2a29f028f27c93cfe444e8f60b67228

Name: SIM 01

Home: Home1

Config panel: /components/sim-agv

Skillset:

```
{
  "dynamic_path_planning": false,
  "register_list": [
    "A0", "A1", "A2", "A3", "A4", "A5", "A6", "A7", "A8", "A9", "B0", "B1", "B2", "B3", "B4", "B5", "B6", "B7", "B8", "B9", "C0", "C1",
    "C2", "C3", "C4", "C5", "C6", "C7", "C8", "C9", "GB0", "GB1", "GB2", "GB3", "GB4", "GB5", "GB6", "GB7", "GB8", "GB9", "P",
    "S0", "PS1", "PS2", "PS3", "PS4", "PS5", "PS6", "PS7", "PS8", "PS9"],
  "skill_descriptions": [
    {
      "id": "agv05_skills.plan_x.NavigateToX",
      "mutexes": [
        "base",
        "plan",
        "traffic"
      ],
      "name": "Navigate To",
      "outcomes": [
        "Done"
      ],
      "params": [
        {
          "description": "Choose a destination station.",
          "name": "station",
          "type": "Station"
        },
        {
          "description": "Use front(0) or rear(1) line sensor, or trackless localization(-1) for station alignment.",
          "name": "align_station_type",
          "type": "int"
        }
      ]
    },
    {
      "id": "agv05_skills.plan_x.NavigateFromX",
      "mutexes": [
        "base",
        "plan",
        "traffic"
      ],
      "name": "Navigate From",
      "outcomes": [
        "Done"
      ],
      "params": [
        {
          "description": "Choose a destination station.",
          "name": "station",
          "type": "Station"
        },
        {
          "description": "Use front(0) or rear(1) line sensor, or trackless localization(-1) for station alignment.",
          "name": "align_station_type",
          "type": "int"
        }
      ]
    }
  ]
}
```

- Copy the entire skillset structure, and then paste it to the Diffchecker (<https://www.diffchecker.com/>) website.

Diffchecker Desktop

The most secure way to run Diffchecker. Get the Diffchecker Desktop app: your diffs never leave your computer!

Get Desktop

Original text

```
1 {
  "dynamic_path_planning": false,
  "register_list": [
    "A0", "A1", "A2", "A3", "A4", "A5", "A6", "A7", "A8", "A9", "B0", "B1", "B2", "B3", "B4",
    "B5", "B6", "B7", "B8", "B9", "C0", "C1", "C2", "C3", "C4", "C5", "C6", "C7", "C8", "C9",
    "GB0", "GB1", "GB2", "GB3", "GB4", "GB5", "GB6", "GB7", "GB8", "GB9", "P",
    "S0", "PS1", "PS2", "PS3", "PS4", "PS5", "PS6", "PS7", "PS8", "PS9"],
  "skill_descriptions": [
    {
      "id": "agv05_skills.plan_x.NavigateToX",
      "mutexes": [
        "base",
        "plan",
        "traffic"
      ],
      "name": "Navigate To",
      "outcomes": [
        "Done"
      ],
      "params": [
        {
          "description": "Choose a destination station.",
          "name": "station",
          "type": "Station"
        },
        {
          "description": "Use front(0) or rear(1) line sensor, or trackless localization(-1) for station alignment.",
          "name": "align_station_type",
          "type": "int"
        }
      ]
    },
    {
      "id": "agv05_skills.plan_x.NavigateFromX",
      "mutexes": [
        "base",
        "plan",
        "traffic"
      ],
      "name": "Navigate From",
      "outcomes": [
        "Done"
      ],
      "params": [
        {
          "description": "Choose a destination station.",
          "name": "station",
          "type": "Station"
        },
        {
          "description": "Use front(0) or rear(1) line sensor, or trackless localization(-1) for station alignment.",
          "name": "align_station_type",
          "type": "int"
        }
      ]
    }
  ]
}
```

Changed text

- Copy the skillset of another AMR and then paste it to the Diffchecker website also. From there you can analyze what's the difference between each AMR's skillset.

How to change speech audio?

- Use Filezilla to check this directory /var/lib/navwiz/media/audio/speech/:

Remote site: /var/lib/navwiz/media/audio/speech_media_	Filename	Filesize	Filetype	Last mod...	Permis...	Owner/...
	rotate_right_b107482c.wav	71,368	WAV FL...	18/6/202...	-rw-r--...	navwiz ...
	rotate_left_ef40f1a8.wav	71,786	WAV FL...	18/6/202...	-rw-r--...	navwiz ...
	obstacle_blocked_37af7672.wav	90,736	WAV FL...	18/6/202...	-rw-r--...	navwiz ...
	obstacle_disabled_5f73fd08.wav	89,174	WAV FL...	18/6/202...	-rw-r--...	navwiz ...

- Prepare the new audio files in .wav format.
- Rename the audio files following the same file name listed in /var/lib/navwiz/media/audio/speech directory.
- Use filezilla to upload the files to replace the original files.

How to clear DFleet mysql log?

- Sometimes the logs get piled up and cause DFleet disk full, to remove this we can go to the VM, log in using root user and delete the old binlog files in /var/lib/mysql

How to draw safety areas using VM for NavWiz 3.2?

- Manually run these commands:
 - `source /opt/ros/noetic/setup.sh`
 - `roslaunch agv05_lidar agv05_lidar`
 - `roslaunch agv05_obstacle_sensor agv05_obstacle_sensor`

How to use 50mm resolution for mapping?

- Go to Robot Configuration page, set MAP_RESOLUTION to 0.05

Environment Variable

#	Identifier	Value	Operation
1	MAP_RESOLUTION	0.05	↑ ↓ 📄 ✕

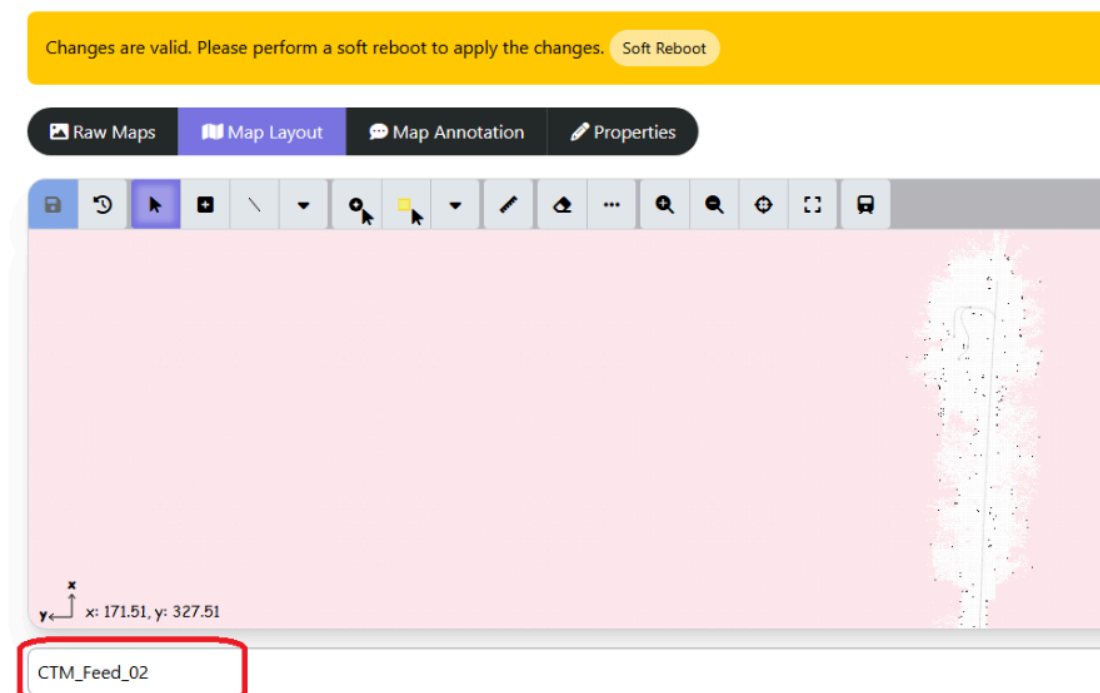


- Submit and softreboot. Then you can start doing mapping.

How to reduce map resolution from 20mm to 50mm (On NavWiz side)

- Identify the active raw map being used:

< Maps Map "DF-CTM_2"



- Go to admin page, and check the raw map file name of the current raw map

← → ↻ ⚠ Not secure 192.168.1.103/admin/systemx/mapocg/

Django administration

Home > Systemx > Raw maps

Select raw map to change

Action: ----- Go 0 of 15 selected

☐	RAW MAP
☐	CTM_Feed_02
☐	CTM_PreFinal_Kiiting

Home > Systemx > Raw maps > CTM_Feed_02

Change raw map

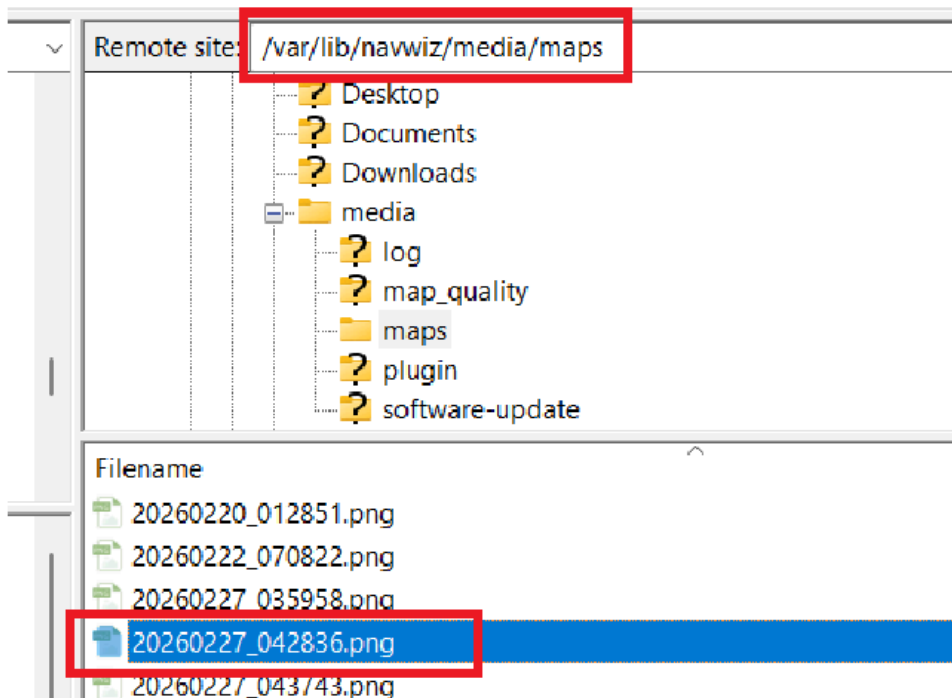
Name: CTM_Feed_02

Metadata: {"frame_id":"map","resolution":0.02,"width":9422,"height":2699,"x0":-3.87,"y0":-23.098}

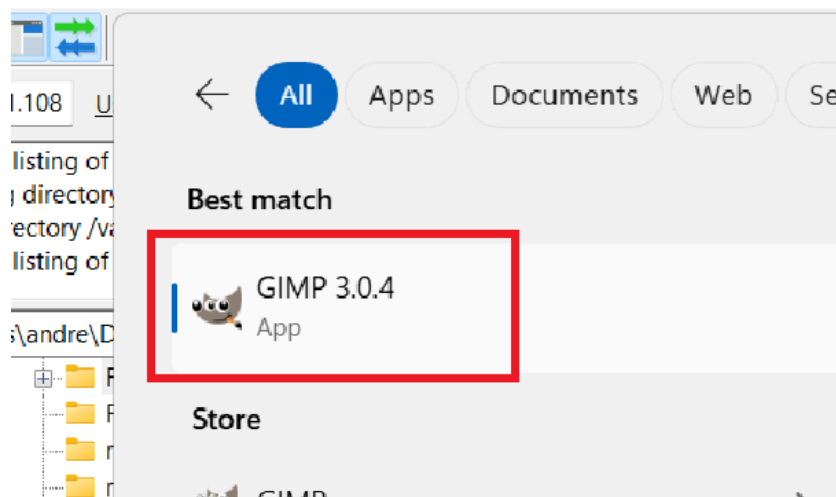
Png file: Currently: maps/20260227_042836.png
Change: Choose File No file chosen

Map: DF-CTM_2

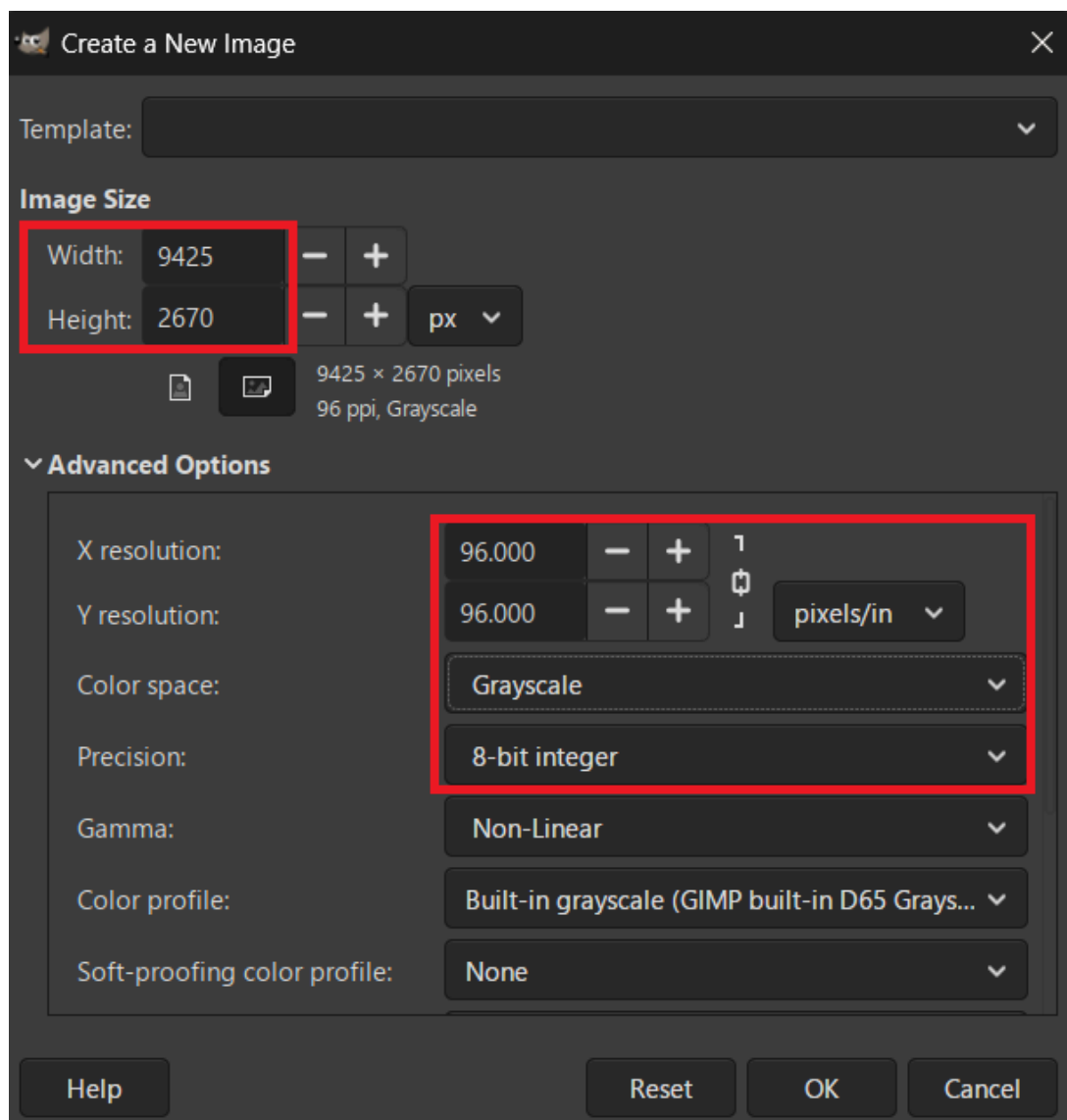
- Extract and save the metadata:
{"frame_id":"map","resolution":0.02,"width":9422,"height":2699,"x0":-3.87,"y0":-23.098}
- Use Filezilla, go to /var/lib/navwiz/media/maps and download the raw image file



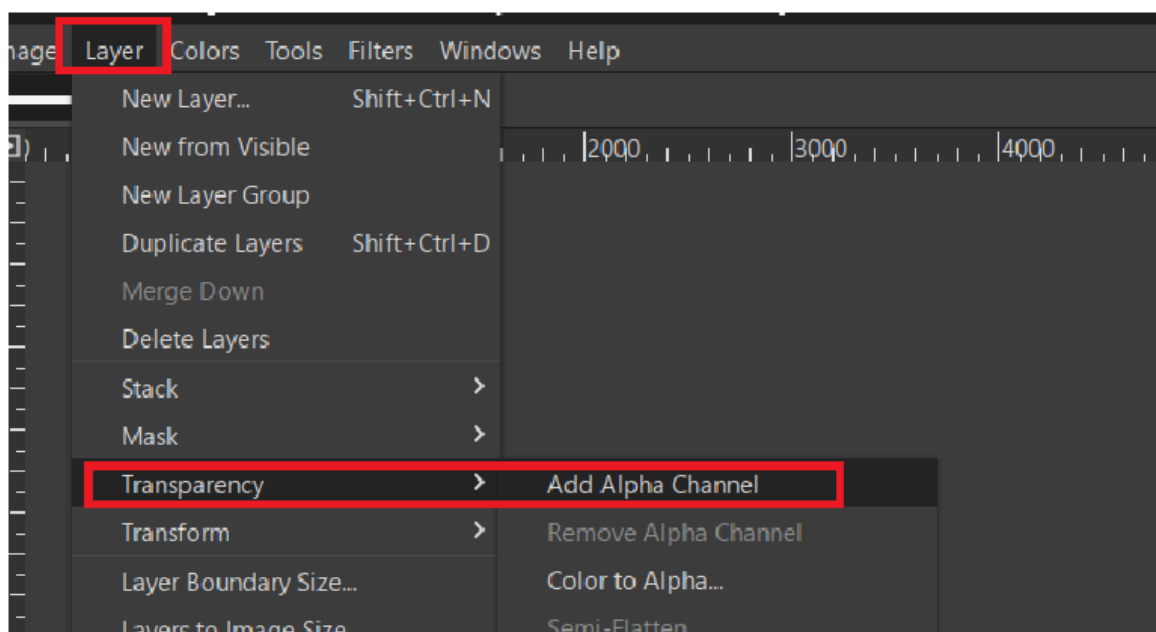
- Now, open GIMP



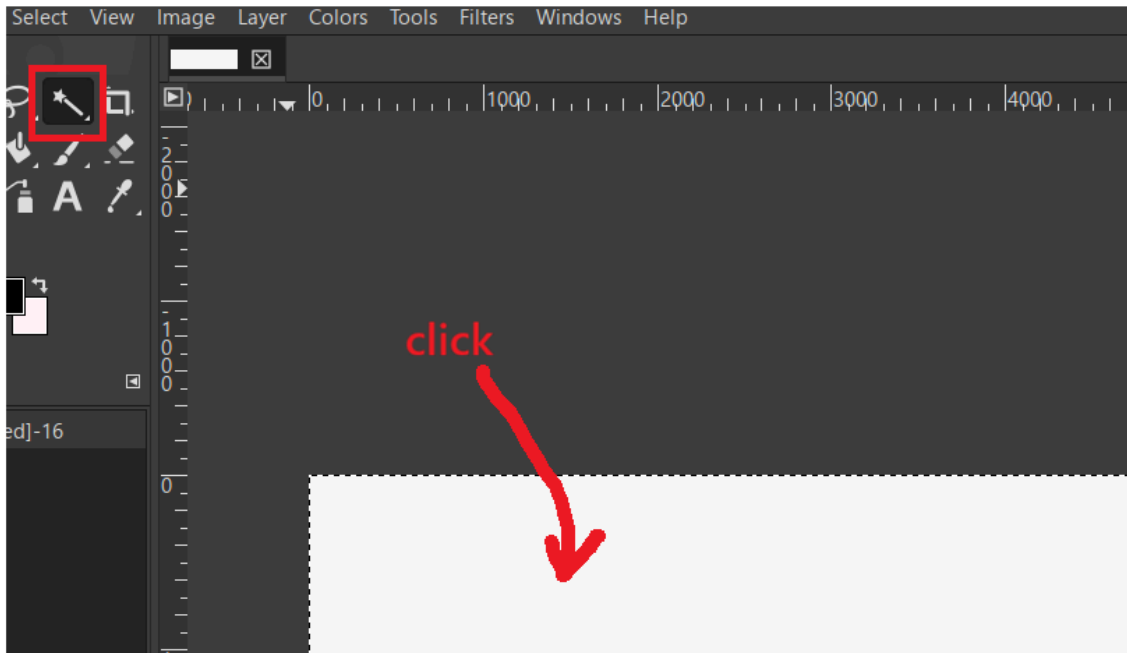
- We need to reduce the resolution from 0.02m/pixel to 0.05m/pixel
 $0.02 / 0.05 = 0.4$
 So we need to reduce the width and height of the current raw map to 40%. From the metadata above, the original width is 9422, original height is 2699
 $9422 * 0.4 = 3768.8$ pixels
 $2699 * 0.4 = 1079.6$ pixels
- Since the pixel is not whole number, we need to add new image with pixel size that can result in whole number:
 - $9425 * 0.4 = 3770$
 - $2670 * 0.4 = 1068$
 - 9422 -> 9425 pixels
 - 2699 -> 2670 pixels



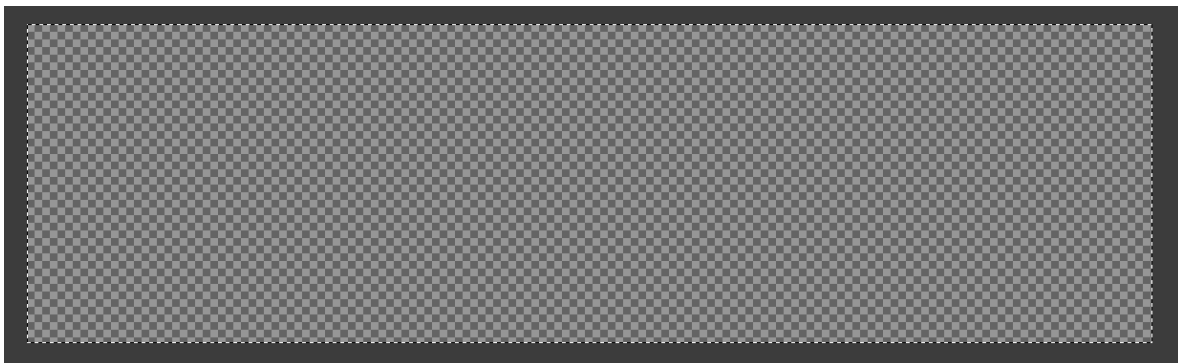
- Add alpha channel to the new image:



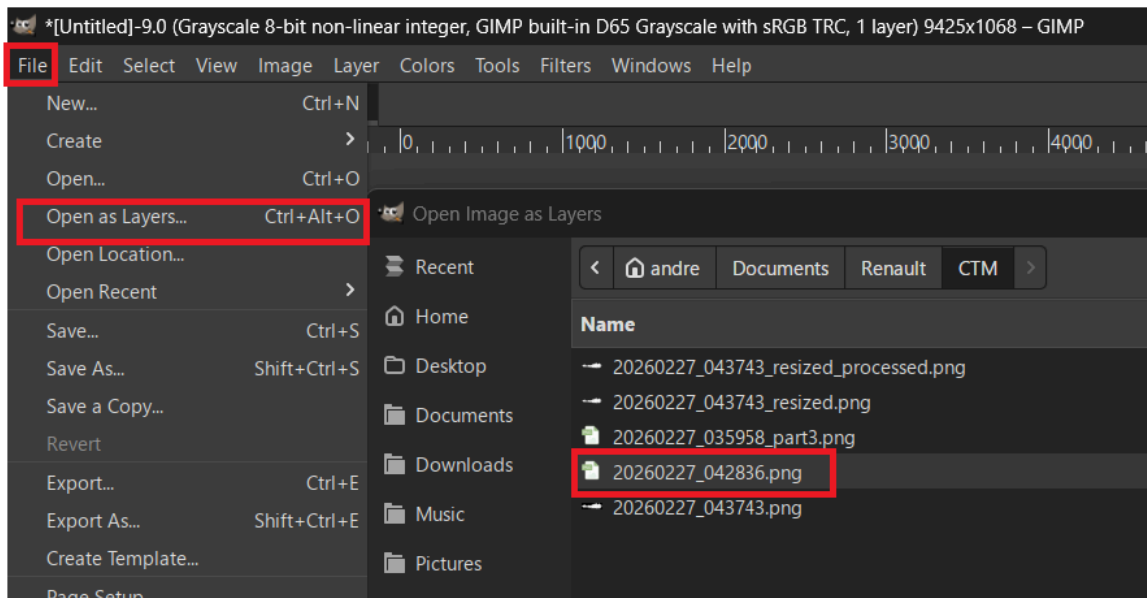
- Use fuzzy tool to select the white image



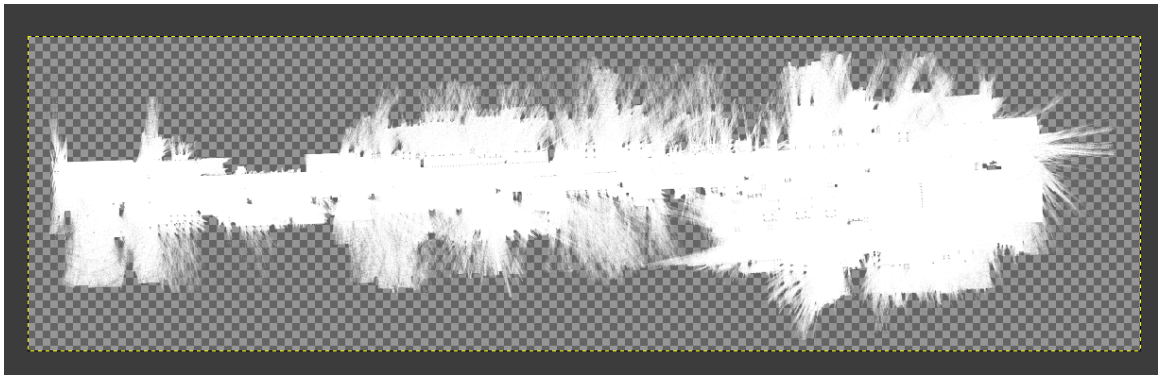
- And then click delete to make the entire image transparent



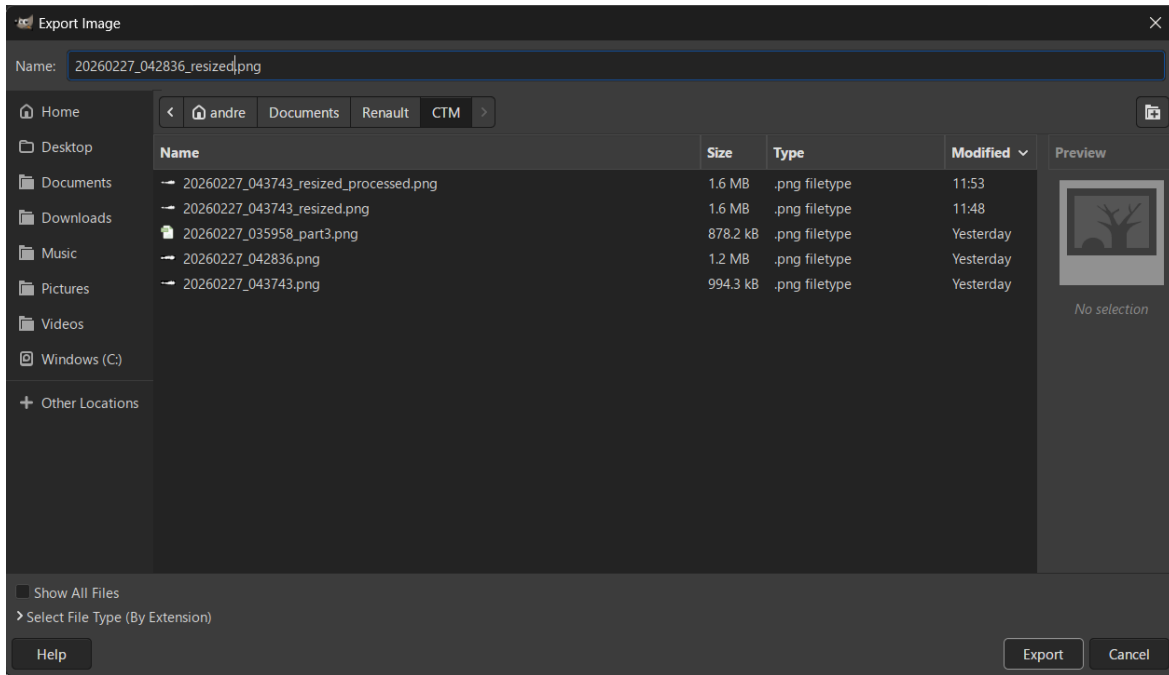
- Open image as layer, and then select the raw map image file you extracted through FileZilla.

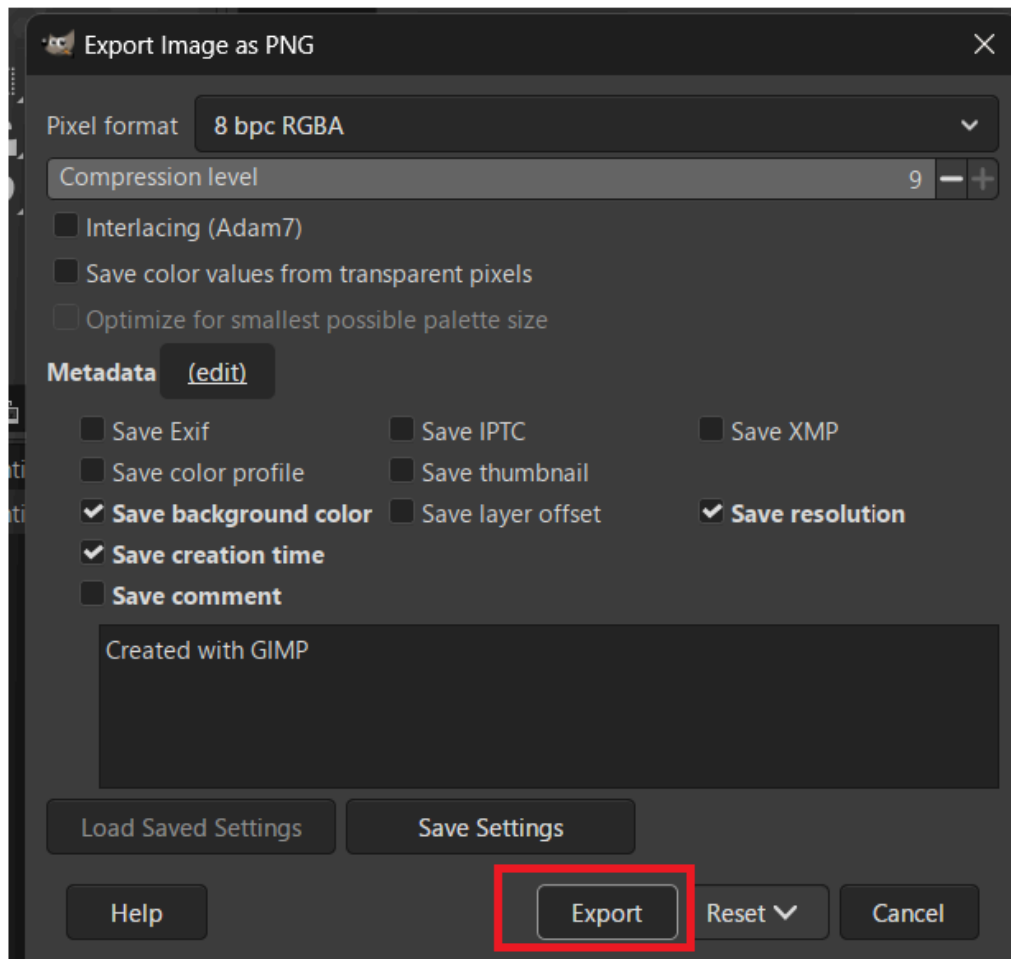


Take note that the image shall have transparent background instead of black background.

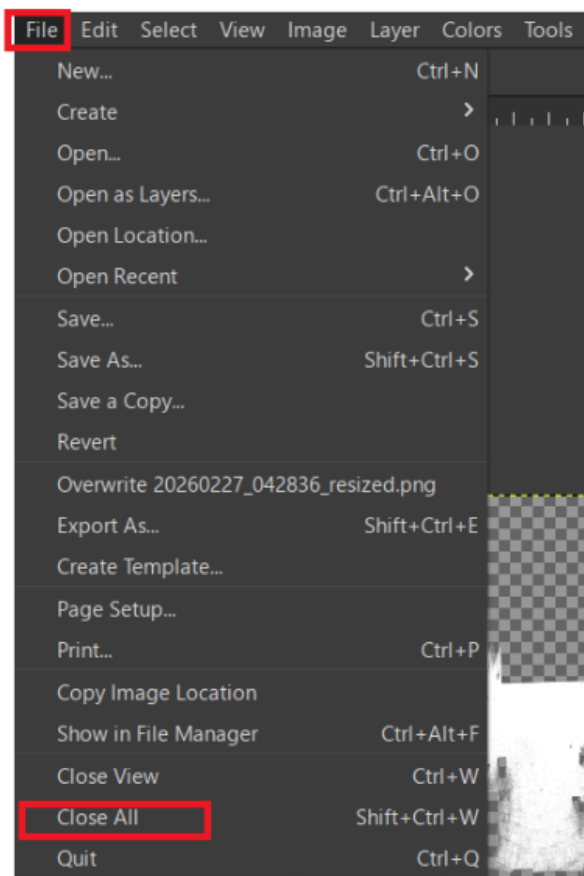


- Export the entire image as a new .png image (my case is something like 20260227_042836_resized.png)

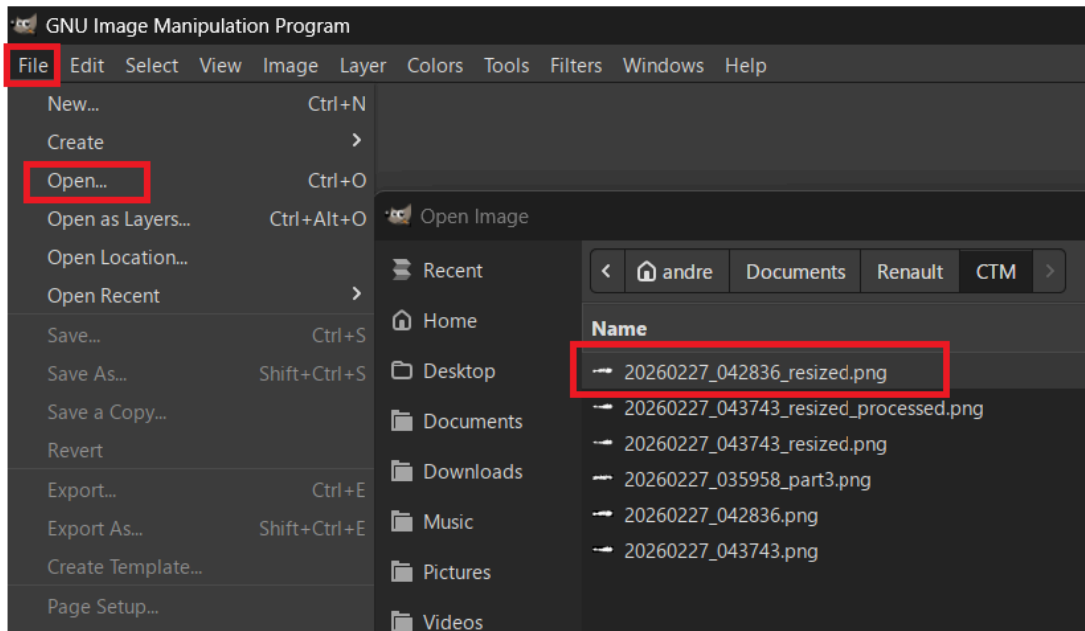




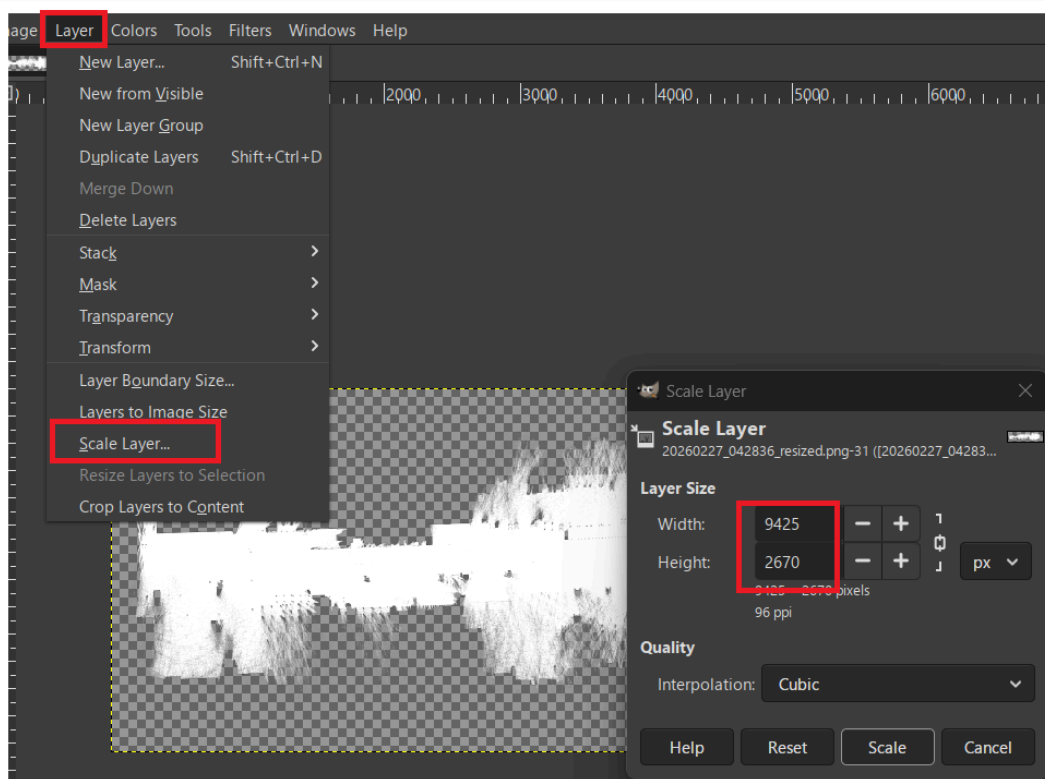
- Close all



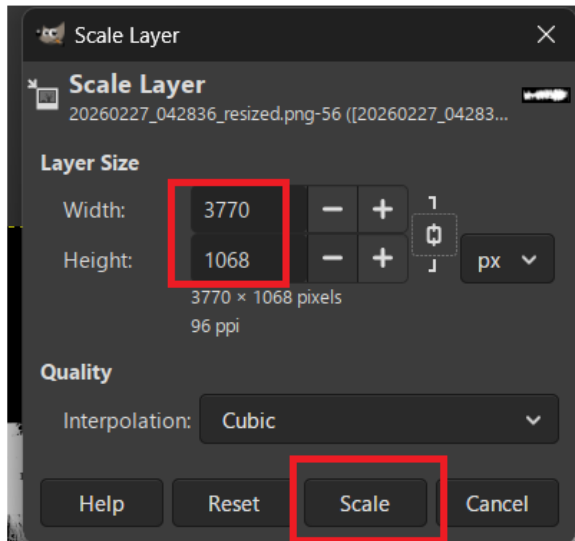
- Open the new resized file:



- Click scale layer and check the size:



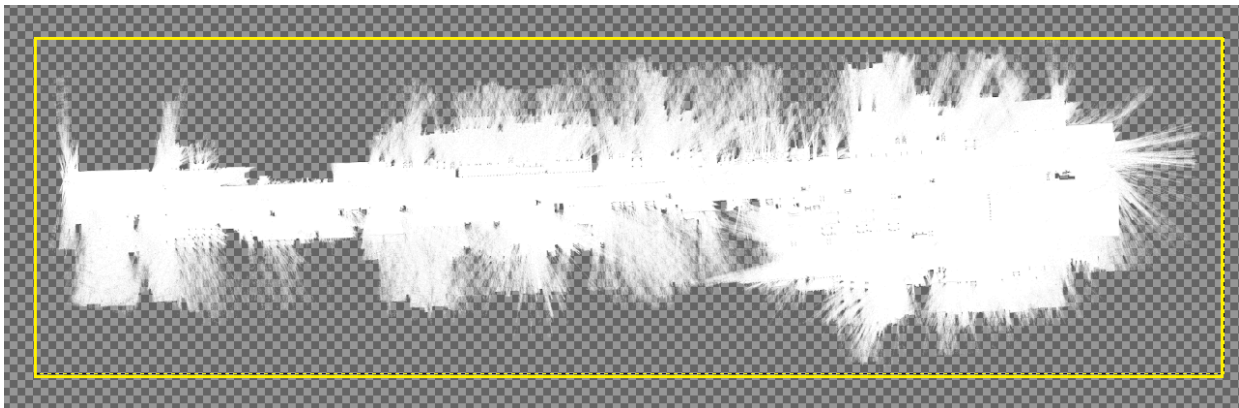
- Now we can scale it to our desired size:



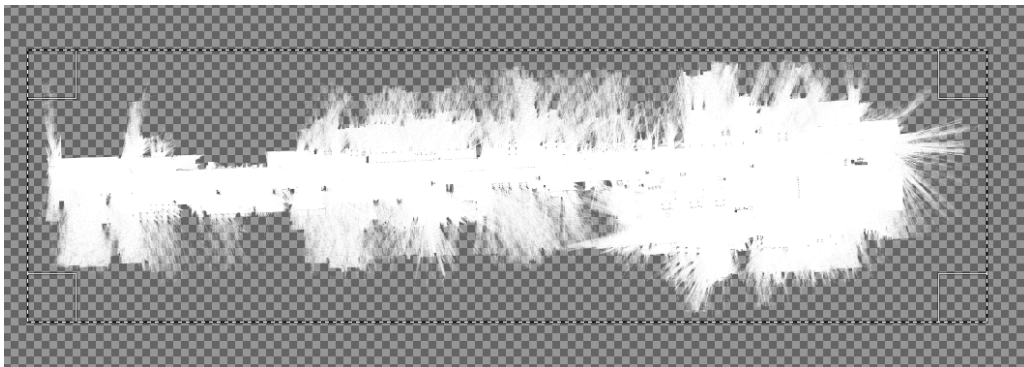
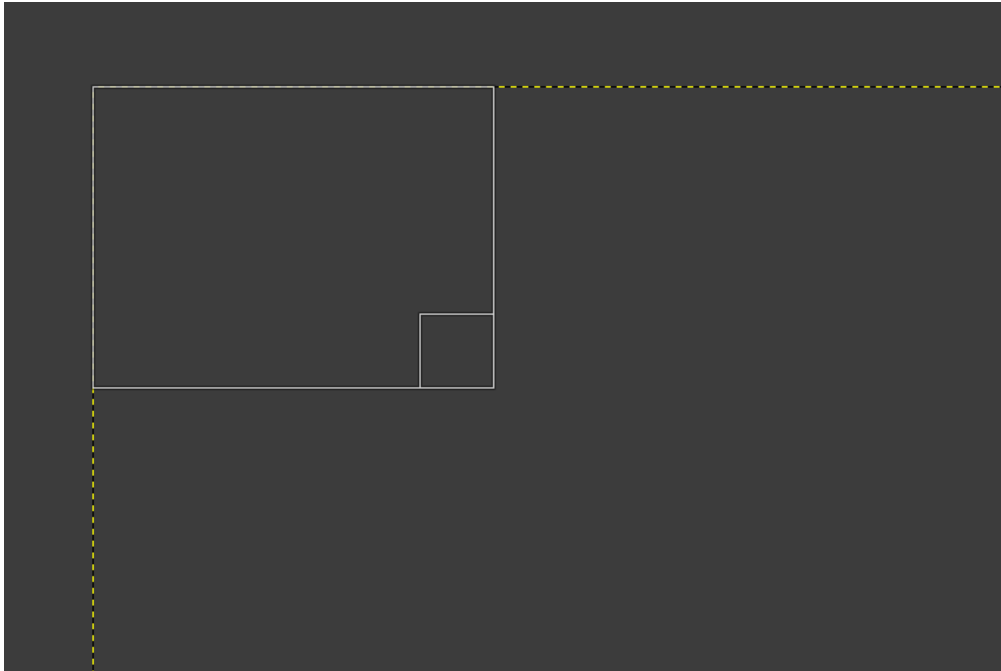
- Click the rectangle selection tool:



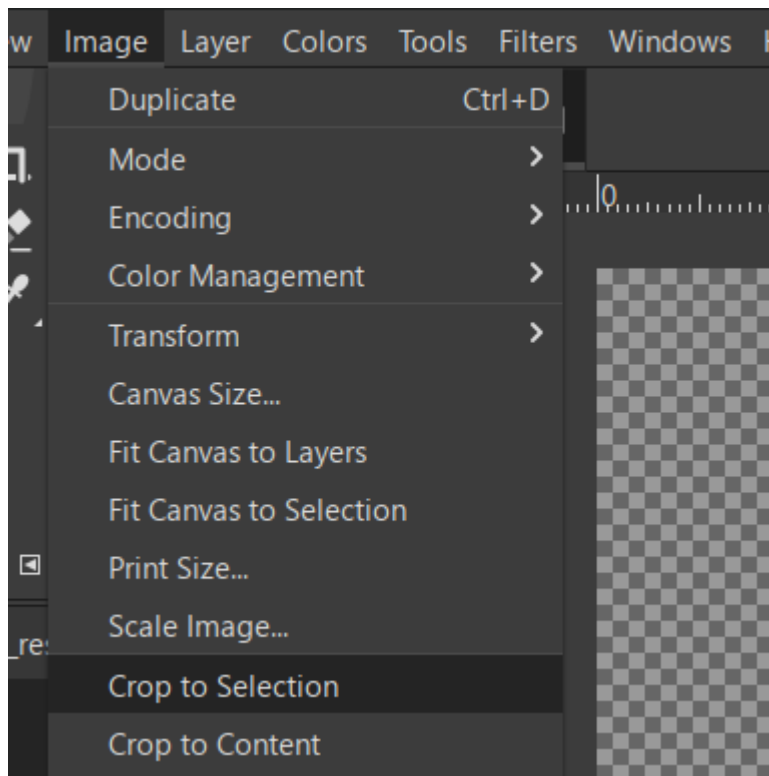
- Select the region highlighted, zoom in closely and ensure your selection region is exactly same size like the rescaled size:



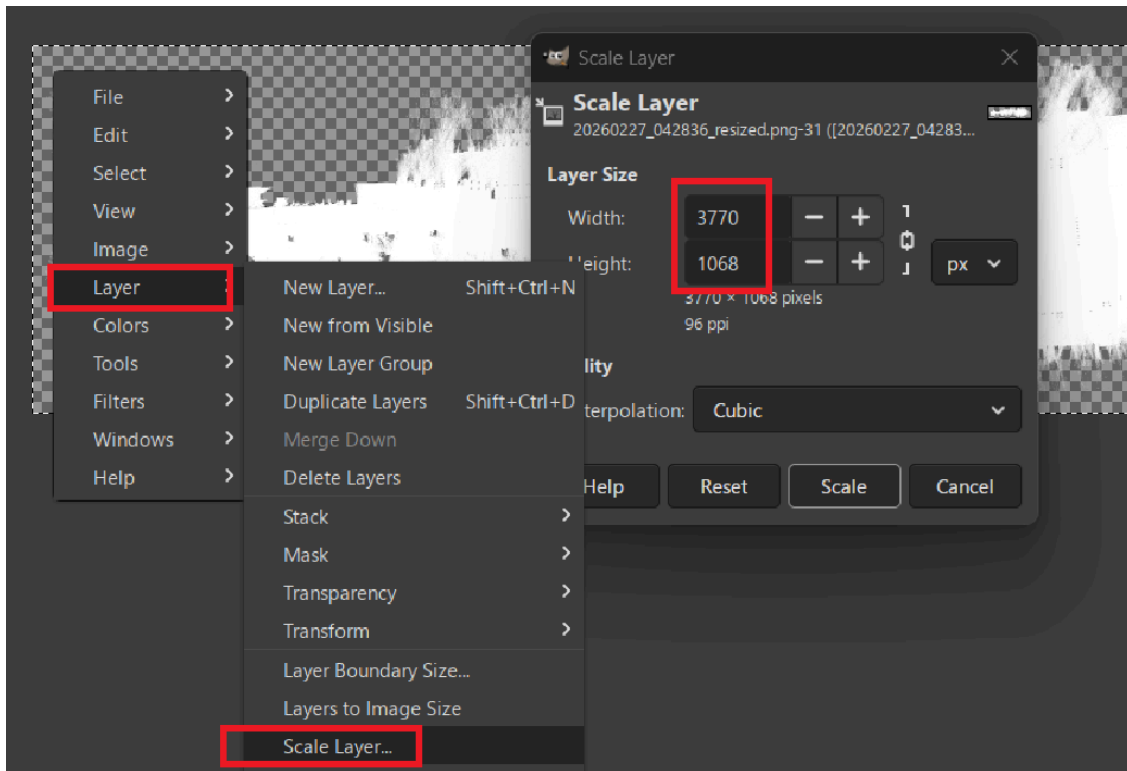
Zoom until the background become black to ensure you didnt miss any pixel:



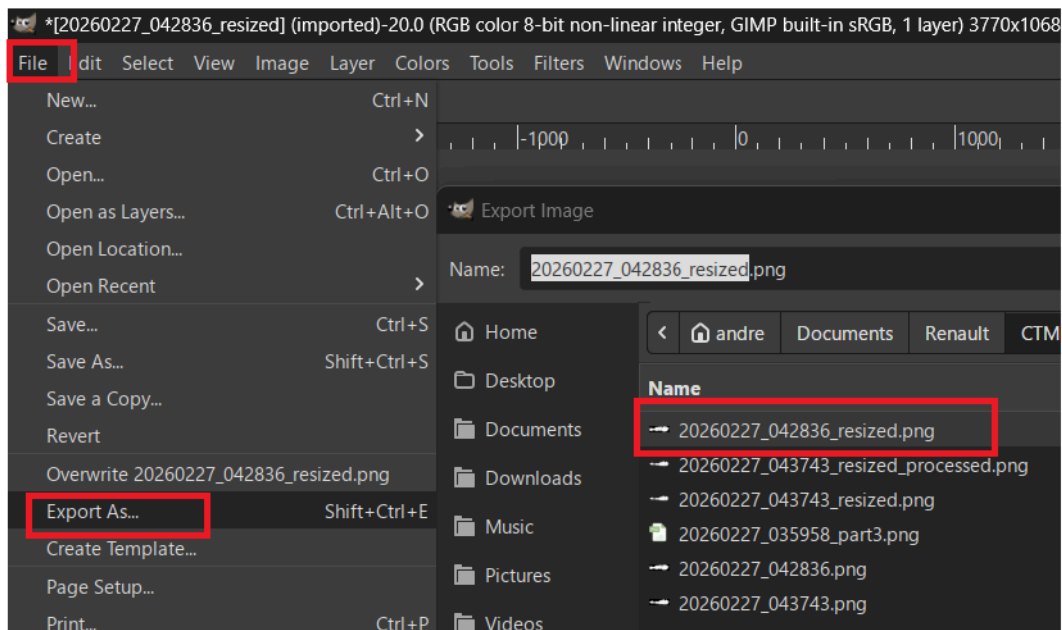
- After the selection, click crop to selection:



- Click right click scale image to ensure the new size is exactly same like what we planned to have:



- Export as again to replace the current file:



- Now, we need to manually add the resized image into raw map database using /admin page



Add raw map

Name:

Metadata:

Png file:

Choose File

No file chosen

Map:

▼

☐ Is named☐ Is autosaved

- Fill in the details. For metadata, reuse the previous one but update the resolution, width and height. Upload the resized map. Assign the raw map to your active map layout. Tick the “Is named”.

Home › Systemx › Raw maps › Add raw map

Add raw map

Name:

Metadata:

(*frame_id:"map",resolution:0.05,width:3770,height:1068,x0:-3.87,y0:-23.098)

Png file:

Choose File | 20260227_0...resized.png

Map:

DF-CTM_2 ▼

☒ Is named

☐ Is autosaved

Save and add another

Save and continue editing

SAVE

- The resized map are in gray colour, we need to darken it, go to the raw map list and edit it using threshold 255:

Darken: Threshold 255

✕Cancel

- Then, go to the map layout and select the new map, switch between the original and the resized map, ensuring there are not many shifts in raw map data. If yes, kindly please fine tune the x0 and y0 values.
-

How to migrate logs from one DFleet to another?

- To copy log files from old server to new server:
 - Ssh into the server
 - **sudo supervisorctl stop all** for both old and new servers, so that there will not be ongoing logs
 - Copy/Sync the files from old server to new server:
 - `sudo rsync -a --progress df@(ip_of_old_server):/var/log/fms/ /var/log/fms/`
 - Example: `sudo rsync -a --progress df@10.77.212.62:/var/log/fms/ /var/log/fms/`
- To copy media files from old server to new server:
 - `sudo rsync -a --progress df@10.77.212.62:/var/lib/fms/media/ /var/lib/fms/media/`

How to serve the DFleet kiosk on port 8080 instead of port 80?

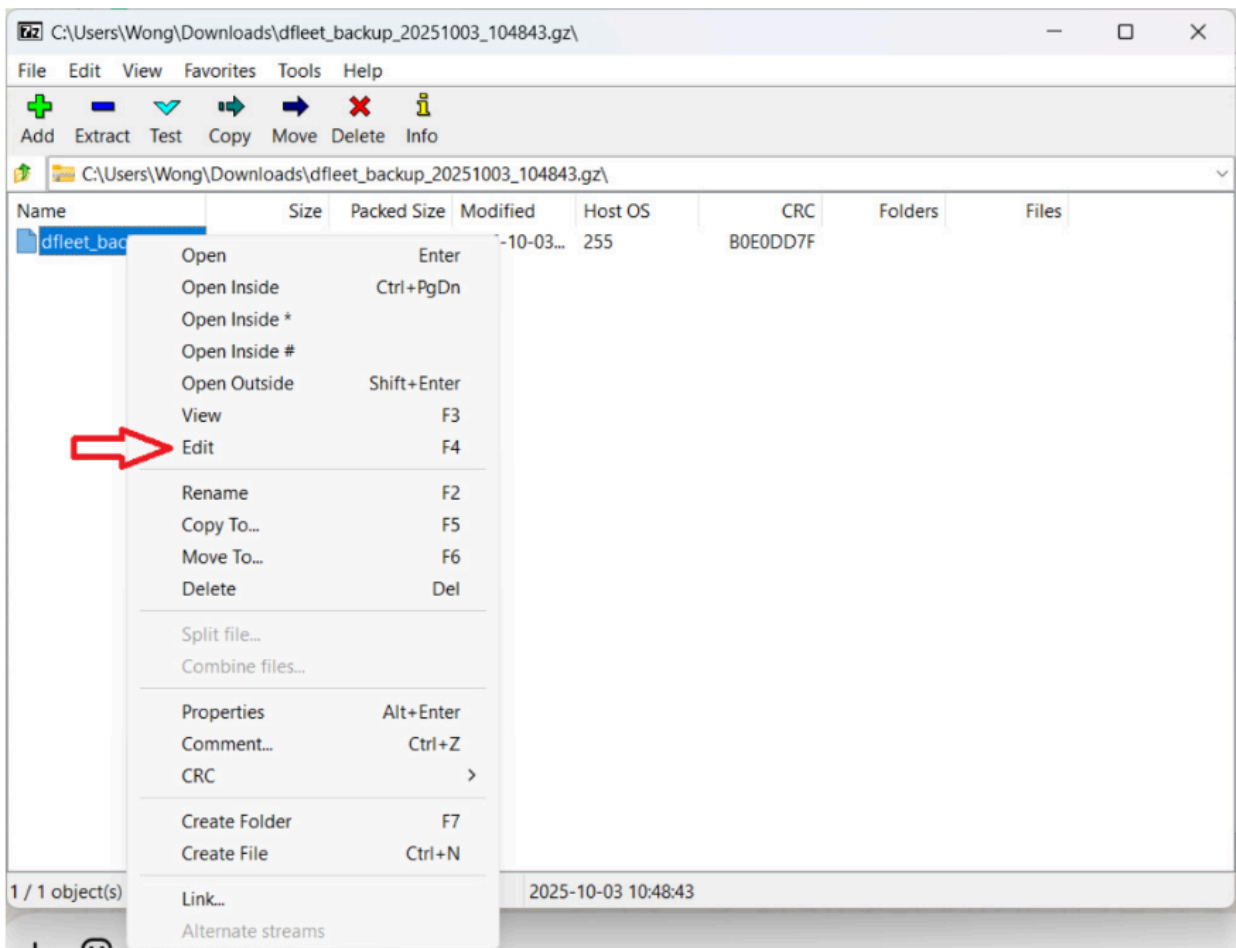
- `sudo vi /etc/nginx/sites-enabled/agvccs`
- add to serve both 80 and 8080

```
server {
    listen      8080 default_server;
    listen      80 default_server;
    listen      [::]:8080 default_server ipv6only=on;
    listen      [::]:80 default_server ipv6only=on;
    server_name localhost;
```

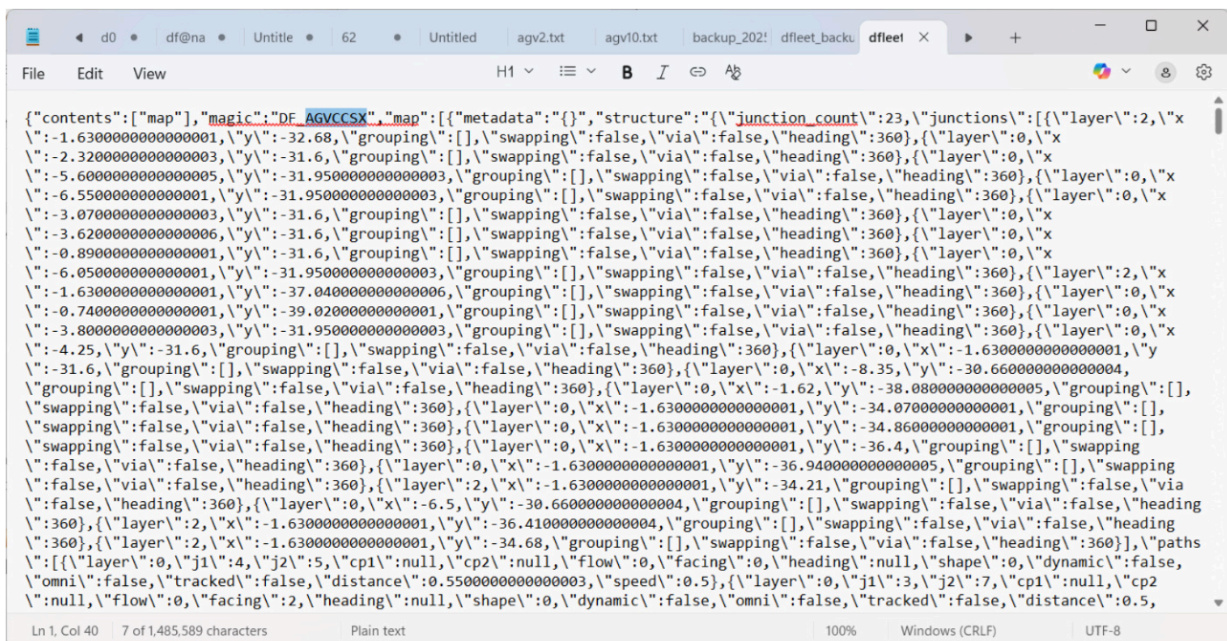
- Reload nginx:
 - `sudo service nginx reload`

How to restore a trackless map from DFleet to Trackless NavWiz?

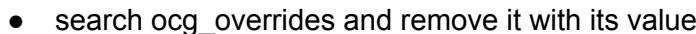
- NEW alternative: use the converter feature on <https://diagnostic.dfautomation.com/>
- Backup the map from DFleet, please backup the maps that you want to convert only.
- Open the DFleet backup.gz file with 7z then right click select edit



- Change AGVCCSX to AGV05X



- Copy the "info" in the last part of NavWiz backup gz file and paste into "info" in the last part of DFleet backup gz file



- Yes to save



- Dont start the NavWiz controller, or you need to kill the agv_mctrl node.
- Change directory into the hardware plugin directory:
 - `cd /var/lib/navwiz/.navwiz_plugin/{your_hardware_plugin}.rootfs`
 - `source opt/ros/melodic/setup.bash`
- Change directory into agv_mctrl lib folder:
 - `cd /var/lib/navwiz/.navwiz_plugin/{your_hardware_plugin}.rootfs/opt/ros/melodic/lib/agv_mctrl`

- Is and you will see there is a file called mctrl_configurator.py
- We can rosrn the file: rosrn agv_mctrl mctrl_configurator.py
- Then you will see the following window:

```
df@ubuntu: /var/lib/navwiz/.navwiz_plugin/zalpha-4.1.0-melodic-20231218_121747.plugin.20240808_155237.rootfs/opt/ros/melodic/lib/agv_mctrl

=====
MCTRL Configurator
=====

Use Arrow key to navigate, and use Space or Enter to click on
the selected line. Use 'q' to exit.

Configuration (click the configuration to configure the parameter)
=====
COM Configuration: /dev/mctrl

[Search now]
```

- Use arrow key to go to the “Search now” and press enter to search for this “dev/mctrl” device:

```
df@ubuntu: /var/lib/navwiz/.navwiz_plugin/zalpha-4.1.0-melodic-20231218_121747.plugin.20240808_155237.rootfs/opt/ros/melodic/lib/agv_mctrl

=====
MCTRL Configurator
=====

Use Arrow key to navigate, and use Space or Enter to click on
the selected line. Use 'q' to exit.

Configuration (click the configuration to configure the parameter)
=====
- COM Configuration: /dev/mctrl

[Search now]
```

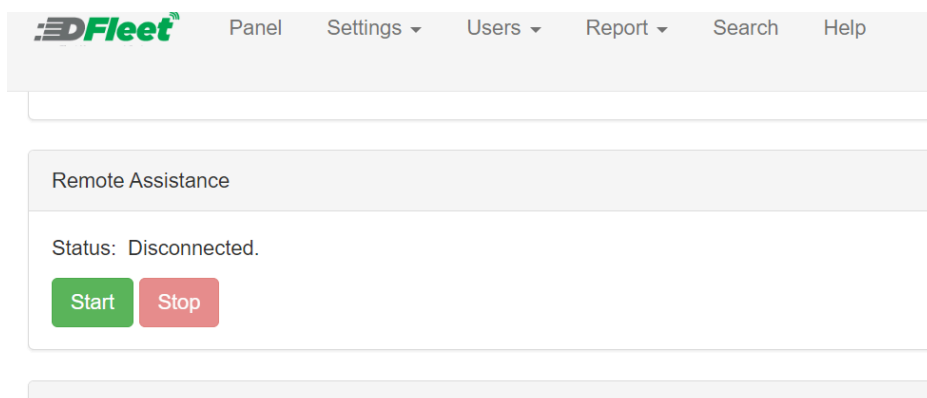
- Then it should show the list of devices found, navigate to the device and press Enter again then you will see the device details.

Network Architecture and Remote Support

How to Remote Support through tunnel

- The onsite engineer or customer should turn on the remote assistance feature. (Take note that the NavWiz or DFleet onsite need to connect to internet)
 - To open remote assistance of DFleet

- Go to Help page
- Press Start button under the Remote Assistance section



- To open remote assistance of NavWiz
 - Navigate to the Help Page on User Panel
 - Click and hold the left mouse button anywhere within the Help page.

- While holding the button, drag the cursor downwards
 - Click “Start” to allow remote access
- Then, the NavWiz or DFleet will automatically tunnel into our office network. You can check the list of links here: <https://connect.dfautomation.com/>
- Take note that, we only have one tunnel for NavWiz, and one tunnel for DFleet. Hence there is only one AGV and one DFleet available to tunnel in at a time.
- So, before turning on remote assistance, make sure currently no other DFleet or NavWiz using the tunnel:

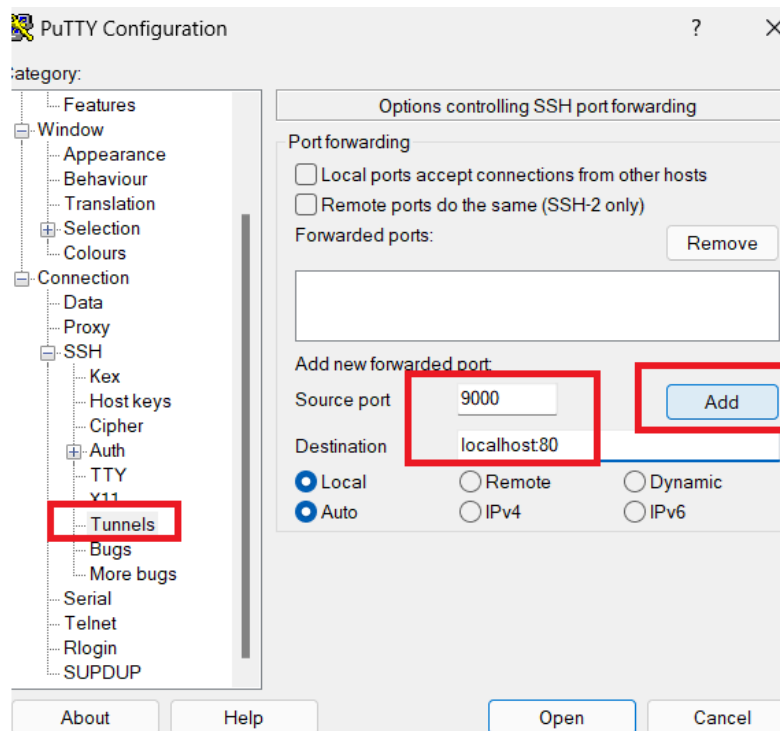
Remote Access Ports

Press F5 to refresh status

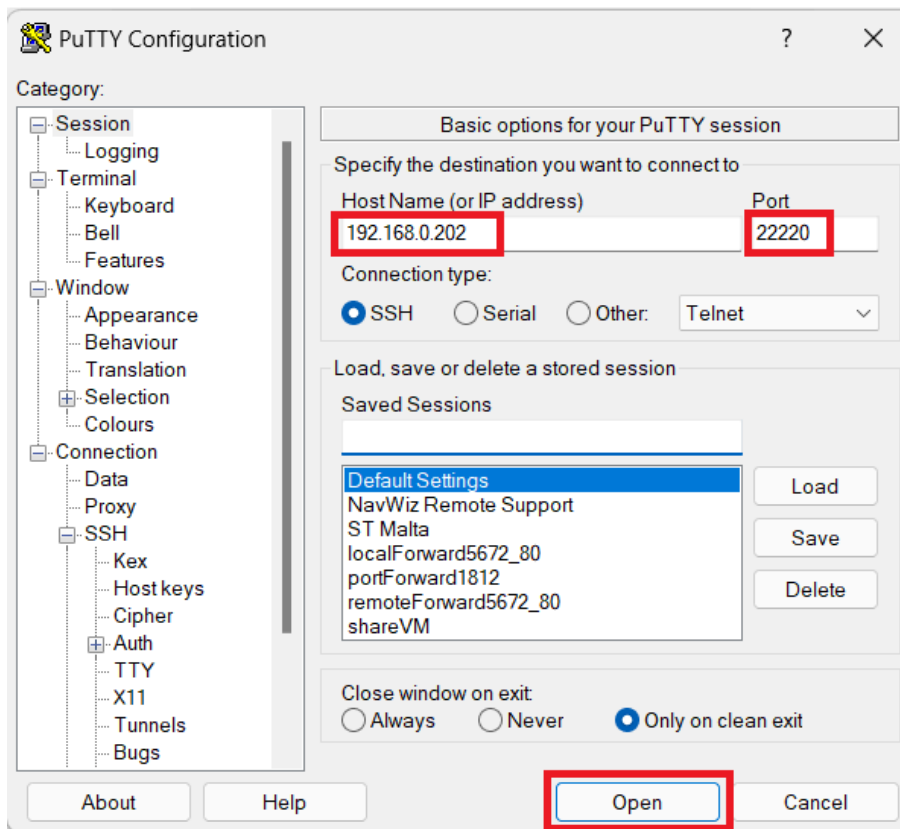
The host for remote access has been changed from 192.168.0.200 to 19

Description	Port	Status
DFleet (common port)	22220	×
NavWiz Lite (common port)	22221	×
NavWiz (common port)	22222	×

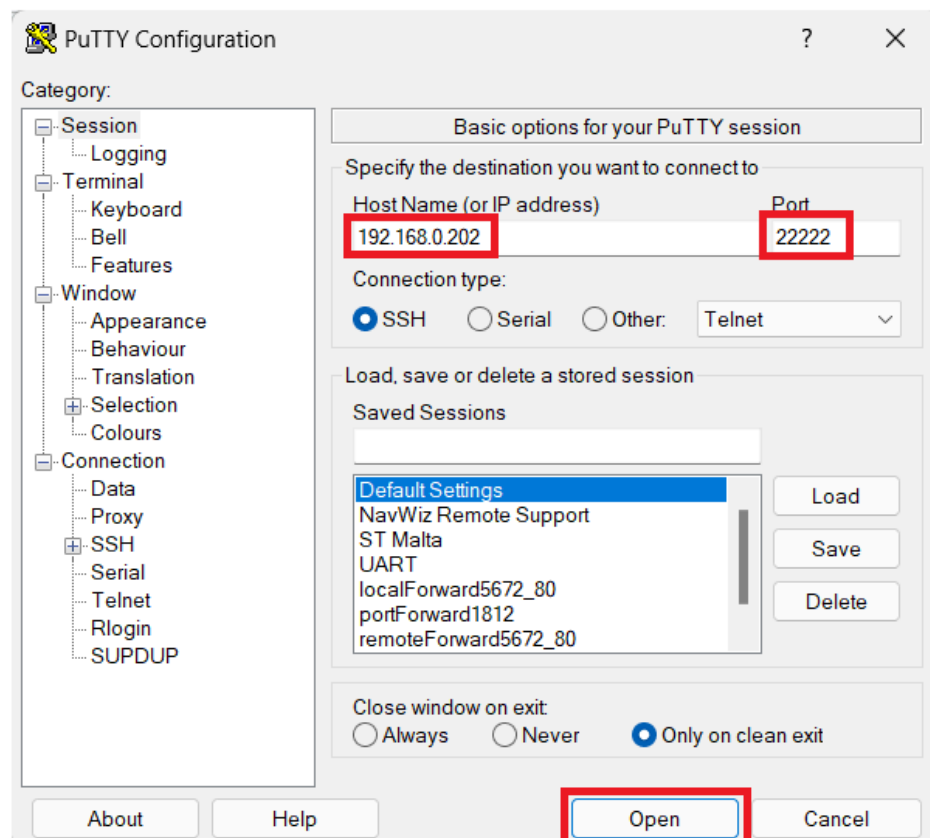
- Then, on your laptop, you should add the local port forwarding of L9000



- Then, use connect to the correct ip and port:
 - DFleet Tunnel: 192.168.0.202 - port 22220



- NavWiz Tunnel: 192.168.0.202 - port 22222



- Then you can access the page using localhost:9000 on your web browser
- If you can access the terminal but cannot access the UI using port 9000:
 - Most likely is because your other application is using port 9000, you should try with other ports such as 9200, 9300 etc
- Remember to ask the onsite engineer or customer to turn off the remote assistance after using.

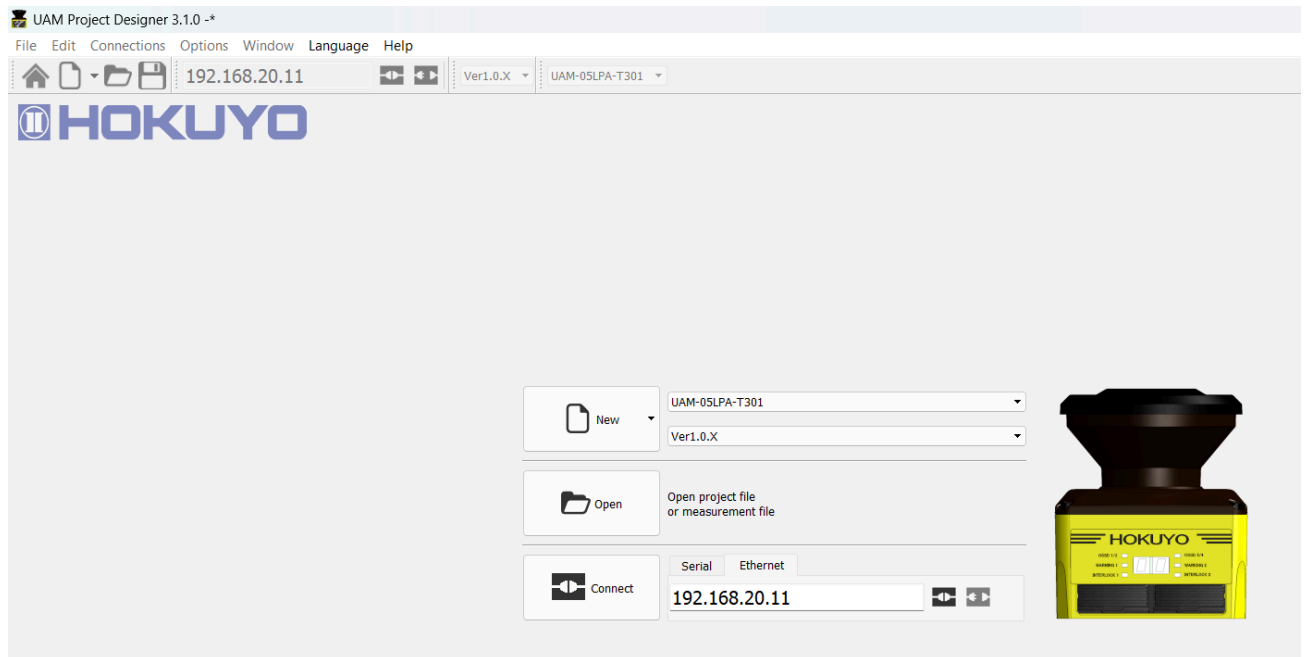
Hardware Configuration and Calibration

UAM Lidar Configuration

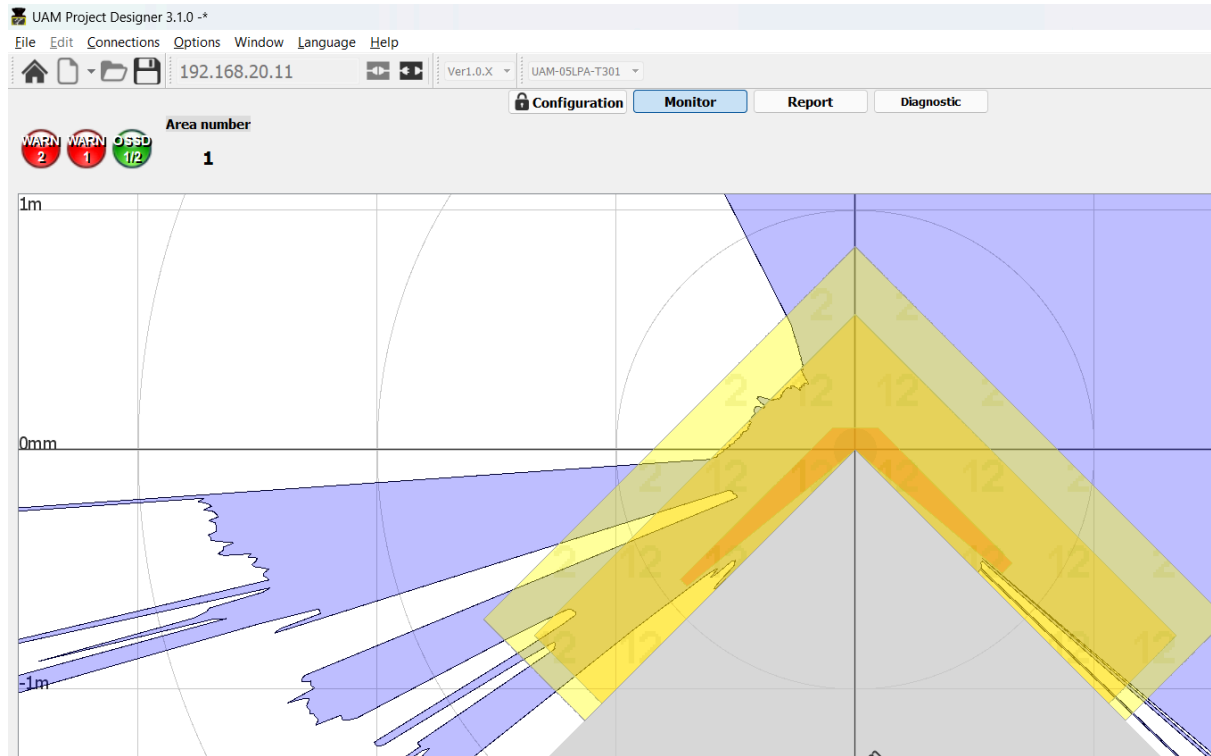
- Once configured using usb, you no longer need to use USB to configure the lidar, you can use LAN cable to configure.
- The IP configured (for omni ST only) are:

```
<launch>
  <arg name="urg_node_front_ip_address" value="$(optenv OMNI_FRONT_LIDAR_IP 192.168.20.11)" />
  <arg name="urg_node_rear_ip_address" value="$(optenv OMNI_REAR_LIDAR_IP 192.168.20.12)" />
```

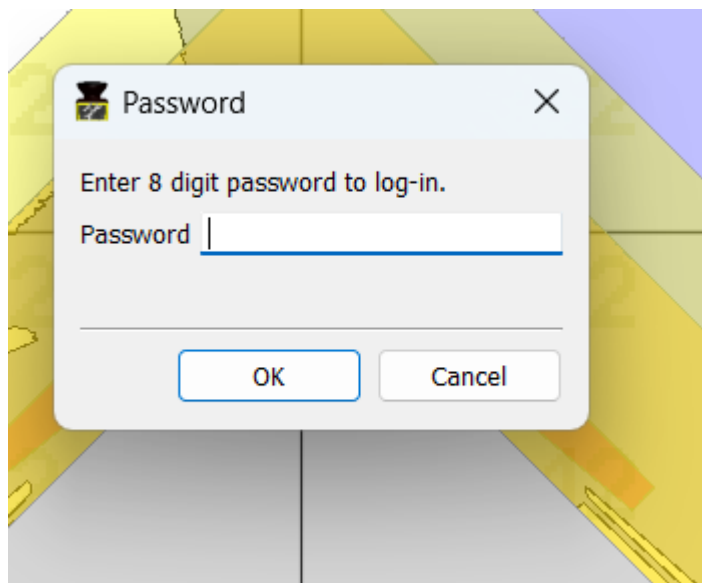
- To configure the UAM lidar, you need UAM Project Designer software.
https://drive.google.com/file/d/1H4NEKCbL2tg6uKFQ5EA1jIQhreohMYWp/view?usp=drive_link
- Connect using Ethernet (if the ip address of the UAM already configured and you already know the ip address), else, connect using USB Cable



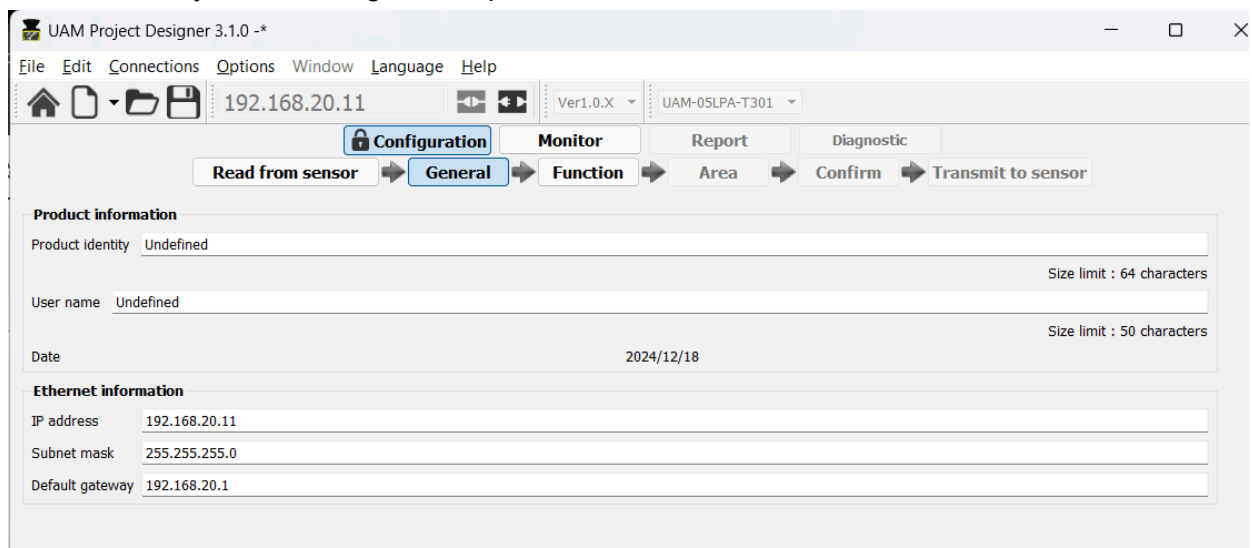
- Click the Configuration button.



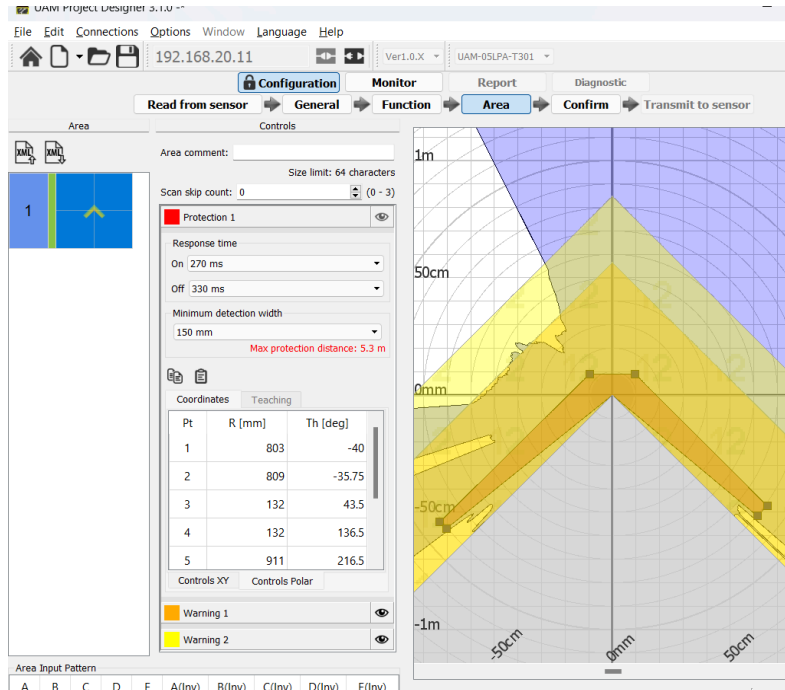
- Password is 12345678



- Here is where you can configure the ip address of the lidar:



- Here is where you can configure the area:



- Then Confirm and Transmit to sensor.

Mapping and Localization Excellence

Mapping Guideline

- Before doing mapping, check the water level of navigation lidars, ensure they are water-leveled or tilted up slightly (should not see floor). If there are multiple AMRs, cross check with another AMR to ensure both AMRs are seeing the same environment.
- Ensure front lidar and rear lidar data are aligned, do the needful to adjust by adjusting the yaw value in agv05_description parameter. Usually you do not need to adjust the x and y offset as it is already defined inside the hardware plugin.

< Parameters

Edit parameter component "/agv05_description"

Lasers Camera Imu

Laser Laser2 Laser3 Laser4 Laser5 Laser6

Laser x offset

0

X offset in meters. (Range: $-0.5 \leq x \leq 0.5$)

Laser y offset

0

Y offset in meters. (Range: $-0.5 \leq x \leq 0.5$)

Laser yaw offset

0

Yaw offset in degrees. (Range: $-90.0 \leq x \leq 90.0$)

Submit

Submit and continue editing

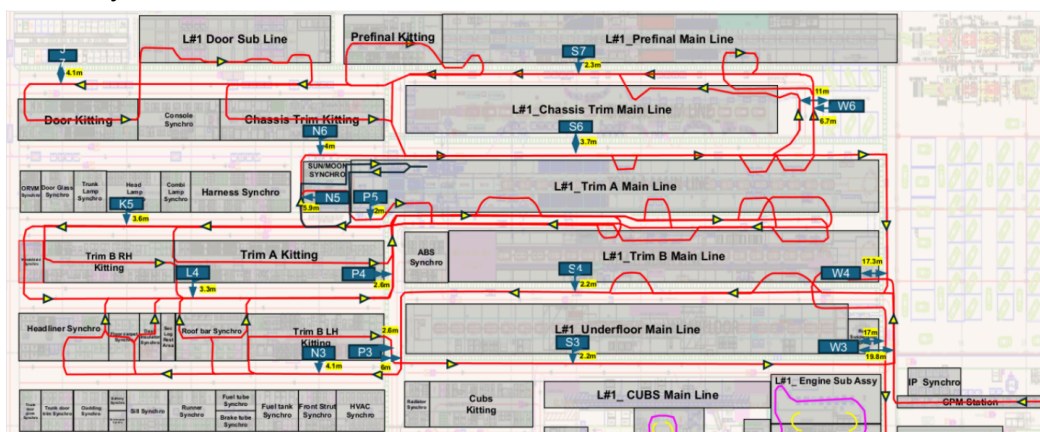
3. Decide whether to use 20mm resolution or 50mm resolution for mapping:
- If the map is small and no map switching and super accurate localization is needed:
 - Proceed with 20mm resolution
 - If the map is very big, you shall use 50mm resolution.

Environment Variable

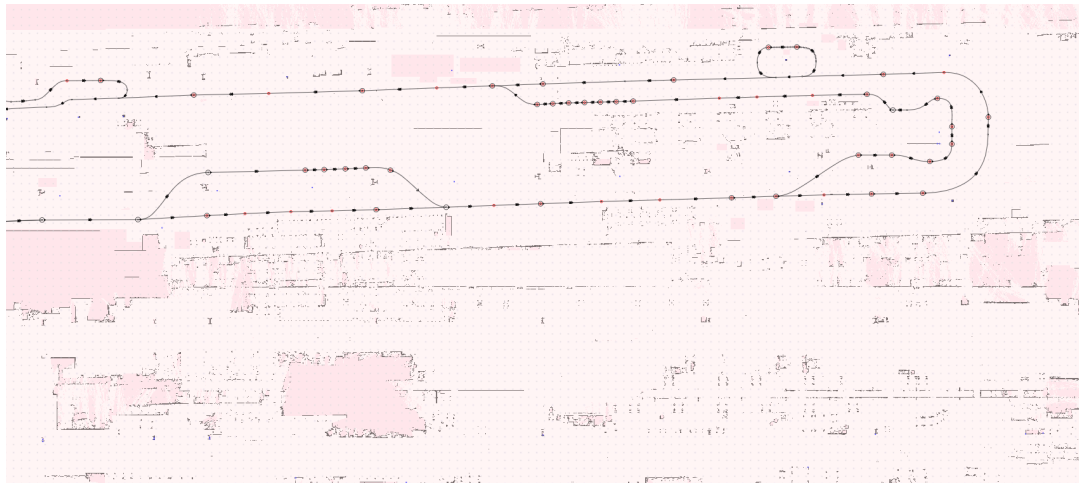
#	Identifier	Value	Operation
1	MAP_RESOLUTION	0.05	   

+

4. For a long corridor or any place without fixed references, place some cardboards or boxes as temporary features. This is necessary to ensure the map is not shifted during the mapping process.
5. Get the layout from the customer, and do mapping according to the orientation of map layout. For example:
- Given layout



- Mapping done



Critical NavWiz Parameters Setup

/agv05_motor

- Configure drive type (either differential or steering or omni etc)

Edit parameter component "/agv05_motor"

Drive type* tricycle_steering_drive (Use tricycle steering drive controller) ▼

Change the drive controller type (soft-reboot required)

- Configure motor fault timeout

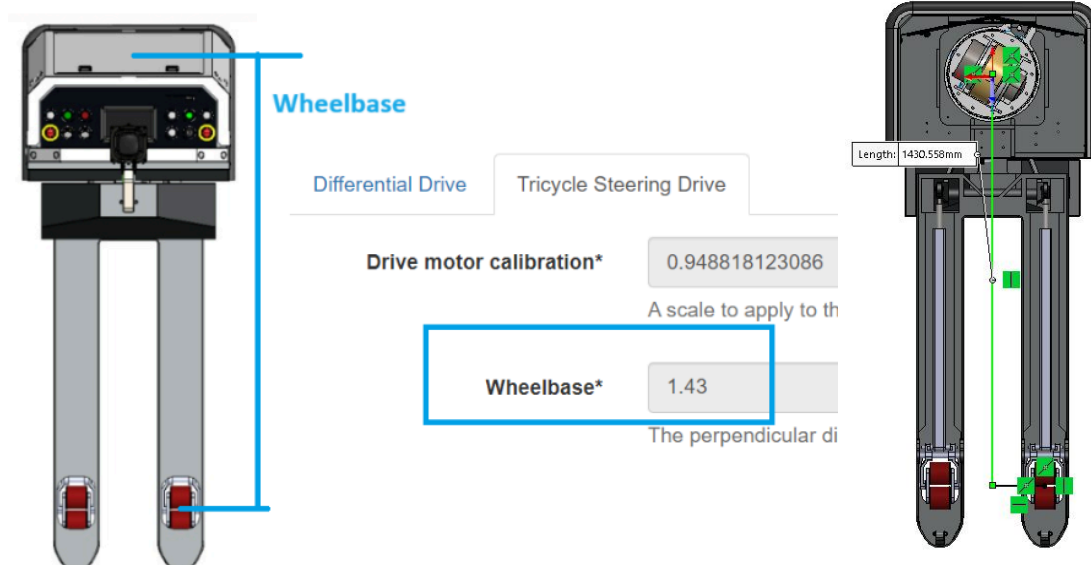
Motor fault trigger
timeout*

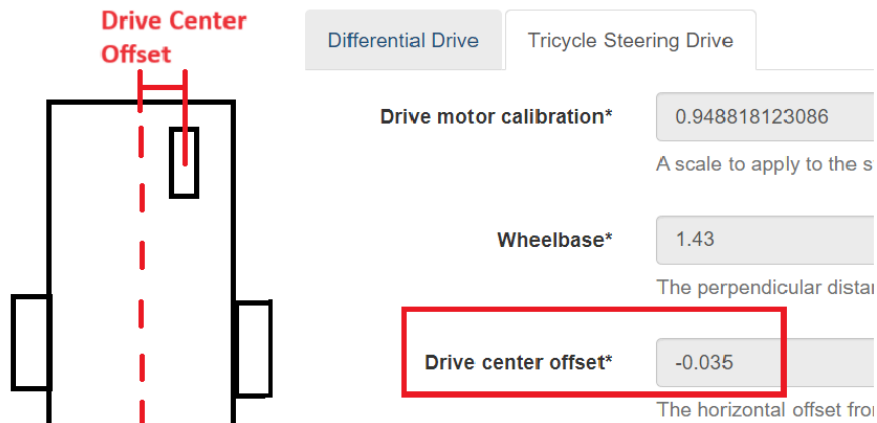
3.0

Motor fault trigger timeout (s). (Range: $0.1 \leq x \leq 10.0$)

- The default is 1 second, which is not enough if there is a delay of 0.5 second in hardware node to release the brake before moving

- (For tricycle steering drive)** Configure wheelbase, steer angle offset and steer angle turn





- **Steer angle turn** is the angle used by the AMR when performing onspot rotation (this is critical if the steering wheel can only turn to a specific angle)
- For wheelbase, if set wrongly, it will affect the turning speed

/agv05_navx

- Edit path follow ahead distance in /agv05_navx/



/agv05_obstacle_sensors

- Check obstacle sensors parameters
 - For lowbed, it does not use a secondary laser, so you need to disable it, else there will be an error.
- Primary obstacle sensors:
 - If have Top lidar, it is top lidar
 - If no top lidar, it is front lidar + rear lidar

/amcl

- The AMCL parameters need to be loaded based on whether the AMR is using top lidar or bottom lidar for navigation.

Edit parameter component "/amcl"

Amcl Reflector Reflector Matching Icp Map Score Inlier Misc

Odom frame id* odom
Which frame to use for odometry.

Base frame id* base
Which frame to use for the robot base.

Global frame id* map
The name of the coordinate frame published by the localization system.

Load profile *
skip (Do not load profile)
profile_1 (Fast recorection and high accuracy with bottom lidar)
profile_2 (Fast recorection and high accuracy with pole lidar)
profile_3 (Slow recorection to reduce swaying movements)
Submit Submit and continue editing

- Select profile_1 and submit if it is using bottom lidar for amcl
 - Select profile_2 and submit if it is using top lidar for amcl
- Reflector intensity threshold needs to be modified based on the type of lidar used, else the reflector detection will be either too sensitive or not able to sense:

Edit parameter component "/amcl"

Amcl Reflector Reflector Matching Icp Map Score Inlier Misc

Reflector intensity threshold* 250.0
Minimum laser intensity when reflected from the reflector surface. (Range: 0.0 ≤ x ≤ inf)

Reflector intensity threshold

- *Minimum laser intensity when reflected from the reflector surface.*
 - **PnF R2000 lidar** : 1,000
 - **Hokuyo UTM30LX** : 10,000
 - **SICK TiM 571/781** : 234
 - **SICK NanoScan3** : 250
- For other parameters related to AMCL, refer to wiki:
<https://wiki.dfautomation.com/product-wiki/amr-system/navwiz/reference/docking-actions>

/marker_localization

- This parameter is only used when the Forward Dock or Reverse Dock actions are used in task templates.

/agv05_state_filter

- This affects the sensitivity of wheel slippage detection, so ensure the values follow Golden Parameter.

Navigation Notes

AMCL

- AMCL only considers true black pixels as a reference obstacle.

AGV Rotate And Search Line Behaviour

- The following is the source code for turn distance calculation:

```
if (line_follow_type_ == Goal::LINE_FOLLOW_ODOM_2D)
{
    float turn_distance = std::abs(goal_.distance);
    turn_distance_ = left_dir_ ? turn_distance : -turn_distance;
}
else
{
    if (nav == Goal::NAV_ROTATE_LEFT || nav == Goal::NAV_ROTATE_RIGHT)
    {
        turn_distance_ = M_PI * 0.5;
    }
    else if (nav == Goal::NAV_UTURN_LEFT || nav == Goal::NAV_UTURN_RIGHT)
    {
        turn_distance_ = M_PI;
    }
    else if (nav == Goal::NAV_ROTATE3Q_LEFT || nav == Goal::NAV_ROTATE3Q_RIGHT)
    {
        turn_distance_ = M_PI * 1.5;
    }
    else
    {
        turn_distance_ = std::abs(goal_.distance);
    }
    turn_distance_ -= config_.turn_search_distance;
}
```

- Based on the source code, the angle where it starts the search line depends on the turn_search_distance parameter.
 - For example, if turn_search distance is set to 0 radian (0 degree), NAV_UTURN_LEFT action will cause the robot to rotate left 180 degrees, then only start to search the line.
 - If turn_search distance is set to 0.5 radian (28 degree), then NAV_UTURN_LEFT action will cause the robot to rotate left 152 degrees, then only start to search the line.
-

Version 4.0 Advanced Features

- Reference:  NavWiz DFleet 4.0 Launching 1.0.pdf

Site Readiness Check

- Preparation:
 - Get the specification sheet of the AMR to understand the dimension of the AMR and the aisle width requirement, floor condition requirement etc.
 - Go through the site readiness checklist and prefilled the requirement:
-

Troubleshooting Guide

Rule of thumb: Always start with these checkings first to avoid unnecessary troubleshooting:

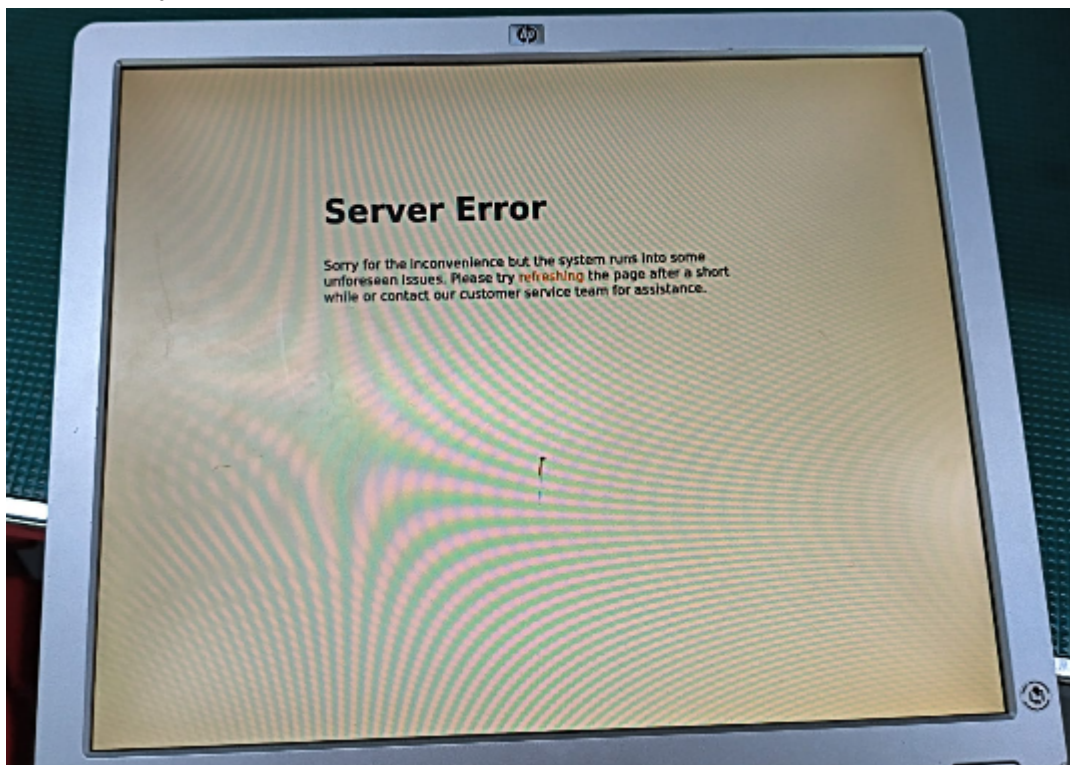
- Check whether NavWiz patch is applied
- Check Robot Configuration
- Check Mainboard firmware version
- Check Network Configuration
- Check Parameters

AMR References:

- Zalpha
 - [Zalpha User Manual](#)
 - IPC UNO2484G Datasheet:
https://advdownload.advantech.com/productfile/Downloadfile4/1-2B2369R/UNO-2484G_User_Manual_En_Ed.FINAL.pdf
-

DFleet Server Error

- This is the symptom of server crashed



- First of all, try to check whether the server still has the ip address using the terminal.
- If able to access, need to get the logs for troubleshooting later:

- /var/log/fms/executor.log
 - /var/log/fms/gunicorn.log
 - /var/log/fms/sockserver.log
- Quick solution to try is to reinstall the patch
 - sudo dpkg -i ./[the patch file]

IPC Hang

- Check syslog
- Check diagnostics for CPU usage and memory usage

Laser Malfunction

- First thing to check is whether the LAN cable is connected and blinking
-

Kiosk is lagging

- This could be due to limited CPU usage.
- To check,

Kiosk/Display Not Showing or Not Working Well

- Sometimes the kiosk may not work, or you may want to manually restart the kiosk.
- For this, you need to enter terminal mode and run this command:
 - sudo service kiosk restart

MCTRL Board not working

- Compare firmware version with another AMR (that is using the same NavWiz version)
- Update the firmware if it is not using the right version, this is because the baud rate could be changed in the newer software.

MSB Calibration Error

- Perform MSB calibration and then observe the Calibrated Max and Calibrated Min, there should not be any weird values that are suddenly very different from others.
- MSB Calibration can fail if the Line Calibration motion didn't finish covering all sensors. This could be due to the track width setting in /agv05_motor.

MSB Line Following Weird Behaviour

- If the AMR does not follow the magnetic tape correctly, perform MSB Calibration according to the procedure in NavWiz User Manual.
- If the calibration succeeded but the MSB Line Following behaviour is still weird, check whether the front and rear MSB are swapped.

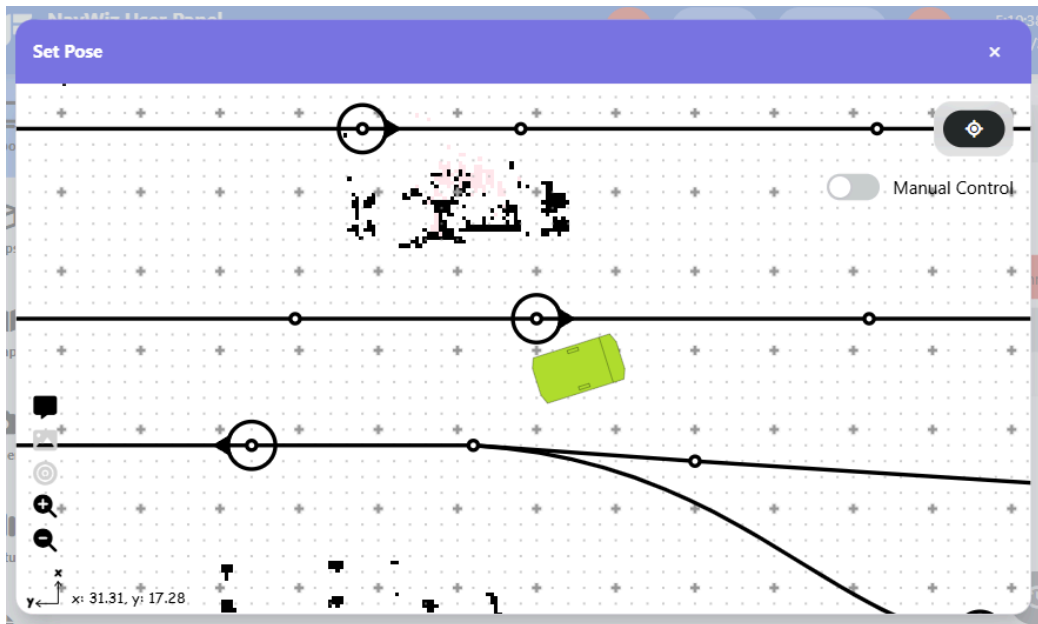
AMR Wobbling During Navigation

- First of all, check the live map and observe whether the localization of the AMR keeps jumping.

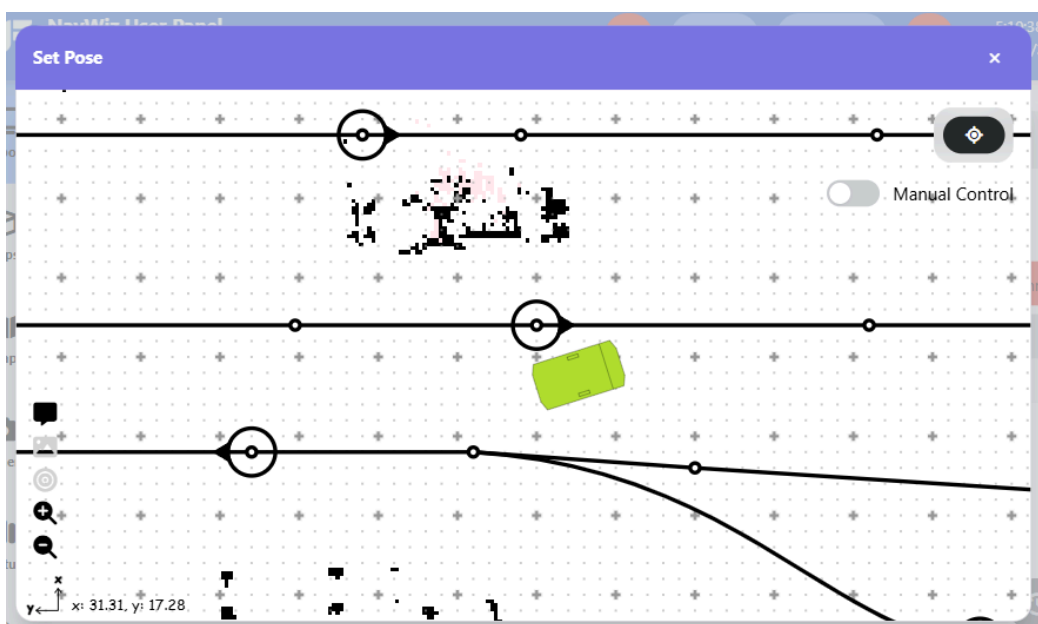
- If the localization keeps jumping, it is due to raw map references not being strong and not correct. Else, it is related to PID values.
-

Navigation Failed Error

- When AMR pops out “Navigation Failed” error, it means the AMR is too far from the planned path.



- This could happen during normal navigation or docking action.
 - When happening during normal navigation, check whether AMR is too far from the planned path (the drawn path).
 - When it happens during docking action, check the live map to see whether AMR is trying to dock something else. An example is AMR doing pallet docking, but it detects the pallet beside it and tries to dock the pallet beside it which is too far from its current position.
- For navigation failed happened during normal navigation, the reasons usually due to:
 - Incorrect starting position
 - Map jump due to no strong reference
- If it was due to map jump (localization of AMR jumped), we can correct it using the Set Pose feature.



- If the AMR is

Motor Fault

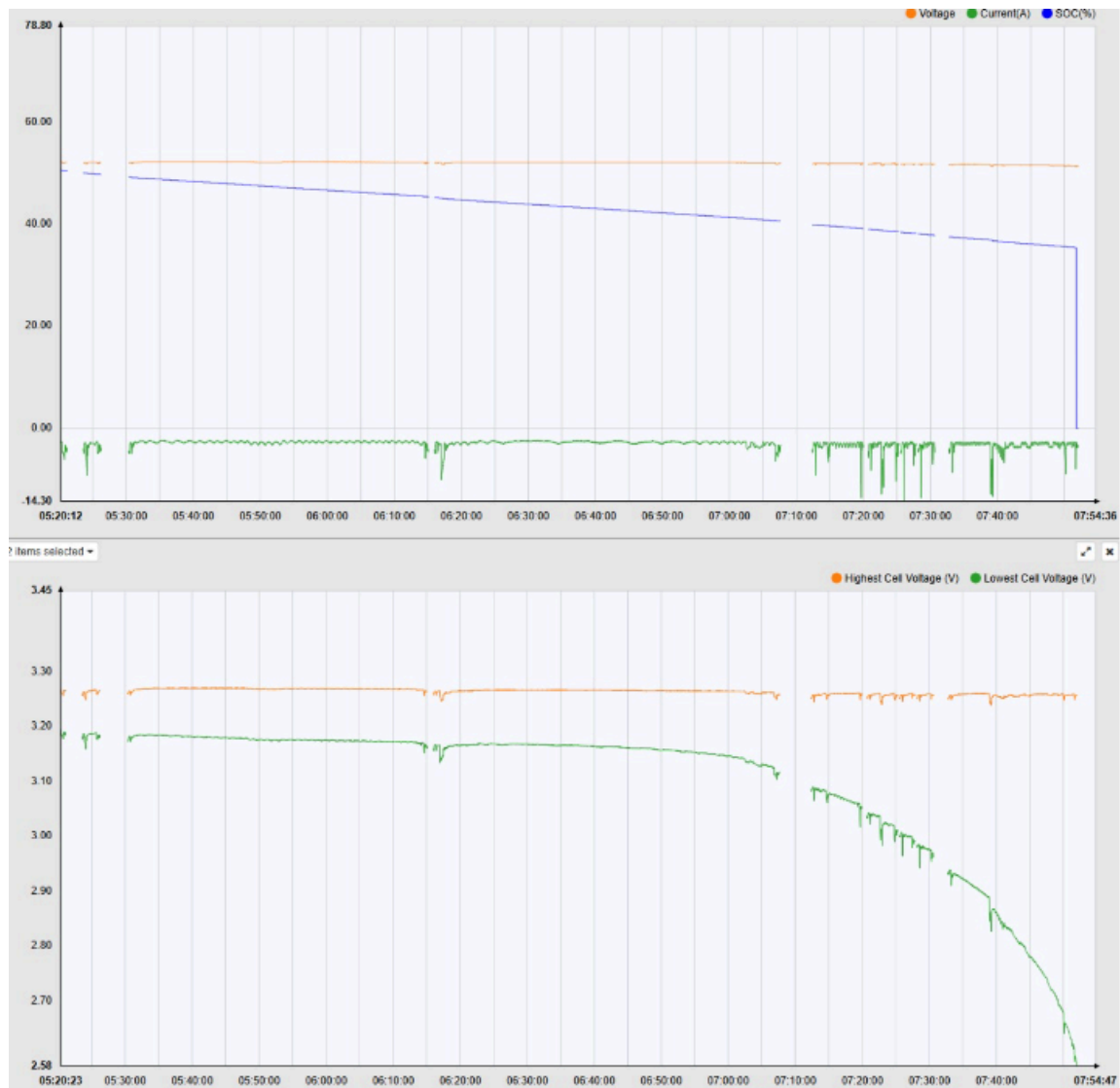
- The Left Speed Index and Right Speed Index in diagnostics are used to check for motor faults.
- In the /agv05_motor parameter, if the Motor fault trigger sensitivity is set to 0.3, and Motor fault trigger timeout is set to 3 , which means if either Left Speed Index or Right Speed Index is less than 0.3 for 3 seconds, then the motor fault will be triggered.
- Another possibility of motor fault is the CAN communication not working, and ensure udev rules is copied to /etc/udev/rules.d if you are using workspace
- Else, check the power supply to the motor. For mabuchi, the brake release should be working.

Battery percentage always showing 0%

- Unlike legacy Zalpha models, most of the new AMR models are getting the data from the Battery Management System..
- Hence, for new AMR models, when battery percentage is showing 0%, it is either:
 - The hardware plugin is not ready with the feature to display battery percentage.
 - The connection between IPC and BMS has issues.

Battery percentage suddenly becomes 0%

- Supposedly each battery has multiple cells inside, if any of the cells is spoiled, it will affect the battery percentage estimation.
- You can verify by checking the highest cell voltage and lowest cell voltage through Diagnostics Visualizer.



Wheel Slippage

- Wheel Slippage means the wheel is rotating but the actual position (amcl) of the robot does not change.
- Wheel Slippage can be triggered also even if the robot is moving (but the robot thought that it is not moving).
- Prerequisites:
 - Wheel slippage thresholds are set by the `/agv05_state_filter` parameter, make sure you use the correct parameter for it.
 - AMCL is crucial part for navigation, ensure you go to the amcl there and load the profile depending on whether you are using top lidar or bottom lidar:

Edit parameter component "/amcl"

Amcl
Reflector
Reflector Matching
Icp
Map Score
Inlier
Misc

Odom frame id*
odom
Which frame to use for odometry.

Base frame id*
base
Which frame to use for the robot base.

Global frame id*
map
The name of the coordinate frame published by the localization system.

Load profile *
skip (Do not load profile)
skip (Do not load profile)
profile_1 (Fast resection and high accuracy with bottom lidar)
profile_2 (Fast resection and high accuracy with pole lidar)
profile_3 (Slow resection to reduce swaying movements)
Submit
Submit and continue editing

- Check floor condition
 - Observe whether the floor condition is really bad and causing the wheel hanging
 - Try it on another flat floor place and try again to observe whether wheel slippage triggered.
- Check lidar mounting and robot configuration:
 - Put your hand close to the front left of the AMR, check the scan data and ensure the scan data shows that your hand is at the front left of the AMR.
 - Then, put your hand close to the rear left of the AMR, check the scan data and ensure the scan data shows that your hand is at the rear left of the AMR.
 - If one of the above is wrong, ensure the mobile robot provider that you selected does match with the actual robot:

Robot

☐ Automatically Start Robot Controller

Robot Type
Trackless

Mobile Robot Capability Provider
zalpha_node/zalpa_ts_bottom

Network Configuration
zalpa_node/zalpa_mg
zalpa_node/zalpa_ts_bottom
zalpa_node/zalpa_ts_top
zalpa_node/zalpa_ts_top_bottom_custom

- If your AMR model is Zalpha, ensure that your ZALPHA_TYPE and ZALPHA_MOTOR configuration are correct:

#	Identifier	Value	Operation
1	ZALPHA_TYPE	lowbed	↑ ↓ 📄 ✕
2	ZALPHA_MOTOR	maxon300	↑ ↓ 📄 ✕

- ZALPHA_TYPE
 - lowbed
 - standard
 - extended
 - ZALPHA_MOTOR
 - maxon100
 - maxon200
 - maxon300
 - mabuchi10
 - mabuchi20

- Use hardware test:

- Move the left and right wheel, make sure they are moving in the correct direction as expected.
 - Observe whether the left distance and right distance change accordingly.
 - If no, meaning the wiring connection might be wrong.
- Observe lidar
 - If one of the lidar is tilted (always scanning the floor), then the AMCL of the robot will not be updated (due to no changes in lidar scanning data), then the robot will think it does not move and hence trigger wheel slippage.
- If from the live map or laser sensor page, the position of the lidar scan is weird, you need to check the URDF.
- Check IMU data
 - If the AMR does have an IMU, check the model from diagnostics status, see whether it is B6 model or A9 model.
 - B6 model is easier to be interrupted by magnetic interference as compared to A9 model
 - Check the diagnostics data to check whether there are jumping values for IMU data.
 - If needed to disable IMU, change the `agv05_motor -> imu_odom_timeout` parameter to 0.

Lidar/Laser Malfunction

- Physically check whether there are any error codes on the lidar
 - If yes, try to restart the lidar
 - If it is Nanoscan3, follow this guide: <https://wiki.dfautomation.com/technical-committee/e-e/hardware/sensor/nanoscan3/how-to-restart-sick-nanoscan3-without-powering-off-the-amr>
 - After restarting the lidar, you may need to soft reboot the NavWiz.
- Sometimes the laser malfunctions are referring to embedded lasers, which could be happening due to hardware node frequency drop, as the embedded lasers are handled by the hardware node.

AMR shutdown

- Check `/var/log/auth.log` and see whether there is something like this:

```
Jun 18 14:59:43 coherent sudo: navwiz : TTY=unknown ; PWD=/ ; USER=root ;
COMMAND=/usr/sbin/poweroff
```
- Then, go to check `/var/log/gunicorn.log` and see whether someone triggered `/boot` API, something like this:

```
[2025-06-18 14:59:10 +0800] [1578] [DEBUG] POST /api/v3/boot
[2025-06-18 14:59:26 +0800] [1578] [DEBUG] POST /api/v3/boot
```
- Then, you can further go to `/var/log/nginx/access.log` to see the HTTP record:

```
192.168.1.68 - - [18/Jun/2025:14:59:41 +0800] "POST /api/v3/boot HTTP/1.1" 499 0
"http://192.168.1.181/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
like Gecko) Chrome/137.0.0.0 Safari/537.36"
```
- How to find out the hostname?
 - `sudo apt install nmap`
 - `nmap -O 192.168.1.68`

Wifi not showing in list

- If it is a hidden network, it will not be visible in the wifi list on the UI.
- To connect to the hidden network, you need to edit the `/etc/wpa_supplicant/wpa_supplicant.conf` and add the network manually.
- The following is an example:


```
network={
```

```

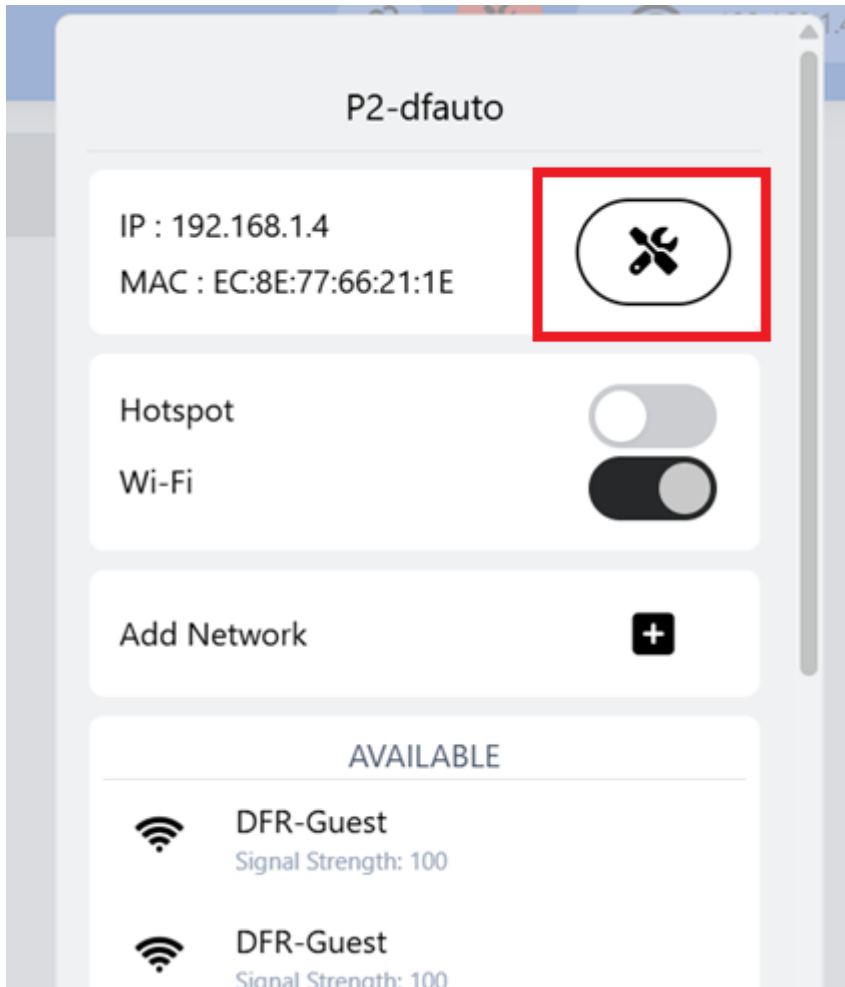
ssid="WIFI_SSID_HERE"
scan_ssid=1
psk="WIFI_PASSWORD_HERE"
bgscan="simple:5:-65:300"
}

```

- Take note that, the scan_ssid=1 is necessary for a hidden network.
- Then, power off the AMR and turn it on again to check.

Wifi not connecting and not showing ip address

- Check static ip configuration



- Check the keep alive parameter whether it is enabled and keep trying to connect to a different wifi.
- Check /var/log/syslog to check log
 - If you see the wpa_supplicant keep trying to associate a connection with the same SSID but with different frequencies (2GHz and 5GHz). For example, something like this:
 Jun 18 17:08:07 amr4 wpa_supplicant[320519]: wlp3s0: SME: Trying to authenticate with 5c:a6:2d:a0:5a:60 (SSID='WFINDUS2' freq=2437 MHz)
 Jun 18 17:08:21 amr4 wpa_supplicant[320519]: wlp3s0: CTRL-Event-SSID-REENABLED id=2 ssid="WFINDUS2"
 Jun 18 17:08:21 amr4 wpa_supplicant[320519]: wlp3s0: SME: Trying to authenticate with 5c:a6:2d:a0:5a:6f (SSID='WFINDUS2' freq=5180 MHz)
 - To fix the wifi card keeps switching between different frequencies of same SSID:
 - Apply the latest linux_wifi patch
 - Add the environment variable below in /etc/default/navwiz-core
 WIFI_FREQ_LIST='2412,2417,2422,2427,2432,2437,2442,2447,2452,2457,2462,2467,2472,2484'
 - Turn off Wifi -> Soft Reboot -> Turn on Wifi

- If you are using PEAP/EAP network:
 - If you are using NavWiz 3.1 and before, check whether you have applied patch for linux_wifi, there should be a new patch that fixes the issue of NavWiz could not connect to PEAP/EAP network without CA cert.
- One more thing to try is:
 - Check whether power_save is turned on:
 - iw dev wlp3s0(change this to your interface) get power_save
 - If it is turned on, try turning it off:
 - sudo iw dev wlp3s0(change this to your interface) set power_save off
 - TAKE NOTE THAT, this is temporary solution for you to test whether this is the root cause, if this solves the problem, you still need to make it permanently turn off.
 - Go to /etc/systemd/system
 - Add a file called disable-powersave.service
 - Add this:


```
[Unit]
Description=Disable Wi-Fi Power Save
After=network.target

[Service]
Type=oneshot
ExecStart=/sbin/iw dev wlp3s0 set power_save off
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```
 - Then you can manually start the service: sudo systemctl start disable-wifi-powersave.service

Connection With DFleet not stable / WiFi connection not stable

- If AMR is in DFleet mode, and frequently showing DFleet Disconnected, it is most likely due to unstable wifi connection.
- To check wifi stability (Especially connection with DFleet), connect the AMR to dfleet, and navigate the AMR to all planned areas.
- Then, get the dfleet backup file and NavWiz diagnostics.
- Upload both of them to <https://diagnostic.dfautomation.com/> or <https://hub.dfautomation.com/diagnostic/>

Navwiz Visualizer

User Guide:

- Upload your diagnostic file in .gz or .csv format.
- Once entered editor page, click on the top button to open config panel
- There are up to maximum 4 panels at one time, each of them can be used to view different data.
- However, the timeline slider is universal for all panels.
- Backup restore file is required for some features to work.

Current limitations:

Limitation of header mapping feature:

- Some versions of diagnostic files have the problem of AGV MSB Front and Rear columns couldn't be differentiated

Diagnostic File*

Choose File 20260409.csv

☒ fast mode

Backup File

Choose File AMR22_backup...60409_172258.gz

Completed Task File

Choose File No file chosen

Completed Transaction File

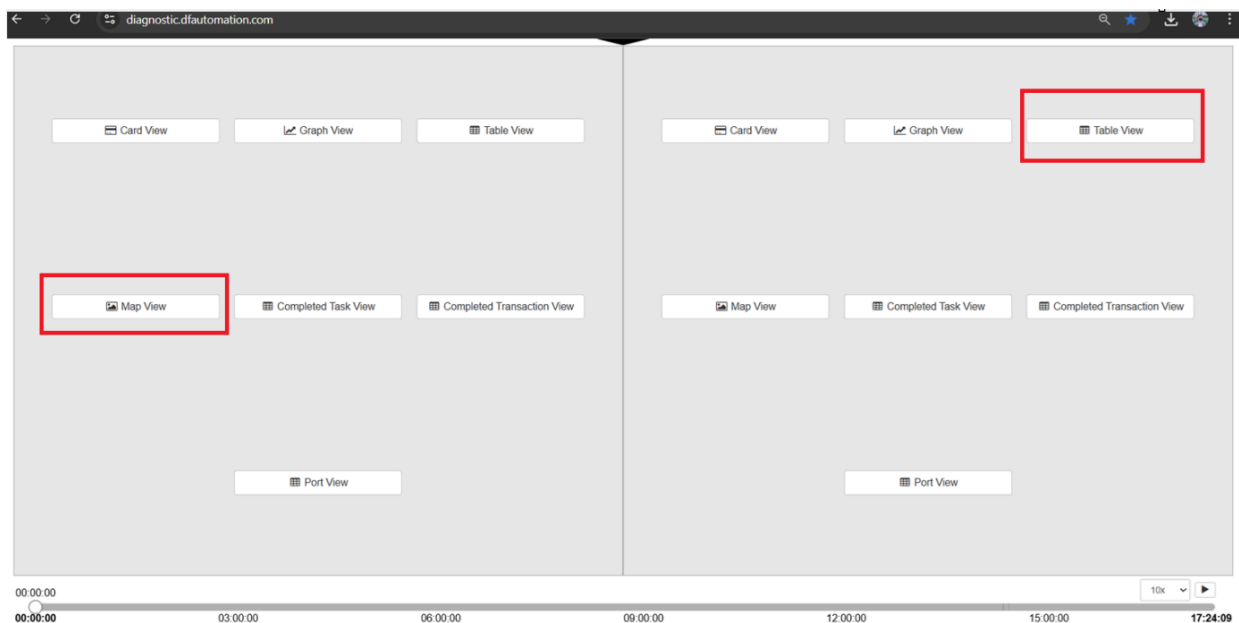
Choose File No file chosen

Carrier Log File

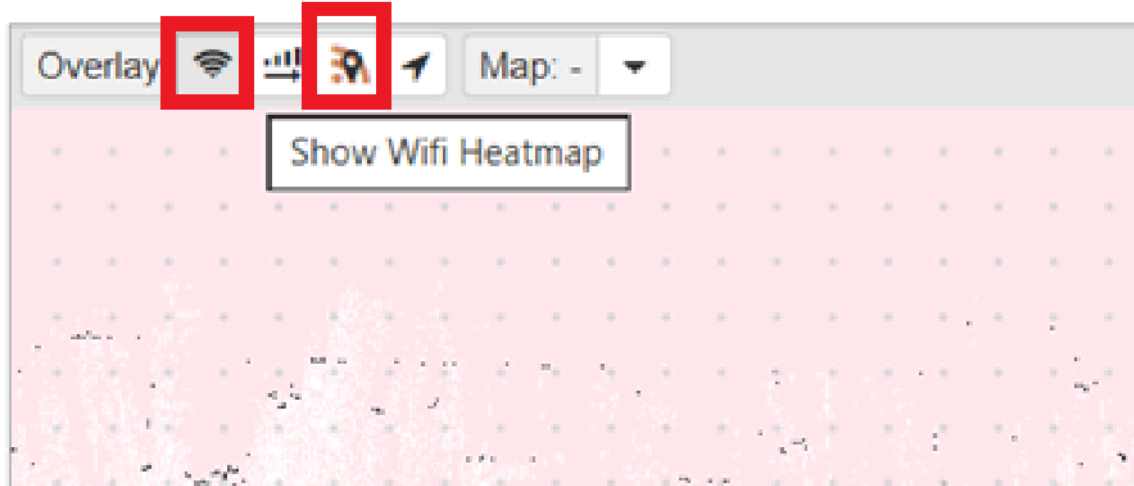
Choose File No file chosen

Upload

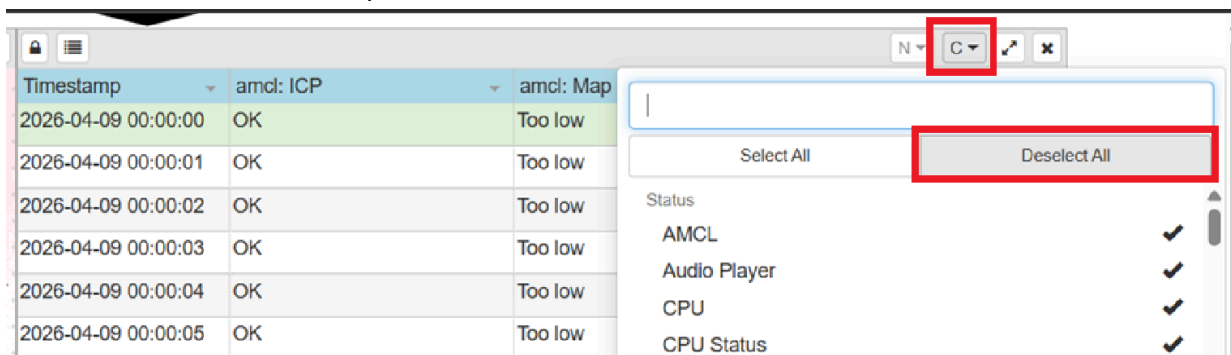
- Then, open one Map View and one Table View



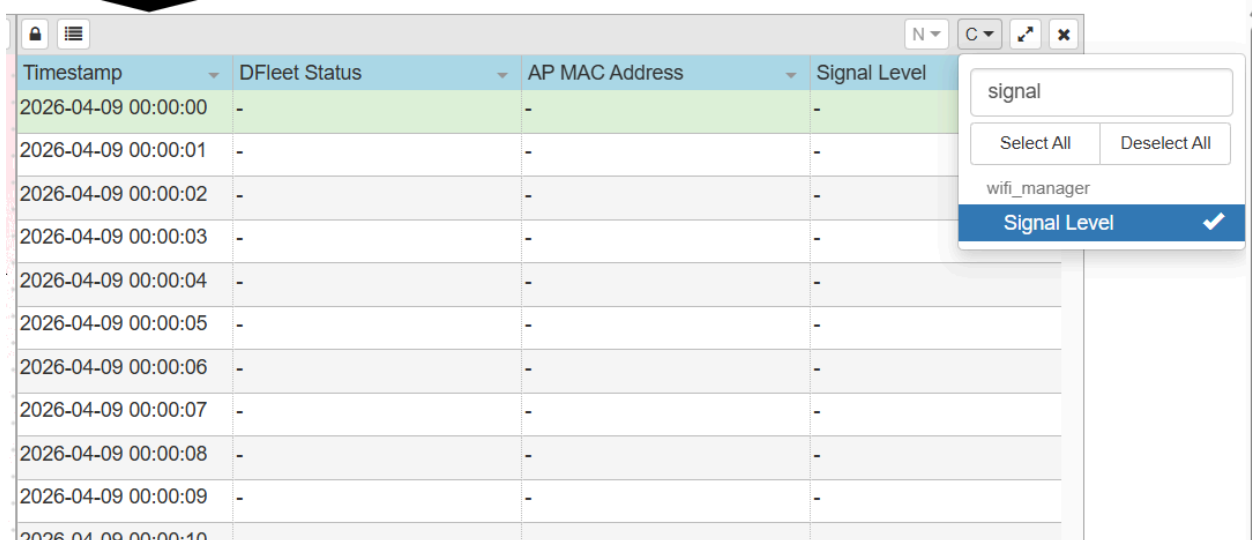
- For Map View, click this 2 buttons:



- For Map View, scroll to zoom in, and make this button active also to make the view keep following robot position.
- For Table View, click the drop down of Column Filter and click Deselect All



- Then, for table view, show these columns:
 - DFleet Status
 - AP MAC Address
 - Signal Level



AMR not connected to DFleet

- Check the WiFi connected by the AMR and DFleet
- From the AMRs, try to ping each other and ping the DFleet

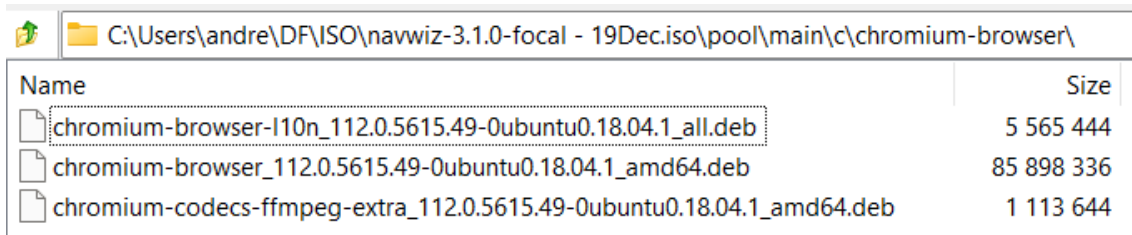
- If there are multiple AMRs within the same DFleet but only one is disconnected, check whether their access points are the same through the Config Panel page.

wlp1s0 (Main Interface)

MAC Address:	E8:62:BE:B7:EE:F9
Wireless:	IEEE 802.11
SSID:	P2-dfauto
Mode:	Managed
Frequency:	5.200 GHz
Access Point:	D2:21:F9:98:2D:9F
Link Speed:	573.5 Mbps
Link Quality:	61/70
Signal Level:	-49 dBm

Kiosk not running but accessible on web browser

- This is most likely due to the chromium browser version not compatible.
- This could happen if someone did `sudo apt update` and `sudo apt upgrade`.
- To fix this, extract the chromium browser debian files from the iso. For example:



Name	Size
chromium-browser-l10n_112.0.5615.49-0ubuntu0.18.04.1_all.deb	5 565 444
chromium-browser_112.0.5615.49-0ubuntu0.18.04.1_amd64.deb	85 898 336
chromium-codecs-ffmpeg-extra_112.0.5615.49-0ubuntu0.18.04.1_amd64.deb	1 113 644

- Install the debian files above to revert the chromium browser version. You should `dpkg -i` the codecs debian first as it is the dependency of the other 2 debian files.

Audio uploaded is chipmunks sound when played on AMR

- This happened because the sample rate of the audio file uploaded does not match with the sample rate of the audio card.
- For example, if the sample rate of the audio card is using 44100kHz, but the sample rate of the audio file is 48000kHz, it will play the audio very fast (which is the chipmunk sound).
- To check the audio rate, can use something like: `cat /proc/asound/card0/pcm0p/sub0/hw_params`

```
df@ubuntu:~$ cat /proc/asound/card0/pcm0p/sub0/hw_params
access: MMAP_INTERLEAVED
format: S16_LE
subformat: STD
channels: 2
rate: 44100 (44100/1)
period_size: 8192
buffer_size: 16384
```

- To convert the sample rate, you can use an online converter.

Traffic Issue: AMR Stops at Junctions

- This happens usually because the path is too long, or too many junctions in between.
- There is a setting in the Traffic Controller setting, called Maximum Buffer Count.



Traffic Controller

Maximum buffer count

The maximum distance and junction count (∩) that can be approved and sent to an AGV (in the navigation plan) ahead of the AGV's current position. Increase this value to reduce intermittent stop due to poor wifi condition, but doing so makes AGV reserve longer path and reduces path sharing efficiency.

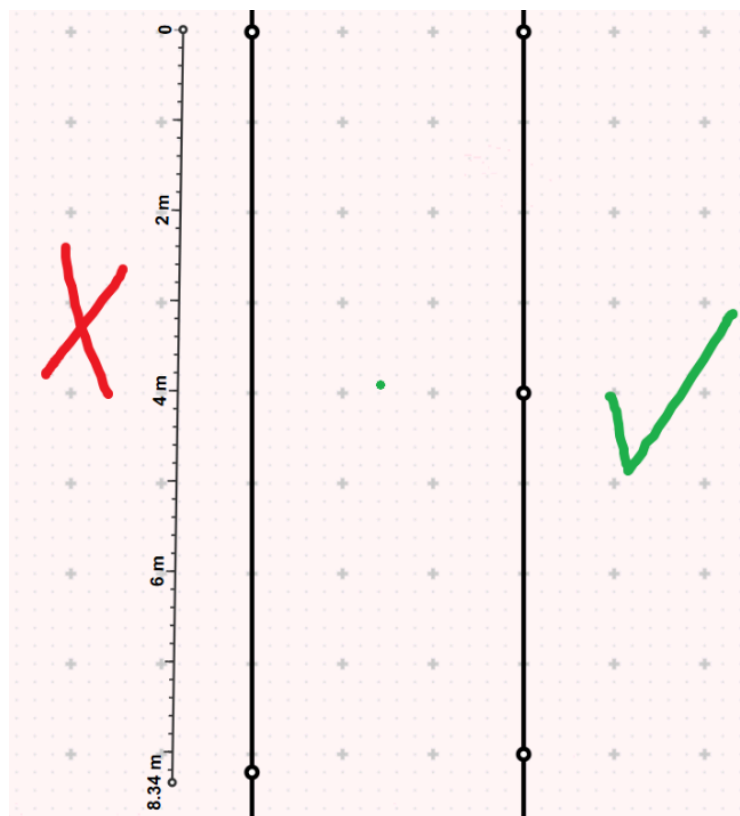
⇒ distance*

default: 15, range: [0.01, 1000000.0]

⇒ junction count*

default: 5, range: [1, 10000]

- Setting the buffer too high will cause the AMR to reserve a very long path. On the other hand, setting the buffer too low will cause the AMR to stop frequently.
- In most cases, it was because the path was too long. The general guide is to make every path to be within 5 meters.



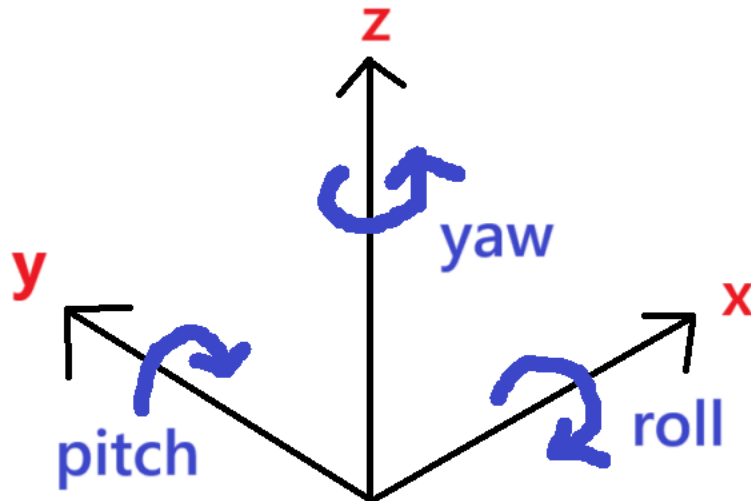
Zalpa Fuse Burnt

- This issue sometimes is not really fuse burnt, sometimes is due to the MCTRL USB cable loosen. Please check the connection.

- Camera Calibration Check

- If the view appears entirely blue, the camera is not calibrated.
- The floor should appear green when calibration is correct.
- Calibration Sequence
 - Calibration must be performed in the following order:
 - Z offset
 - Roll and pitch offsets
 - Yaw offset

- X and Y offsets
- Axis Definitions



- Roll: Rotation around the X-axis (forward direction)
- Pitch: Rotation around the Y-axis (left direction)
- Yaw: Rotation around the Z-axis (upward direction)
- Verification Steps
 - In the camera calibration view, ensure the floor appears green.
 - Place a vertical flat surface facing the camera at the center of the castor wheels, ensuring it is symmetrical on both the left and right sides.
 - Trigger pallet detection in the camera view, then open the map view.
 - Verify that:
 - The green straight line is perpendicular to the robot's centerline (symmetry axis)
 - The intersection point aligns with the robot base
 - These checks confirm correct calibration for Steps 3 and 4.
- Pallet Detection Parameters
 - Refer to the official documentation for parameter configuration: [Docking Actions Wiki](#)